CONCENTRATION OF MEASURE
fall term 2015
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Suggested literature

- **Ledoux "The Concentration of Measure Phenomenon"** Very general, may be hard to read
- **Dubhashi, Panconesi "Concentration of Measure for the Analysis of Randomized Algorithms"** Quite elementary, from the computer science point of view, mainly describes martingale method, has lots of application examples (to CS)
- **Boucheron, Lugosi, Massart "Concentration Inequalities"** The best book is mainly dealing with concentration on product spaces (examples like log concave measures or distributions over convex bodies are covered scarcely).
- **Lecture notes from Professor Alexander Barvinok class** on measure concentration (math 710, winter 2005). Available here: http://www.math.lsa.umich.edu/ barvinok/total710.pdf.

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Contents

1. 09/08 Concentration on S^{n-1}, Levy isoperimetric inequality

The first real application of the concentration of measure phenomenon appeared in the Vitaly Milman's proof of the Dvoretzky theorem.

Theorem 1.1 (Dvoretzky)**.** *Let X be a normed space, $\dim X = n$. Then for any $\varepsilon > 0$, there exists a Hilbert space norm $\|.\|_2$ and a subspace $E \subset \mathbb{R}^n$ of $\dim E > c(\varepsilon)\log(n)$, such that for any $x \in E$*

$$(1-\varepsilon)\|x\|_2 \leq \|x\|_X \leq (1+\varepsilon)\|x\|_2.$$

Informally, it says that all the sections of the dimensions up to $c\log(n)$ of every convex body in $\mathbb{R}^n$ are almost ellipsoidal. Original proof of the theorem (by Dvoretzky) included heavy computations of certain integrals over the Grassmanian $G_{\dim E,n}$. Milman found a completely different proof making use of the following phenomenon:

Let $A \subset X$ and X is a metric probability space $(X, d, \mathbb{P})$, $\mathbb{P}(A) = \frac{1}{2}$. For every $\varepsilon > 0$ let $A_\varepsilon = \{x \in \mathbb{R}^n : d(x, A) \leq \varepsilon\}$. It might happen that $\mathbb{P}(A_\varepsilon) = 1 - o(1)$ when $\varepsilon \to 0$.

Clearly, this does not happen always (even in high dimension).

For example, if K is the standard unit cube $[0,1]^n \subset \mathbb{R}^n$ with the Lebesgue measure m on it, and A is its half defined as $A = \{x \in \mathbb{R}^n : x_1 \leq \frac{1}{2}\}$, then $A_\varepsilon = \{x \in \mathbb{R}^n : x_1 \leq \frac{1}{2} + \varepsilon\}$ and $m(A_\varepsilon) = \frac{1}{2} + \varepsilon$, i.e. far from one for epsilon close to zero.

However, the fact holds true for the crucial case S^{n-1} (with the geodesic distance d and uniform measure $\mathbb{P}$ on the surface).

Lemma 1.2. *Let $A = \{x \in S^{n-1} : x_1 \leq 0\}$. Then $\mathbb{P}(A_\varepsilon)$ is exponentially close to 1. Precisely,*

$$\mathbb{P}(A_\varepsilon) = 1 - C \cdot \exp(-\varepsilon^2(n-2)/2)$$

for some absolute constant $C > 0$.

Remark 1.3. This lemma together with the exact value of the constant C can be obtained by explicit computation using measure of the sphere formula. We will follow an alternative way.

Proof. Clearly,

$$A_\varepsilon = \{x \ \in S^{n-1} : |\theta| < \frac{\pi}{2} - \varepsilon\}$$

(θ is the angle between x and normal to the hyperplane $\{x_1 = 0\}$). Let $k := n - 2$. Let us denote $m(S(\theta))$ to be an induced measure of the sphere in $\mathbb{R}^{n-1}$ of radius θ. Then, integrating over slices,

$$\mathbb{P}(A_\varepsilon^c) = \frac{\int_0^{\pi/2-\varepsilon} m(S(\theta))d\theta}{2\int_0^{\pi/2} m(S(\theta))d\theta} = \frac{\int_0^{\pi/2-\varepsilon} \sin^k \theta d\theta}{2\int_0^{\pi/2} \sin^k \theta d\theta} = \frac{\int_\varepsilon^{\pi/2} \cos^k \theta d\theta}{2\int_0^{\pi/2} \cos^k \theta d\theta}.$$

Let us estimate numerator and denominator separately:

$$\int_0^{\pi/2} \cos^k \theta d\theta \geq \int_0^{1/\sqrt{k}} \cos^k \theta d\theta \geq \frac{1}{\sqrt{k}} \cos^k \left(\frac{1}{\sqrt{k}}\right) \geq$$

$$\geq \frac{1}{\sqrt{k}} \left(1 - \frac{1}{2k}\right)^k \geq \frac{c}{\sqrt{k}}.$$

(we used the inequality $\cos\varphi \geq 1 - \varphi^2/2$).

$$\int_\varepsilon^{\pi/2} \cos^k \theta d\theta = \frac{1}{\sqrt{k}} \int_{\varepsilon\sqrt{k}}^{\pi\sqrt{k}/2} \cos^k \left(\frac{\varphi}{\sqrt{k}}\right) d\varphi \leq$$

$$\leq \frac{1}{\sqrt{k}} \int_{\varepsilon\sqrt{k}}^{\pi\sqrt{k}/2} \exp\left(-\frac{\varphi^2}{2}\right) d\varphi \leq$$

$$\leq \frac{1}{\sqrt{k}} \int_{\varepsilon\sqrt{k}}^{\infty} \exp\left(-\frac{\varepsilon^2 k + (\varphi - \varepsilon\sqrt{k})^2}{2}\right) d\varphi \leq$$

$$\leq \frac{C}{\sqrt{k}} \exp\left(-\frac{\varepsilon^2 k}{2}\right),$$

where in the first inequality we used that $\cos\varphi \leq \exp(-\varphi^2/2)$, which can be easily checked by Taylor series comparison. Therefore,

$$\mathbb{P}(A_\varepsilon^c) \leq C' \exp(-\varepsilon^2 k/2).$$

$\square$

Standard isoperimetric inequality claims that a ball has the lowest surface area measure among all the bodies of the same volume. Somewhat similarly, Levy isoperimetric inequality shows that the measure of the ε-neighborhood is minimized in the case of the spherical cap, which is a ball in the geodesic metric:

Theorem 1.4. *[Levy isoperimetric inequality] Let $\mu, \epsilon \in (0,1)$. Let $A \subset S^{n-1}$ be a set such that $\mathbb{P}(A) = \mu$. Then $\mathbb{P}(A_\varepsilon) \geq \mathbb{P}(C(\mu)_\varepsilon)$, where $C(\mu)$ is a spherical cap of measure μ.*

Corollary 1.5. *If $A \subset S^{n+2}$ and $\mathbb{P}(A) = \frac{1}{2}$, then for any $\varepsilon > 0$*

$$\mathbb{P}(A_\varepsilon) \geq 1 - C\exp\left(-\frac{\varepsilon^2 n}{2}\right).$$

Corollary 1.6. *Let $f : S^{n-1} \to \mathbb{R}$ be an 1-Lipschitz function, and $k = n-2$. Then there exists $M \in \mathbb{R}$, such that*

$$\mathbb{P}(|f(x) - M| > \varepsilon) \leq \exp\left(-\frac{\varepsilon^2 k}{2}\right).$$

Proof. Let M be a median of f and let $A = \{x : f(x) \leq M\}$. If $y \in S^{n+1}$ is such that $f(y) > M + \varepsilon$, then $y \notin A_\varepsilon$ (by 1-Lip condition). Hence, by Theorem 1.4

$$\mathbb{P}(f(y) > M + \varepsilon) \leq C \exp(-\varepsilon^2 k/2).$$

Similarly, we can get the same estimate for $\mathbb{P}(f(y) < M - \varepsilon)$. □

This is one example of the geometric method of proving concentration. If the solution to the isoperimetric problem is known and we need only concentration for Lipschitz functions, then the geometric method yields the optimal results. This is very helpful in geometric functional analysis, as one mainly deals with linear functionals, radial functions and norms, all of which are Lipschitz functions. Unfortunately, the collection of metric probability spaces for which the solution of the isoperimetric problem is known is very limited. Moreover, there are many natural cases where concentration is expected and would be very helpful, but the function under consideration is not Lipschitz or the Lipschitz constant is too high. Let us consider some of them.

Example 1.7 (Bin packing problem). Let $X_1, X_2, \dots X_n \in [0,1]$ be i.i.d. random variables representing the weights of the balls. We want to put these balls into the minimal number of bins so that the weight of every bin is at most 1. The function $\psi(n) =$ "minimal number of the bins needed" considered on the product probability space $[0,1]^n$ is integer valued, hence, it is not continuous, and of course, not Lipschitz.

Example 1.8 (Erdos-Renyi graphs). Let $G = G(n,p)$ be the graph with n vertices and with every possible edge appearing independently with the probability p. Let the function $t_G(n) =$ "number of triangles in G" considered on the probability space $\{0,1\}^{\binom{n}{2}}$ (a point in this space encodes the random set of edges of G) with Hamming metric on it. However, if we consider all graphs that differs by one edge ($\mathrm{dist}(X,Y) = 1$), the difference between number of triangles can be of order n ($|t_X(n) - t_Y(n)| \leq n-2$). So the Lipschitz constant of $t_G(n)$ is at least $n-2$, which is too high to get an efficient concentration.

Example 1.9. Let A be $n \times n$ fixed matrix with 0-1 entries. Let g_{ij} be independent standard gaussian variables and let G be $n \times n$ matrix defined by

$$G_{ij} = \begin{cases} g_{ij} & \text{if } a_{ij} = 1, \\ 0 & \text{if } a_{ij} = 0. \end{cases}$$

Metric space is the product space $\mathbb{R}^{n^2}$ endowed with the gaussian measure. Consider the function $F(G) = |\det G|$. This function is a high-degree polynomial, so it is far from being Lipschitz.

For many years Lipschitz condition was required to apply concentration. Then transportation method (by Talagrand, Marton, ...) and information theory approach (by Ledoux, Massart, ...) were developed to overcome this restriction.

In this class we will start with the linear functions case, then continue with the Lipschitz case and then we will talk about more general cases when concentration of measure takes place.

2. 09/10 Subgaussian random variables

We introduce an important class of random variables with strong tail decay properties. This class contains the normal variables, as well as all bounded random variables.

Definition 2.1. *Let $v > 0$. A random variable ξ is called v-subgaussian if there exists a constant C such that for any $t > 0$*

$$\mathbb{P}(|\xi| > t) \leq Ce^{-vt^2}.$$

A random variable ξ is called centered if $\mathbb{E}\xi = 0$.

If the parameter v is an absolute constant, we call a v-subgaussian random variable subgaussian. We shall assume that the random variable ξ is non-degenerate, i.e. $\mathrm{Var}(\xi) > 0$.

The subgaussian condition can be formulated in a number of different ways.

Theorem 2.2. *Let X be a random variable. The following conditions are equivalent:*

(1) *X is subgaussian;*
(2) $\exists a > 0\ \mathbb{E}e^{aX^2} < +\infty$ (ψ_2-condition)*;*
(3) $\exists B, b > 0\ \forall \lambda \in \mathbb{R}\ \ \mathbb{E}e^{\lambda X} \leq Be^{\lambda^2 b}$ (Laplace transform condition)*;*
(4) $\exists K > 0\ \forall p \geq 1\ \ (\mathbb{E}|X|^p)^{1/p} \leq K\sqrt{p}$ (moment condition).

Moreover, if X is a centered random variable, (3) can be rewritten as
(3)′ $\exists b' > 0\ \forall \lambda \in \mathbb{R}\ \mathbb{E}e^{\lambda X} \le e^{\lambda^2 b'}$.

Proof. The proof is a series of elementary calculations.
(1) ⇒ (2) Let $a < v$. By the integral distribution formula,

$$\mathbb{E}e^{aX^2} = 1 + \int_0^\infty 2ate^{at^2} \cdot \mathbb{P}(|X| > t)\, dt \le 1 + \int_0^\infty 2at \cdot Ce^{-(v-a)t^2}\, dt < +\infty.$$

(2) ⇒ (3) Let λ be any real number. Then

$$\mathbb{E}e^{\lambda X} = \mathbb{E}e^{\lambda X - aX^2} e^{aX^2} \le \sup_{t \in \mathbb{R}} e^{\lambda t - at^2} \cdot \mathbb{E}e^{aX^2} \le Be^{\lambda^2/4a}.$$

(3) ⇒ (4) Set $\lambda = \sqrt{p}$. Replacing, as before, the the function by its supremum, we get

$$\mathbb{E}|X|^p \le \sup_{t>0} t^p e^{-\sqrt{p}\, t} \cdot \mathbb{E}e^{\sqrt{p}|X|} \le \left(\frac{\sqrt{p}}{e} \right)^p \cdot Ce^{pb}.$$

(4) ⇒ (1) Assume first $t \ge eK$. Choose p so that $\frac{K\sqrt{p}}{t} = e^{-1}$.

$$\mathbb{P}(|X| > t) \le \frac{\mathbb{E}|X|^p}{t^p} \le \left(\frac{K\sqrt{p}}{t} \right)^p = e^{-p} = e^{-vt^2},$$

where $v = e^{-2}K^{-2}$. This proves (1) for $t \ge eK$. Setting $C = e$ automatically guaranties that (1) holds for $0 < t < eK$ as well.
(3)′ We will assume that (3) holds with $B > 1$ since otherwise the statement is trivial. Assume first that X is symmetric. For large values of λ, we can derive (3) with constant $B = 1$ by changing the parameter b. Indeed, set $\lambda_0 = \sqrt{2a}$ and choose $\bar{b} > 0$ so that $Be^{\lambda_0^2 b} \le e^{\lambda_0^2 \bar{b}}$. This guarantees that (3) holds for all λ such that $|\lambda| \ge \lambda_0$ with $B = 1$ and b replaced by $\bar{b}$.

If $\lambda^2 \le 2a$, then by Holder's inequality and the ψ_2-condition,

$$\mathbb{E}e^{\lambda X} = \mathbb{E}\frac{1}{2}(e^{\lambda X} + e^{-\lambda X}) \le \mathbb{E}e^{\lambda^2 X^2/2} \le \left(\mathbb{E}e^{aX^2} \right)^{\lambda^2/2a} \le \exp\left(c\frac{\lambda^2}{2a} \right).$$

Finally, we set $b' = \max(c/2a, \bar{b})$.

In the general case, we use a simple symmetrization. Let X' be an independent copy of X. Then by Jensen's inequality,

$$\mathbb{E}e^{\lambda X} = \mathbb{E}e^{\lambda(X - \mathbb{E}X')} \le \mathbb{E}e^{\lambda(X - X')},$$

where $X - X'$ is a symmetric subgaussian random variable. □

Remark 2.3. The ψ_2-condition turns the set of centered subgaussian random variables into a normed space. Define the function $\psi_2 : \mathbb{R} \to \mathbb{R}$ by $\psi_2(t) = \exp(t^2) - 1$. Then for a non-zero random variable set

$$\|X\|_{\psi_2} = \inf\{s > 0 \mid \mathbb{E}\psi_2(X/s) \le 1\}.$$

The subgaussian random variables equipped with this norm form an Orlicz space.

The characterization of the subgaussian property proven above allows to establish a concentration inequality for a linear combination of independent centered subgaussian random variables. Note that a linear combination of independent Gaussian random variables is Gaussian. We prove below that a linear combination of independent subgaussian random variables is subgaussian. This was originally proved by Bernstein and rediscovered many times, so the theorem below appears as Chernoff inequality, Hoeffding inequality, or Azuma inequality in different books and papers.

Theorem 2.4. *Let $X_1, \dots, X_n$ be independent centered subgaussian random variables. Then for any $a_1, \dots, a_n \in \mathbb{R}$*

$$\mathbb{P}\left(\left|\sum_{j=1}^{n} a_j X_j\right| > t\right) \le 2\exp\left(-\frac{ct^2}{\sum_{j=1}^{n} a_j^2}\right).$$

Proof. Set $v_j = a_j / \left(\sum_{j=1}^{n} a_j^2\right)^{1/2}$. We have to show that the random variable $Y = \sum_{j=1}^{n} v_j X_j$ is subgaussian. Let us check the Laplace transform condition $(3)'$. For any $\lambda \in \mathbb{R}$

$$\begin{aligned}\mathbb{E}\exp\left(\lambda \sum_{j=1}^{n} v_j X_j\right) &= \prod_{j=1}^{n} \mathbb{E}\exp(\lambda v_j X_j)\\ &\le \prod_{j=1}^{n} \exp(\lambda^2 v_j^2 b) = \exp\left(\lambda^2 b \sum_{j=1}^{n} v_j^2\right) = e^{\lambda^2 b}.\end{aligned}$$

The inequality here follows from $(3)'$. Note that the fact that the constant in front of the exponent in $(3)'$ is 1 plays the crucial role here. □

Theorem 2.4 can be used to give a very short proof of a classical inequality due to Khinchin.

Theorem 2.5 (Khinchin)**.** *Let $X_1, \dots, X_n$ be independent centered subgaussian random variables of unit variance. For any $p \ge 1$ there exist $A_p, B_p > 0$ such that the inequality*

$$A_p \left(\sum_{j=1}^{n} a_j^2\right)^{1/2} \le \left(\mathbb{E}\left|\sum_{j=1}^{n} a_j X_j\right|^p\right)^{1/p} \le B_p \left(\sum_{j=1}^{n} a_j^2\right)^{1/2}$$

holds for all $a_1, \dots, a_n \in \mathbb{R}$.

Proof. Without loss of generality, assume that $\left(\sum_{j=1}^n a_j^2\right)^{1/2} = 1$.

Let $p \geq 2$. Then by Hölder's inequality

$$\left(\sum_{j=1}^n a_j^2\right)^{1/2} = \left(\mathbb{E}\left|\sum_{j=1}^n a_j X_j\right|^2\right)^{1/2} \leq \left(\left|\mathbb{E}\sum_{j=1}^n a_j X_j\right|^p\right)^{1/p},$$

so $A_p = 1$. By Theorem 2.4, $Y = \sum_{j=1}^n a_j X_j$ is a subgaussian random variable. Hence,

$$(\mathbb{E}|Y|^p)^{1/p} \leq C\sqrt{p} =: B_p.$$

In the case $1 \leq p \leq 2$ it is enough to prove the inequality for $p = 1$. As before, by Hölder's inequality, we can choose $B_p = 1$. Applying Khinchin's inequality with $p = 3$, we get

$$\mathbb{E}|Y|^2 = \mathbb{E}|Y|^{1/2} \cdot |Y|^{3/2} \leq (\mathbb{E}|Y|)^{1/2} \cdot \left(\mathbb{E}|Y|^3\right)^{1/2} \leq (\mathbb{E}|Y|)^{1/2} \cdot B_3^{3/2} \left(\mathbb{E}|Y|^2\right)^{3/4}.$$

Hence,

$$B_3^{-3}\left(\mathbb{E}|Y|^2\right)^{1/2} \leq \mathbb{E}|Y|. \qquad \square$$

The proof above shows that one can take $B_p = C\sqrt{p}$ for $p > 2$. This is the asymptotically optimal dependence on p as $p \to +\infty$. It can be easily checked by considering the case when $X_1, \ldots, X_n$ are independent $N(0,1)$ random variables. We will discuss the optimal value of the constant A_1 next time.

3. 09/15 Szarek's theorem, Bennett and Bernstein inequalities

Let $\varepsilon_1, \ldots, \varepsilon_n$ denote independent symmetric Rademacher random variables (taking values ± 1 with probability $1/2$ each).

Theorem 3.1 (Kahane)**.** *Let Y be a Banach space. For any $p \in [1, +\infty)$ there exist constants $A_p, B_p > 0$ such that for any points $a_1, \ldots, a_n \in Y$*

$$A_p(\mathbb{E}\|\sum_{j=1}^n a_j\varepsilon_j\|_Y^2)^{1/2} \leq (\mathbb{E}\|\sum_{j=1}^n a_j\varepsilon_j\|_Y^p)^{1/p} \leq B_p\mathbb{E}\|\sum_{j=1}^n a_j\varepsilon_j\|_Y^2)^{1/2}.$$

(Proof will be given later in this course, as it relies on Borell lemma for log-concave functions.)

Theorem 3.2 (Szarek)**.** *The optimal constant A_1 in Kahane's theorem is $A_1 = \frac{1}{\sqrt{2}}$.*

Remark 3.3. Considering $Y = \mathbb{R}$ and $(a_1, \ldots, a_n) = (1, 1, 0, \ldots, 0)$ we see that $A_1 \le \frac{1}{\sqrt{2}}$, so we just need to show that $\frac{1}{\sqrt{2}}$ will always work.

Remark 3.4. The proof provided is the modern one, given by Juval Filmus in 2011.

Proof. Consider the set of vectors $\{-1, 1\}^n$ as a group under coordinate-wise multiplication. Then let us consider characters on this group (homomorphisms $\varphi : \{-1, 1\} \to \mathbb{T}$, where $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$). There are 2^n distinct characters of the type

$$\varphi_J(\varepsilon_1 \ldots, \varepsilon_n) = \prod_{j \in J} \varepsilon_j \text{ for every } J \subset \{1, \ldots, n\}.$$

Note that these homomorphisms form an orthonormal system in l_2-space of functionals on $\{-1, 1\}^n$. The domain has exactly 2^n points, hence l_2 space on it is 2^n dimensional, hence these characters form an orthonormal basis of $L_2(\{-1, 1\}^n)$. For any function $f : \{-1, 1\}^n \to \mathbb{C}$ we can define Fourier coefficients with respect to the orthonormal system$\{\varphi_J\}_{J \subset \{1, \ldots, n\}}$, namely

$$\widehat{f}(J) := \langle f, \varphi_J \rangle = \mathbb{E}(f \varphi_J).$$

Let us fix some $a_1, \ldots a_n \in Y$ and set

$$Z := \| \sum_{j=1}^{n} \varepsilon_j a_j \|_Y.$$

Note that Z is an even function on $\{-1, 1\}^n$ and φ_J is odd for the sets J of odd cardinality. Hence,

$$\widehat{Z}(J) = 0, \text{ if } |J| \text{ is odd.}$$

Now, let us define a flip of k-th coordinate in Z: for any $k \in \{1, \ldots, n\}$ set

$$\varepsilon_j^{(k)} = \begin{cases} -\varepsilon_j, & \text{if } j = k \\ \varepsilon_j, & \text{otherwise,} \end{cases}$$

and let

$$Z_k := \| \sum_{j=1}^{n} \varepsilon_j^{(k)} a_j \|_Y.$$

Finally, set

$$W = \sum_{k=1}^{n} Z_k.$$

Fourier coefficients of these functions can be easily evaluated:

$$\widehat{Z}_k(J) = \begin{cases} -\widehat{Z}(J), & \text{if } k \in J \\ \widehat{Z}(J), & \text{if } k \notin J, \end{cases}$$

hence

$$\widehat{W}(J) = \widehat{Z}(J)(n - 2|J|).$$

Now we will estimate scalar product $\langle Z, W\rangle$ above and below and then compare estimates. First, use Parseval equality to obtain an upper estimate. Denote $[n] = \{1, \dots, n\}$. We have

$$\langle Z, W\rangle = \sum_{J\subset[n]} \widehat{Z}(J)\widehat{W}(J) = \sum_{J\subset[n]} \widehat{Z}^2(J)(n - 2|J|) \leq$$

(here we use the fact that all odd Fourier coefficients are zeros)

$$\leq n\widehat{Z}^2(\emptyset)+\sum_{J\subset[n]} \widehat{Z}^2(J)(n-4) = 4\widehat{Z}^2(\emptyset)+(n-4)\sum_{J\subset[n]} \widehat{Z}^2(J) = 4(\mathbb{E}Z)^2+(n-4)\|Z\|_2^2.$$

Note that $\mathbb{E}Z = \|Z\|_1$ as Z is non-negative. On the other side,

$$\langle Z, W\rangle = \mathbb{E}(\|\sum_{j=1}^n \varepsilon_j a_j\|_Y \cdot (\sum_{k=1}^n \|\sum_{j=1}^n \varepsilon_j^{(k)} a_j\|_Y)) \geq$$

(by triangle inequality in the second term)

$$\geq \mathbb{E}(\|\sum_{j=1}^n \varepsilon_j a_j\|_Y \cdot \|\sum_{k=1}^n \sum_{j=1}^n \varepsilon_j^{(k)} a_j\|_Y) = (n-2)\mathbb{E}(\|\sum_{j=1}^n \varepsilon_j a_j\|_Y^2 = (n-2)\|Z\|_2^2.$$

Comparing these two estimates, we get

$$(n-2)\|Z\|_2^2 \leq 4\|Z\|_1^2 + (n-4)\|Z\|_2^2$$

or, equivalently,

$$\frac{1}{\sqrt{2}}\|Z\|_2 \leq \|Z\|_1 \leq \|Z\|_p$$

for every $p \geq 1$ by the basic property of the probability l_p-spaces. □

In the second part of class we will discuss inequalities that provide some bounds for the tails of (weighted) sums of i.i.d. random variables.

Theorem 3.5 (Bennett inequality)**.** *Let $X_1, \dots X_n$ be i.i.d. random variables, such that*

(1) $|X_k| \leq 1$ *a.s.*

(2) $\mathbb{E}X_k = 0$
(3) $\mathbb{E}X_k^2 = \delta > 0$

Then for any $t > 0$ and $a \in \mathbb{R}^n$

$$\mathbb{P}(|\sum_{j=1}^n a_j X_j| > t) \le \begin{cases} 2\exp\left(-\frac{t^2}{2e\delta\|a\|_2^2}\right), & \text{if } t \le t_* \\ 2\exp\left(-\frac{t}{4\|a\|_\infty}\log\frac{t\|a\|_\infty}{\delta\|a\|_2^2}\right), & \text{if } t > t_*, \end{cases}$$

where $\frac{t_*\|a\|_\infty}{\delta\|a\|_2^2} = e$.

Although the tail estimate looks complicated, its shape is natural. We can actually guess it from the classical limit theorems.

For simplicity, consider the case $a_1 = a_2 = \cdots = a_n = 1/\sqrt{n}$, so $\|a\|_2 = 1$. If n is large, and δ is not too small, then by the Central Limit Theorem the tail of $\frac{1}{\sqrt{n}}\sum_{j=1}^n X_j$ should be close to that of a normal random variable with expectation 0 and variance δ, which decays as $\exp\left(-\frac{t^2}{2\delta}\right)$.

On the other hand, if δ is very small, the Central Limit Theorem does not provide a good approximation. In such regime, Poissonian Limit Theorem takes over. For example, if $X_1, \dots X_n$ are i.i.d. centered Bernoulli random variables with variance $\delta = \lambda/n$, then the asymptotics of the tail of $\sum_{j=1}^n X_j$ should be close to a Poisson random variable with parameter λ. The tail of a Poisson random variable

$$\sum_{n \ge t} \frac{e^{-\lambda}\lambda^n}{n!}$$

behaves like the first term, since the terms in the series decay faster than a geometric progression. The order of the first term is

$$\frac{e^{-\lambda}\lambda^t}{t!} \sim \left(\frac{\lambda e}{t}\right)^t \sim \exp\left(-t\log\frac{t}{e\delta n}\right)$$

(we used Stirling's formula here). To compare this to Bennett's inequality, note that in this case, the sequence of coefficients is $a_1 = \cdots = a_n = 1$, so $\|a\|_\infty = 1$ and $\|a\|_2 = \sqrt{n}$.

Bennett's inequality asserts that the only possible types of the tail behavior are those resembling the Gaussian and Poissonian tail distribution.

Proof. WLOG, assume that $\|a\|_\infty = 1$. Let $\lambda > 0$. Set

$$\varphi(\lambda) = \mathbb{E}(\exp(\lambda\sum_{j=1}^n a_j X_j)) = \prod_{j=1}^n \mathbb{E}(\exp(\lambda a_j X_j)).$$

For any $x \in \mathbb{R}$,

$$e^x \leq 1 + x + \frac{x^2}{2} e^{|x|}$$

(Taylor's formula with the Lagrange remainder yields $e^x \leq 1+x+\frac{e^\theta}{2}x^2$ for some $\theta \in (0, x)$.) So, we have

$$\mathbb{E}\exp(\lambda a_j X_j) \leq 1 + \mathbb{E}(\lambda a_j X_j) + \mathbb{E}\left[\frac{\lambda^2 a_j^2 X_j^2}{2}\exp(\lambda |a_j||X_j|)\right]$$

Then, (using $\mathbb{E}X_k = 0$)

$$\mathbb{E}\exp(\lambda a_j X_j) \leq 1 + \mathbb{E}\left[\frac{\lambda^2 a_j^2 X_j^2}{2} e^\lambda\right] = 1 + \frac{\lambda^2 a_j^2 \delta}{2} e^\lambda.$$

Hence,

$$\varphi(\lambda) \leq \prod_{j=1}^{n}(1 + \frac{\lambda^2 a_j^2 \delta}{2} e^\lambda) \leq \prod_{i=1}^{n} \exp(\frac{\lambda^2 a_j^2 \delta}{2} e^\lambda) =$$

$$= \exp(\frac{\lambda^2 e^\lambda \delta}{2} \|a\|_2^2).$$

By Chebyshev's inequality,

$$\mathbb{P}(\sum_{j=1}^{n} a_j X_j > t) = \mathbb{P}(e^{\lambda \sum_{j=1}^{n} a_j X_j} > e^{\lambda t}) \leq \frac{\varphi(\lambda)}{e^{\lambda t}} = \exp(-\lambda t + \frac{\lambda^2 e^\lambda \delta}{2}\|a\|_2^2).$$

And now we want to choose λ, mininizing the expression.

Case 1. Assume that $0 < \lambda < 1$. Then

$$RHS \leq \exp(-\lambda t + \frac{\lambda^2 e \delta \|a\|_2^2}{2}).$$

This expression attains the minimal value for

$$\lambda_* := \frac{t}{e\delta \|a\|_2^2}.$$

The condition $\lambda_* < 1$ implies the restriction $t < t_*$.

Case 2. If $\lambda > 1$, (use $\lambda < e^\lambda$)

$$RHS \leq \exp(\lambda t + \frac{\lambda e^{2\lambda}\delta}{2}\|a\|_2^2) = \exp(-\lambda[t - \frac{e^{2\lambda}\delta \|a\|_2^2}{2}]).$$

The choice

$$\lambda_* := \frac{1}{2}\log \frac{t}{\delta \|a\|_2^2}$$

gives us

$$RHS \leq \exp(-\lambda_* \frac{t}{2}) = \exp(-\frac{t}{4}\log \frac{t}{\delta \|a\|_2^2}).$$

This proves the estimate for $\mathbb{P}(\sum_{j=1}^n a_j X_j > t)$. To derive the estimate for $\mathbb{P}(|\sum_{j=1}^n a_j X_j| > t)$, we apply the previous result to $\mathbb{P}(\sum_{j=1}^n (-a_j) X_j > t)$. Combining these two bounds results in the coefficient 2 in front of the exponential terms. □

Note that we can modify the condition in Case 1 by requiring $\lambda < \lambda_0$ for some $\lambda_0 > 0$. This will result in different numerical coefficients in the inequality, and a different switching point t_*. Case 2 actually works for any $\lambda > 0$.

Exponential type behavior is the third type of tail decay frequently appearing in concentration estimates. It does not correspond to any stable law. To see how it appears, consider a linear combination of i.i.d. exponential random variables. Its density can be written explicitly. The behavior of this linear combination close to its expectation is described by the central limit theorem. On the other hand, the tail decay far away from the expectation cannot be better that that of one term of the linear combination having the largest coefficient. In other words, it should be subgaussian around the expectation and subexponential far away from it.

Definition 3.6. *We will call a random variable X K-subexponential if*

$$\forall t > 0 \ \mathbb{P}(|X_i| \geq t) \leq 2\exp(-t/K).$$

As before, admitting some abuse of terminology, we will call such random variable subexponential if $K = O(1)$ and we are not interested in the explicit dependence on K. Subexponential random variables are also called ψ_1 varaibles.

Theorem 3.7 (Bernstein inequality)**.** *Let $X_1, \ldots, X_n$ be independent random variables, such that*

(1) $X - 1, \ldots, X_n$ *are subexponential;*
(2) $\mathbb{E}X_k = 0$
(3) $\mathbb{E}X_k^2 = \delta > 0$

Then, for any $t > 0$ and $a_1, \ldots, a_n \in \mathbb{R}$

$$\mathbb{P}(|\sum_{j=1}^n a_j X_j| > t) \leq 2\exp\left(-\frac{ct^2}{\delta\|a\|_2^2} \wedge \frac{c't}{\|a\|_\infty}\right)$$

We will not prove Bernstein's inequality as the proof is similar to that of Bennett's one. It is also based on using the Laplace transform and optimizing over λ. Notice that subexponential tail is the heaviest one for which such approach is possible.

If random variables in the sum are Rademacher, there is even better tail bound:

Theorem 3.8 (Montgomery-Smith). *For every* $a_1, \dots, a_n \in \mathbb{R}$ *and every* $t > 0$

$$\mathbb{P}(\sum_{j=1}^{n} \varepsilon_j a_j \geq c_1 K(a_j, t)) \leq \exp(-t^2)$$

and

$$\mathbb{P}(\sum_{j=1}^{n} \varepsilon_j a_j \geq c_2 K(a_j, t)) \geq \exp(-t^2),$$

where c_1 *and* c_2 *are some positive constants and* $K(\cdot)$ *(*K*-functional) describes the optimal rate of decay, namely,*

$$K(a,t) = \min(\|b\|_1 + t\|d\|_2 : a = b + d).$$

K-functional appears in the theory of interpolation of Banach spaces, for instance in the Marcinkiewicz theorem in harmonic analysis. For the spaces ℓ_1 and ℓ_2 it can be easily estimated. The idea is to split the set of variables $\{a_1, \dots, a_n\}$ into two disjoint parts. If a_i's are ordered such that $|a_1| \geq |a_2| \geq \ldots \geq |a_n|$, then this is a split on t^2-level:

$$K(a,t) \sim \sum_{i=1}^{t^2} |a_i| + t \cdot (\sum_{i=t^2+1}^{n} a_i^2)^{1/2}.$$

This formula gives a clue why the Montgomery-Smith theorem holds. We can bound $\sum_{j=1}^{n} \varepsilon_j a_j$ in two ways. First, this quantity does not exceed $\|a\|_1$. Second, we can estimate $\mathbb{P}(\sum_{j=1}^{n} \varepsilon_j a_j \geq s)$ using Hoeffding's inequality, and such bound will involve $\|a\|_2$. This means that we can optimize this estimate by splitting the largest coordinates of a and using the first bound for them, and applying Hoeffding's inequality to the rest. Montgomery-Smith's theorem claims that this approach yields the optimal tail bound if the splitting is chosen appropriately.

4. 09/17 Feige's inequality, Johnsson-Lindenstrauss inequality

We motivated by the article "On Sums of Independent Random Variables with Unbounded Variance, and Estimating the Average Degree in a Graph" by Uriel Feige about an algorithm of finding the average degree of vertices in some graph. The paper also presents an algorithm for finding edges of the graph which belong to many shortest paths. If too many paths go through the same edge, we consider such edge "overloaded" and removing or reweighing it would make the graph more "homogeneous". We will discuss the main probabilistic tool in Feige's paper: Markov-type (i.e. all moments higher then the first could be infinite) concentration inequality for sums of random variables.

Theorem 4.1 (Feige, 2005)**.** *Let $X_1, \dots X_n \geq 0$ be independent random variables with $\mathbb{E}X_i \leq 1$ for all $i = 1, \dots n$. Let $S_n = \sum_{i=1}^n X_i$. Then for every $\delta > 0$ there is a constant $C = C(\delta) \in (0,1)$, such that*

$$\mathbb{P}\{S_n \geq \mathbb{E}S_n + \delta\} \leq C(\delta).$$

Original proof is heavily combinatorial, we will present a last year proof by Fedor Nazarov based on Bernstein's method.

Proof. Define $Y_j := \mathbb{E}X_j - X_j$. Then $Y_j \leq 1$ and $\mathbb{E}Y_j = 0$. Then, let $Y = \sum_j Y_j$. We have to show that

$$p := \mathbb{P}\{Y \geq -\delta\} \geq c(\delta) > 0.$$

Case 1: Assume that $\mathbb{E}e^Y \leq 2$. By Jensen's inequality,

$$1 = \exp(\mathbb{E}Y/2) \leq \mathbb{E}\exp(Y/2) =$$

$$= \mathbb{E}\exp(Y/2)\mathbb{1}_{(-\infty,-\delta]}(Y) + \mathbb{E}\exp(Y/2)\mathbb{1}_{(-\delta,\infty]}(Y) \leq$$

(by Cauchy-Swartz in the second term)

$$\leq e^{-\delta/2}(1-p) + (\mathbb{E}e^Y)^{1/2}\sqrt{p} \leq e^{-\delta/2}(1-p) + \sqrt{2}\sqrt{p}.$$

So,

$$e^{-\delta/2}p - \sqrt{2}\sqrt{p} + 1 - e^{-\delta/2} \leq 0.$$

Left hand side is a continuous function of p, and it is positive at $p = 0$. Hence, $p \geq c$ for some $c > 0$.

Case 2: Assume that $\mathbb{E}e^Y > 2$.

First, let us find $t \geq 0$ such that $Ee^{tY} = 2$. Clearly, $t \in (0,1)$.

Second, consider the function

$$f(x) := e^x - x - 1.$$

There exists the constant $K > 0$, such that $f(2x) \leq Kf(x)$ for all $x < 1$ (indeed, for any compact interval $[-a, 1]$ we can find some constant K

with this property, and since near $-\infty$ the function $f(x)$ behaves like a linear function, we choose parameter a big enough and extend it to all $x \leq 1$ with the same K.)

As $Y_i \leq 1$ for every $i = 1, \ldots, n$ and $t \in (0,1)$, we have $tY_j \in (-\infty, 1]$. Moreover, $\mathbb{E}(Y_j) = 0$, so, taking $x := tY_j$ and applying the estimate $f(2x) \leq Kf(x)$ and then taking expectation of both sides, we get

$$\mathbb{E}e^{2tY_j} - 1 \leq K(\mathbb{E}e^{tY_j} - 1).$$

Sincet $1 + Ks \leq (1+s)^K$ for al $s > 0$,

$$\mathbb{E}e^{2tY_j} \leq 1 + K(\mathbb{E}e^{tY_j} - 1) \leq (\mathbb{E}e^{tY_j})^K.$$

Hence,

$$\mathbb{E}e^{2tY} = \prod_{j=1}^{n} \mathbb{E}e^{2tY_j} \leq (\prod_{j=1}^{n} \mathbb{E}e^{tY_j})^K = 2^K.$$

Set $\varphi(s) := e^s - 2^{-K-1}e^{2s} - 1$. Then $\mathbb{E}\varphi(tY) \geq 2 - \frac{1}{2} - 1 = \frac{1}{2}$.

$$\frac{1}{2} \leq \mathbb{E}\varphi(tY)\mathbb{1}_{(-\infty,0]}(Y) + \mathbb{E}\varphi(tY)\mathbb{1}_{(0,+\infty)}(Y) \leq$$

(as for $s \leq 0$ we have $\varphi(s) \leq 1 - 1 = 0$ and for $s > 0$ the function $\varphi(s)$ is bounded)

$$\leq \mathbb{E}\varphi(tY)\mathbb{1}_{(0,+\infty)}(Y) \leq \sup_{s>0} \varphi(s) \cdot \mathbb{P}\{Y > 0\} \leq cp.$$

□

The next lemma is extremely useful in applications. It allows to project a finite set of the points in $\mathbb{R}^N$ onto much smaller space almost preserving their pairwise distances.

Lemma 4.2 (Johnsson-Lindenstrauss)**.** *Let $x_1, \ldots, x_n \in \mathbb{R}^N$. Let $\varepsilon > 0$ and $k \geq c \log n/\varepsilon^2$. Then there exists a linear mapping $T : \mathbb{R}^N \to \mathbb{R}^k$, such that for every $i \neq j$*

$$1 - \varepsilon \leq \frac{\|Tx_i - Tx_j\|_2}{\|x_i - x_j\|_2} \leq 1 + \varepsilon.$$

Proof. Let A be a $k \times N$ matrix with centered subgaussian unit variance entries. Set

$$V := \{\frac{x_i - x_j}{\|x_i - x_j\|_2} : x_i \neq x_j\}.$$

Note that V is a subset of a unit sphere $V \subset S^{n-1}$, $|V| \leq n^2$. So it is enough to show that with high probability for any $v \in V$

$$1 - \varepsilon \leq \frac{\|Av\|_2}{\sqrt{k}} \leq 1 + \varepsilon.$$

Let $v \in V$. Then by Hoeffding's inequality,

$$X_i = (Av)_i = \sum_{i=1}^{N} a_i v_i$$

is a subgaussian random variable with $\mathbb{E}X_i^2 = \|v_i\|_2^2 = 1$.

Set $Y_i := X_i^2 - 1$. Then $\mathbb{E}Y_i = 0$. Let $a \in \mathbb{R}$. If $a < 0$ we have that $\mathbb{E}e^{aY_i} \leq e^a$. If $a > 0$, then $\mathbb{E}e^{aY_i} \leq e^{-a}\mathbb{E}e^{aX_i^2} < +\infty$ for some $a > 0$. Hence, $\mathbb{E}e^{aY_i}$ is bounded function on a on some interval around zero, so Y_i are sub-exponential random variables and we can use Bernstein inequality for sub-exponential random variables to estimate

$$\mathbb{P}\{|\sum_{i=1}^{k} Y_i| > t\} \leq 2\exp(-\frac{ct}{1} \wedge \frac{ct^2}{k}).$$

Choose $t := \varepsilon k$. Then

$$\mathbb{P}\{|\sum_{i=1}^{k} X_i^2 - k| > \varepsilon k\} \leq 2\exp(-c'\varepsilon^2 k).$$

By the union bound,

$$\mathbb{P}\{\exists v \in V : |\|Av\|_2^2 - k| > \varepsilon k\} \leq |V| \cdot 2\exp(-c'\varepsilon^2 k),$$

If $k \geq c \log n/\varepsilon^2$,

$$|V| \cdot 2\exp(-c'\varepsilon^2 k) \leq 2n^2 \exp(-c'\varepsilon^2 k) \ll 1.$$

Assume that $|\|Av\|_2^2 - k| \leq \varepsilon k$ for all $v \in V$. Then it is easy to show that

$$\left| \frac{1}{\sqrt{k}} \|Av\|_2 - 1 \right| \leq \varepsilon.$$

This means that the statement of the lemma holds for $T = (1/\sqrt{k})A$. $\square$

We have shown that he ℓ_2 dimension reduction is very efficient. However, for many optimization problems, the ℓ_1 metric is preferable to ℓ_2 since the ℓ_1-norm minimization can be formulated as a linear programming problem, which makes powerful linear programming algorithms avalable. The problem whether dimension reduction is possible in ℓ_1 was open for a long time. A negative solution was obtained by B. Brinkman and M. Charikar in 2003. There is an n point subset of ℓ_1 which cannot be embedded in ℓ_1^k with a small distortion unless $k \geq cn$. The embedding in this result does not have to be linear, so the dimension reduction in ℓ_1 fails in a very strong sense. Later a simple construction of such non-embeddable set was found by J. Lee and A. Naor (see the paper in supplementary materials).

Now, we will discuss a result showing that for the sums of i.i.d. random variables, the bounds obtained by Bernstein's method are essentially the best possible.

Theorem 4.3 (Cramer's Theorem, Part 1)**.** *Let X be a random variable, such that for all $\lambda \in \mathbb{R}$ $\mathbb{E}e^{\lambda X} < +\infty$. Let $X_1, \dots X_n$ be i.i.d. copies of X, and set $S = \sum_{i=1}^n X_i$. Then*

- *for any $a > \mathbb{E}X$ we have $\mathbb{P}\{\frac{1}{n}S > a\} \leq \exp(-I(a) \cdot n)$*
- *for any $a < \mathbb{E}X$ we have $\mathbb{P}\{\frac{1}{n}S < a\} \leq \exp(-I(a) \cdot n)$,*

where the function $I : \mathbb{R} \to [0, +\infty]$ is defined by

$$I(a) = \sup_{t \in \mathbb{R}} (ta - \log \mathbb{E} \exp(tX)).$$

Proof. Let $a > \mathbb{E}X$. Using the exponential form of Markov inequality (with some parameter $t > 0$),

$$\mathbb{P}(S/n > a) = \mathbb{P}(\exp(tS) > \exp(nta)\} \leq e^{-nta}\mathbb{E}\exp(tS) =$$

$$= e^{-nta} \prod_{i=1}^n \mathbb{E}\exp(tX_j) = \exp(-n(ta - \Lambda(t)),$$

where $\Lambda(t) := \log \mathbb{E}\exp(tX)$. Then, optimization over t implies that we should take

$$I(a) := \sup_{t>0} (ta - \Lambda(t)).$$

When $a > \mathbb{E}X$ (i.e. in the case we consider),

$$(ta - \Lambda(t))'(0) = a - \mathbb{E}X > 0$$

and moreover this function is concave, so

$$\sup_{t \in \mathbb{R}} (ta - \Lambda(t)) = \sup_{t>0} (ta - \Lambda(t)) = I(a).$$

We can obtain the second inequality (for $a < \mathbb{E}X$) considering $-X_i$ instead of X_i for all $i = 1, \dots, n$.

$\square$

5. 09/22 Cramer's thm. Concentration for quadratic forms

Last time in Part 1 of Cramer's theorem we defined *rate function*

$$I(a) := \sup_{t \in \mathbb{R}} (ta - \Lambda(t)),$$

where a is a parameter and $\Lambda(t)$ is the log-moment generating function of X,

$$\Lambda(t) := \log \mathbb{E} \exp(tX).$$

The rate function is the *Legendre transform* of Λ. The notion of Legendre trensform appears frequently in convex geometry and functional analysis. For example, if $\Lambda(t)$ is some function defining an Orlitz norm, then $I(a)$ defines the norm in its dual. In particular, it is easy to see that if $\Lambda(t) = \frac{1}{p} t^p$ on the interval $[0, +\infty)$, then $I(a) = \frac{1}{q} a^q$ and $1/p + 1/q = 1$.

Theorem 5.1 (Cramer's Theorem, Part 2)**.** *Let X be a random variable, such that for all $\lambda \in \mathbb{R}$ $\mathbb{E} e^{\lambda X} < +\infty$. Let $X_1, \dots X_n$ be i.i.d. copies of X, and set $S = \sum_{i=1}^n X_i$.*

Assume that $\mathbb{P}(X = y) < 1$ for any $y \in \mathbb{R}$ (X is not concentrated at one point). Also, assume that $[\alpha, \beta]$ is the smallest interval such that $\mathbb{P}(X \in [\alpha, \beta]) = 1$ (α and β are allowed to be infinite).

Then for any $a \in (\alpha, \beta)$ and any $\varepsilon > 0$,

$$\mathbb{P}\left(\left| \frac{1}{n} S - a \right| < \varepsilon \right) \geq \left(1 - \frac{\Lambda''(\varphi(a))}{n \varepsilon^2} \right) \cdot \exp(-n[I(a) - \varphi(a)\varepsilon]),$$

where $\varphi = (\Lambda')^{-1}$.

Remark 5.2. Part 2 of Cramer's theorem ensures that the concentration result that we got in Part 1 is essentially the best possible:

$$\mathbb{P}\left(\left| \frac{1}{n} S - a \right| < \varepsilon \right) \gtrsim \exp(-nI(a)).$$

Indeed, the functions Λ, φ, and I depend only on the distribution of X. For a sufficiently small ε, $\varphi(a)\varepsilon \ll I(a)$, and so the exponential term is close to $\exp(-nI(a)$. Also, if n is sufficiently large, then the coefficient in front of the exponential term is close to 1.

Before we start the proof, let us discuss some properties of Λ and I that will be useful for the proof.

(1) $\Lambda \in C^\infty(\mathbb{R})$. Logarithmic function is nice, so it is enough to check that $\mathbb{E} e^{tX}$ is infinitely differentiable. It follows from Lebesgue dominated convergence theorem (one has to write the

derivative as a limit and check the existence of an integrable majorant). Then for $\forall t \in \mathbb{R}$

$$\Lambda'(t) = \frac{\mathbb{E}Xe^{tX}}{\mathbb{E}e^{tX}}.$$

(2) for any $c \in (\alpha, \beta)$, $I(c)$ is well-defined and non-negative.

For every $c < \beta$ take $\varepsilon > 0$ such that $c + \varepsilon < \beta$, then

$$e^{-ct}\mathbb{E}e^{tX} \geq e^{-ct}\mathbb{E}e^{tX}\mathbb{1}_{(c+\varepsilon,\infty)}(X) \geq e^{\varepsilon t}\mathbb{P}(X > c + \varepsilon).$$

Hence,

$$\lim_{t\to+\infty} e^{-ct}\mathbb{E}e^{tX} = +\infty,$$

and, similarly, for every $c > \alpha$

$$\lim_{t\to-\infty} e^{-ct}\mathbb{E}e^{tX} = +\infty.$$

Therefore, taking logarithm,

$$\lim_{t\to\infty}(ct - \Lambda(t)) = -\infty.$$

So we proved that $I(c) \in [0, +\infty)$ for every $c \in (\alpha, \beta)$ (rate function attains its maximum value, it is finite and at least zero, as $\Lambda(0) = 0$ by definition).

(3) Properties of $\Lambda'(t)$.

Let $t \in \mathbb{R}$. Define a random variable Y_t by

$$\mathbb{P}\{Y_t \in E\} = \frac{1}{\mathbb{E}e^{tX}}\mathbb{E}(e^{tX}\mathbb{1}_E(X))$$

for any Borel set E on $\mathbb{R}$. Then $\Lambda'(t) = \mathbb{E}Y_t$. In particular, $\mathbb{E}Y_t \in (\alpha, \beta)$ (average always lies between lower and upper bound, and Y_t is defined on the domain of X).

We will show below that $\mathbb{E}(Y_t)$ can take any value in the interval (α, β) if t is chosen appropriately. This is the main reason for the introduction of these random variables. To investigate the behavior of S/n around the point a, we will replace the random variables $X_1, \ldots, X_n$ by the new variables $(Y_t)_1, \ldots, (Y_t)_n$ centered at a.

$$\Lambda''(t) = \left(\frac{\mathbb{E}Xe^{tX}}{\mathbb{E}e^{tX}}\right)' = \frac{\mathbb{E}X^2e^{tX}}{\mathbb{E}e^{tX}} - \left(\frac{\mathbb{E}Xe^{tX}}{\mathbb{E}e^{tX}}\right)^2 = \mathrm{Var}(Y_t).$$

Since X is not concentrated at a single poit, $\mathrm{Var}(Y_t) > 0$. Hence, $\Lambda''(t) > 0$, which means that $\Lambda'(t)$ is a strictly increasing function from $\mathbb{R}$ to (α, β). Let us show that it is onto (and hence, has an inverse on (α, β)). It is the same as to show that $\lim_{t\to\infty} \Lambda'(t) = \beta$ and $\lim_{t\to-\infty} \Lambda'(t) = \alpha$.

Let

$$f_t(y) = \frac{e^{ty}}{\mathbb{E}e^{tX}}.$$

Then for any $y < \beta$ it goes to zero as t grows: $f_t(y) \to_{t\to+\infty} 0$. Therefore, for any $a < \beta$,

$$\frac{\mathbb{E}Xe^{tX}\mathbb{1}_{(-\infty,a]}(X)}{\mathbb{E}e^{tX}} \to_{t\to+\infty} 0$$

and so $\Lambda'(t) \geq a$ for any $t > t(a)$ (large enough). We proved that Λ' is onto, and it is increasing, hence, it has nice inverse function.

(4) Description of the optimal t in the definition of $I(a)$.

Set $\varphi = (\Lambda')^{-1}$. Then φ is an infinitely differentiable increasing function from (α, β) to $\mathbb{R}$.

Let $a \in (\alpha, \beta)$. Let $f(t) = at - \Lambda(t)$. Then $f'(t) = a - \Lambda'(t)$. Note that

$$t_{\max}(a) = \varphi(a)$$

and

$$I(a) = f(t_{\max}) = a\varphi(a) - \Lambda(\varphi(a)).$$

Proof. (of Theorem 5.1) Let $a \in (\alpha, \beta)$ and $\varepsilon > 0$. Let $t \in \mathbb{R}$. For every measurable function $F : \mathbb{R}^n \to \mathbb{R}$

$$\mathbb{E}(F(Y_t)_1, \ldots, (Y_t)_n) = \frac{1}{(\mathbb{E}e^{tX})^n}\mathbb{E}e^{tS}F(X_1, \ldots, X_n) =$$

$$= \exp(-n\Lambda(t)) \cdot \mathbb{E}e^{tS}F(X_1, \ldots X_n).$$

Let us take an optimal $t := \varphi(a)$. Then $\mathbb{E}Y_t = \Lambda'(\varphi(a)) = a$. Then, if we denote $Z := \sum_{j=1}^n (Y_t)_j$,

$$\mathbb{P}\{|\frac{1}{n}S - a| < \varepsilon\} = \mathbb{E}\mathbb{1}_{(a-\varepsilon,a+\varepsilon)}(\frac{1}{n}\sum_{j=1}^{n} X_j) =$$

$$= \exp(n\Lambda(t)) \cdot \mathbb{E}e^{-tZ}\mathbb{1}_{(a-\varepsilon,a+\varepsilon)}(Z/n) \geq$$

$$\geq \exp(n\Lambda(t)) \exp\big(-tn(a+\varepsilon)\big) \cdot \mathbb{P}(|\frac{1}{n}Z - a| < \varepsilon) \geq$$

$$\geq \exp(-n(ta - \Lambda(t)) - nt\varepsilon) \cdot \left(1 - \frac{\mathrm{Var}(Z)}{(n\varepsilon)^2}\right) =$$

$$= \left(1 - \frac{\Lambda''(\varphi(a))}{n\varepsilon^2}\right) \cdot \exp(-n[I(a) - \varphi(a)\varepsilon])$$

(we used Chebyshev inequality to pass to estimate probability via variance and linearity of variance in the last step).

$\square$

Remark 5.3. The Cramer theorem provides us with a precise result if we can compute or at least estimate the functions Λ, φ and I. Unfortunately, they can be computed only for some "easy" random variables (like Rademacher or uniform case). That is why people frequently use less precise, but more tractable estimates instead of Cramer's theorem.

Note also, that light tails of the distribution are essential for the possibility to apply the Laplace transform method. It can be extended to the case of exponential random variable (as it is enough to have all moments on some interval instead of the whole $\mathbb{R}$), but heavy tails would be impossible to treat this way.

Except from the generalization to the heavy-tailed variables (see Latala's paper on Ctools about this), another natural question to ask is what if we want to estimate something more complicated than linear combination of X_i. Two possible candidates for a generalization of a linear function are (1)quadratic forms and (2)martingales. We're going to consider they both, and we start with the quadratic forms.

Concentration of quadratic forms.

Let $f(X) = X^T AX$ defined on $\mathbb{R}^n$, where A is a symmetric $n \times n$ matrix.

We will use Laplace transform method again, so we will need the existence of exponential moments of $X^T AX$, so we have to assume that the random variable X has light tails (at least, with subgaussian decay, otherwise the exponential moment is infinite).

Notation:

$$\|A\| = \max_{x \in S^{n-1}} \|Ax\|_2 \text{ (operator norm)}$$

$$\|A\|_F = (\sum_{i,k} |a_{ik}|^2)^{1/2} \text{ (Frobenius or Hilbert-Smidt norm)}$$

Theorem 5.4 (Hanson-Wright)**.** *Let $X_1, \dots, X_n$ be independent centered subgaussian random variables. Consider a random vector $X^T = (X_1, \dots, X_n)$. Then for any $t > 0$*

$$\mathbb{P}(|X^T AX - \mathbb{E}X^T AX| > t) \leq 2\exp\left(-c\left[\frac{t^2}{\|A\|_F^2} \wedge \frac{t}{\|A\|}\right]\right).$$

We will prove this theorem next time, and now let us show why it works in the particular case of a diagonal matrix A.

Let $A = \operatorname{diag}(a_1, \dots, a_n)$. Then

$$X^T AX = \sum_{i=1}^{n} a_{ii} X_i^2;$$

$$X^T AX - \mathbb{E}X^T AX = \sum a_{ii}(X_i^2 - \mathbb{E}X_i^2)$$

Note that $Y_i = X_i^2 - \mathbb{E}X_i^2$ is a centered subexponential random variable (as the square of a subgaussian variable, for the rigorous proof see the proof of Johnson-Lindenstrauss Lemma). So, we can use Bernstein inequality for Y_i to get

$$\mathbb{P}\{|X^T AX - \mathbb{E}X^T AX| > t\} \leq 2\exp\left(-c\left[\frac{t^2}{\sum a_{ii}^2} \wedge \frac{t}{\max|a_{ii}|}\right]\right) \leq$$

$$\leq 2\exp\left(-c\left[\frac{t^2}{\|A\|_F^2} \wedge \frac{t}{\|A\|}\right]\right).$$

Thus for a diagonal matrix, Hanson-Wright theorem is am easy consequence of Bernstein's inequality.

6. 09/24 Hanson-Wright thm, Frobenius norms concentrarion

Theorem 6.1 (Hanson-Wright)**.** *Let $X_1, \dots, X_n$ be independent centered subgaussian random variables. Consider a random vector $X^T = (X_1, \dots, X_n)$. Then for any $t > 0$*

$$\mathbb{P}(|X^T AX - \mathbb{E}X^T AX| > t) \leq 2\exp\left(-c\left[\frac{t^2}{\|A\|_F^2} \wedge \frac{t}{\|A\|}\right]\right).$$

Proof. Let us separate the diagonal and off-diagonal parts of the matrix.

$$\mathbb{P}\{|X^T AX - \mathbb{E}X^T AX| > t\} \leq \mathbb{P}\{|\sum_{i=1}^{n} a_{ii} X_i^2 - \sum_{i=1}^{n} a_{ii}\mathbb{E}X_i^2| > \frac{t}{2}\} +$$

$$+\mathbb{P}\{|\sum_{i\neq k} a_{ik} X_i X_k| > \frac{t}{2}\}.$$

Note that we already proved that the first term is bounded as required by Hanson-Wright theorem (this is the diagonal case considered in the last part of the previous class). So we need to estimate only the second term. We will do it with the help of Laplace transform with some parameter $\lambda > 0$ (that we optimize later, as usual).

Consider the function

$$f(\lambda) = \mathbb{E}\exp(\lambda \sum_{j \neq k} a_{jk} X_j X_k).$$

We know the tools for linear combinations of independent random variables, so it would be convenient to condition on one of two $X's$ in each product, but it is impossible to do straightforward (as they are clearly dependent). So, we start with the following decoupling procedure (by Bourgain-Tzafriri).

Let $\delta_1, \dots, \delta_n$ be independent Bernoulli 0-1 random variables with $\mathbb{E}\delta_i = 1/2$. Then $\mathbb{E}_\delta \delta_j (1-\delta_k) = 1/4$. Then

$$f(\lambda) = \mathbb{E}\exp(\lambda \sum_{j \neq k} a_{jk} 4\mathbb{E}_\delta \delta_j (1-\delta_k) X_j X_k) \leq$$

(by Jensen inequality)

$$\leq \mathbb{E}\mathbb{E}_\delta \exp(4\lambda \delta_j (1-\delta_k) a_{jk} X_j X_k).$$

Let $J = \{j \in \{1, \dots, n\}, \delta_j = 1\}$. Set

$$f_J(\lambda) = \mathbb{E}[\exp(4\lambda \sum_{k \notin J} (\sum_{j \in J} a_{jk} X_j) X_k) | J].$$

Note that all X_j in the inner sum and X_k in the outer sum are different, hence, independent. Moreover, conditioned on J inner sum is fixed, so we can apply Hoeffding inequality to the outer sum:

$$f_J(\lambda) \leq \mathbb{E}\left[\exp\left((4\lambda)^2 \sum_{k \notin J} (\sum_{j \in J} a_{jk} X_j)^2\right) | J\right].$$

To deal with this new quadratic form, let us recall that a standard gaussian random variable $g \sim N(0,1)$ has the Laplace transform $\mathbb{E}\exp(Cg) = \exp(C^2/2)$. Hence, for the k independent $N(0,1)$ random variables $g_1, \dots, g_k$, $\mathbb{E}\exp(\sum_{i=1}^k C_i g_i) = \exp(\sum_{i=1}^k C_i^2/2)$. Therefore, we can rewrite the inequality as

$$f_J(\lambda) \leq \mathbb{E}[\exp(C\lambda \sum_{k \notin J} (\sum_{j \in J} a_{jk} X_j) g_k) | J],$$

where $g_k \sim N(0,1)$ for every $k \notin J$.

Now we can interchange the sums and apply the same observation again:

$$f_J(\lambda) \leq \mathbb{E}[\exp(C\lambda \sum_{j \in J} (\sum_{k \notin J} a_{jk} g_k) X_j) | J] \leq$$

(with $g_j \sim N(0,1)$ for every $j \in J$)

$$\leq \mathbb{E}\left[\exp\left(C'\lambda^2 \sum_{j\in J}(\sum_{k\notin J} a_{jk}g_k)^2\right)|J\right].$$

This last sum can be calculated precisely: let $P_J : \mathbb{R}^n \to \mathbb{R}^n$ be the coordinate projection onto J. Let

$$B = P_J A(I - P_J)$$

and $g = (g_1, \dots, g_n)^T$, then

$$\sum_{j\in J}(\sum_{k\notin J} a_{jk}g_k)^2 = \|Bg\|_2^2.$$

Let $s_1(B) \geq s_2(B) \geq \dots \geq s_n(B) \geq 0$ be singular values of B (by definition $s_j = (\lambda_j(BB^T))^{1/2}$, where λ_j is the j-th eigenvalue of the symmetric matrix BB^T). Gaussian random vectors are rotationally invariant, so

$$f_J(\lambda) \leq \mathbb{E}[\exp(C\lambda \sum_{j=1}^n s_j(B)\tilde{g}_j^2)|J],$$

if we change the original basis to the right singular vectors basis by an orthogonal transformation. Since $\mathbb{E}e^{tg^2} = (1-2t^2)^{-1/2}$ when $t \in (0,1)$, we get

$$f_J(\lambda) \leq \mathbb{E}[\prod_{j=1}^n (1 - 2(C\lambda s_j(B))^2)^{-1/2}|J],$$

if $C\lambda s_j(B) \leq 1$. Assume that λ is chosen such that $\lambda \leq (2Cs_1(B))^{-1}$ (then the same condition is satisfied for all s_j).

As $(1-x)^{1/2} \leq e^x$ for $x \in (0,1)$, we have that

$$f_J(\lambda) \leq \mathbb{E}[\exp(c\lambda^2 \sum_{j=1}^n s_j^2(B))|J],$$

if $\lambda \leq \tilde{C}/s_1(B)$.

Note that

$$\sum_{j=1}^n s_j^2(B) = \sum_{j=1}^n \lambda_j(BB^T) = \mathrm{Tr}(BB^T) = \sum_{j,k=1}^n b_{jk}^2 = \|B\|_F^2 \leq \|A\|_F^2$$

and $s_1(B) = \|B\| \leq \|A\|$.

So,

$$f_J(\lambda) \leq \mathbb{E}[e^{c\lambda^2\|B\|_F^2}|J] \leq \mathbb{E}[e^{c\lambda^2\|A\|_F^2}|J] = \exp(c\lambda^2\|A\|_F^2).$$

and the condition $\lambda \leq C/\|B\|$ follows from the stronger one $\lambda \leq C/\|A\|$ (we used that norm is submultiplicative and norm of projection is 1).

Note that left hand side depends on J and right hand side doesn't, so we can the integrate inequality over all choices of J to get

$$f(\lambda) = \mathbb{E}\exp(\lambda \sum_{j\neq k} a_{jk} X_j X_k) \leq \exp(c\lambda^2 \|A\|_F^2) \quad \text{if } 0 < \lambda \leq C/\|A\|.$$

Hence,

$$\begin{aligned}\mathbb{P}\{\sum_{i\neq k} a_{ik} X_i X_k > \frac{t}{2}\} &= \mathbb{P}\{\exp(\lambda \sum_{i\neq k} a_{ik} X_i X_k) > \exp(\lambda t/2)\} \\ &\leq \exp(c\lambda^2 \|A\|_F^2 - \lambda t/2).\end{aligned}$$

In the last step we used Markov inequality together with the estimate for $f(X)$. Combining this with the same inequality to the matrix $-A$ yields

$$\mathbb{P}\{\sum_{i\neq k} a_{ik} X_i X_k > \frac{t}{2}\} \leq 2\exp(c\lambda^2 \|A\|_F^2 - \lambda t/2) \quad \text{whenever } 0 < \lambda \leq C/\|A\|.$$

To complete the proof, it remains to optimize by λ. Taking $\lambda_{\max} = t/4c\|A\|_F^2$ (it provides the minimal value on the parabola $c\lambda^2\|A\|_F^2 - \lambda t/2$), we get that

$$\mathbb{P}\{|\sum_{i\neq k} a_{ik} X_i X_k| > \frac{t}{2}\} \leq \exp\left(-\frac{c' t^2}{\|A\|_F^2}\right)$$

whenever $\lambda_{\max} \leq C/\|A\|$. If $\lambda_{\max} > C/\|A\|$, setting $\lambda = \lambda_{\max}$, we obtain

$$\mathbb{P}\{|\sum_{i\neq k} a_{ik} X_i X_k| > \frac{t}{2}\} \leq \exp\left(-\frac{c'' t}{\|A\|}\right).$$

This completes the proof of the Hanson-Writht inequality. □

Corollary 6.2. *Let $X_1, \ldots, X_n$ be independent centered subgaussian vectors with variance 1. Let A be an $n \times n$ (fixed) matrix. Then for $\forall \tau > 0$,*

$$\mathbb{P}\{|\|AX\|_2 - \|A\|_F| > \tau\} \leq 2\exp\left(-c\frac{\tau^2}{\|A\|^2}\right).$$

Remark 6.3. Note that $\|A\|_F$ is not the expectation of $\|AX\|_2$, however (as it follows from this theorem), it is very close to the expectation.

Proof. We have,

$$\|AX\|_2^2 = \langle AX, AX\rangle = \langle A^T AX, X\rangle \quad \text{and} \quad \mathbb{E}\langle A^T AX, X\rangle = \|A\|_F^2.$$

By the Hanson-Wright inequality, for any $t > 0$,

$$\mathbb{P}\{|\|AX\|_2^2 - \mathbb{E}\langle A^T AX, X\rangle| > t\} \le 2\exp\left(-c\left[\frac{t^2}{\|A^T A\|_F^2} \wedge \frac{t}{\|A^T A\|}\right]\right) =$$

$$= 2\exp\left(-c\left[\frac{t^2}{\|A\|^2\|A\|_F^2} \wedge \frac{t}{\|A\|^2}\right]\right),$$

as for any two matrices $\|UV\|_F \le \|U\| \cdot \|V\|_F$ (can be checked directly).

Taking

$$t := \varepsilon\|A\|_F^2,$$

we get

$$\mathbb{P}\{\left|\frac{\|AX\|_2^2}{\|A\|_F^2} - 1\right| > \varepsilon\} \le 2\exp\left(-c\frac{\|A\|_F^2}{\|A\|^2}(\varepsilon^2 \wedge \varepsilon)\right).$$

- $\varepsilon \in (0,1)$

  We use that

  $$\left|\frac{\|AX\|_2^2}{\|A\|_F^2} - 1\right| \ge \left|\frac{\|AX\|_2}{\|A\|_F} - 1\right|$$

  to get

  $$\mathbb{P}\{\left|\frac{\|AX\|_2}{\|A\|_F} - 1\right| > \varepsilon\} \le \exp\left(-c\frac{\varepsilon^2\|A\|_F^2}{\|A\|_2}\right).$$

  Substituting back $\tau := \varepsilon\|A\|_F$,

  $$\mathbb{P}\{|\|AX\|_2 - \|A\|_F| > \tau\} \le 2\exp(-c\tau^2/\|A\|^2).$$

- $\varepsilon \ge 1$.

  In this case

  $$\left|\frac{\|AX\|_2^2}{\|A\|_F^2} - 1\right| \ge \left(\frac{\|AX\|}{\|A\|_F} - 1\right)^2,$$

  hence,

  $$\mathbb{P}\{\left|\frac{\|AX\|_2}{\|A\|_F} - 1\right| > \sqrt{\varepsilon}\} \le \exp\left(-c\frac{\varepsilon\|A\|_F^2}{\|A\|_2}\right).$$

  Substituting back $\tau := \sqrt{\varepsilon}\|A\|_F$, we get

  $$\mathbb{P}\{|\|AX\|_2 - \|A\|_F| > \tau\} \le 2\exp(-c\tau^2/\|A\|^2)$$

in this case as well. $\square$

7. 09/29 Martingales and Azuma inequality

Last time we proved the following corollary from he Hanson-Wright theorem.

Theorem 7.1. *Let $X_1, \dots, X_n$ be independent centered subgaussian vectors with variance 1. Let A be an $n \times n$ (fixed) matrix. Then for any $t > 0$*

$$\mathbb{P}\{|\|AX\|_2 - \|A\|_F| > t\} \le 2\exp(-c\frac{t^2}{\|A\|^2}). \tag{1}$$

Remark 7.2. This result implies that the expectation of $\|AX\|_2$ is close to over Frobenius norm of A. Indeed, (1) shows that the random variable

$$Y = \frac{\|AX\|_2 - \|A\|_F}{\|A\|}$$

is subgaussian. Hence, Y has bounded first moment, $\mathbb{E}|Y| \le C$, and

$$\|A\|_F - C\|A\| \le \mathbb{E}\|AX\|_2 \le \|A\|_F + C\|A\|.$$

Typically, $\|A\|$ is much smaller than $\|A\|_F$. For instance, if A is a random ± 1 matrix, then $\|A\|_F = n$, while $\|A\| = O(\sqrt{n})$ with high probability.

There is also an anti-concentration corollary from Hanson-Wright theorm:

Theorem 7.3. *Let X and A be as above. For any $y \in \mathbb{R}^n$*

$$p := \mathbb{P}\{\|AX - y\|_2 < \frac{\|A\|_F}{2}\} \le 2 \cdot \exp\left(-C\frac{\|A\|_F^2}{\|A\|^2}\right).$$

Remark 7.4. The quantity $\rho(A) = \|A\|_F^2/\|A\|^2$ is called the *stable rank.* If $A = P$ is an orthogonal projection, then $\|P\| = 1$ and $\|P\|_F^2 = \operatorname{Tr}(P^T P) = \operatorname{Tr} P = \operatorname{rank}(P)$, so $\rho(A) = \operatorname{rank}(A)$. However, the stable rank is stable under all small perturbations, unlike $\operatorname{rank} P$ (a small perturbation of a projection makes it full-dimensional).

Proof. Let X' be independent copy of X. Then

$$p^2 = \mathbb{P}\{\|AX - y\|_2 < \frac{\|A\|_F}{2} \,\&\, \|AX' - y\|_2 < \frac{\|A\|_F}{2}\} \le$$
$$\le \mathbb{P}\{\|AX - AX'\|_2 < \|A\|_F\}.$$

Note that $X - X'$ is a vector with independent subgaussian coordinates of variance 2. Hence,

$$p^2 \le \mathbb{P}\left(\left\|A\left(\frac{X - X'}{\sqrt{2}}\right)\right\|_2 < \frac{\|A\|_F}{\sqrt{2}}\right) \le$$

$$\leq \mathbb{P}\left(\left|\left\|A\left(\frac{X-X'}{\sqrt{2}}\right)\right\|_2 - \|A\|_F\right| > \left(1-\frac{1}{\sqrt{2}}\right)\cdot\|A\|_F\right) \leq$$

(by the Theorem 7.1)

$$\leq 2\exp\left(-\tilde{c}\frac{\|A\|_F^2}{\|A\|^2}\right).$$

□

Let us now discuss some examples of other types of random variables that could also concentrate (under certain conditions).

Example 7.5. Let Y be a normed space. Let $X_1, \ldots X_n$ be independent centered random variables in Y. Let $Z = \|\sum_{j=1}^n X_j\|_Y$. Is Z concentrated?

There are different approaches to this problem. If $Y = \mathbb{R}$, we can split the sum into positive and negative parts and then apply Bennet inequality. Alternatively we can estimate the Y-norm on a net (on a unit sphere in dual space), and take a union bound over the elements of the net (however we need very strong concentration conditions at every point to make this work in combination with the union bound).

Example 7.6. *(Bin Occupancy)* Suppose that n balls are put randomly into m bins. Let Z denote the number of empty bins. Let Z_j be an indicator of the event that j-th bin is empty. Then $Z = \sum_{j=1}^n Z_j$ and

$$\mathbb{E}Z = \sum_{j=1}^n \mathbb{E}Z_j = m(1-\frac{1}{m})^n \approx me^{-n/m}.$$

The interesting regime is when $m \sim n$ (or $n/m = r \sim 1$).

Note that Z_j's are not independent, so we cannot directly apply Bennet inequality in this case. In the rest of this class we will introduce the *martingale approach* that provides good concentration results for these and other problems.

Definition 7.7 (martingale)**.** *Let $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and let $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_n$ be a filtration. Let M be a $\mathcal{F}_n$-measurable random variable. Then the sequence*

$$M_j = \mathbb{E}[M_n|\mathcal{F}_j]$$

is called a martingale.

Theorem 7.8. *(Azuma's inequality) Let $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_n$ be a filtration. Let M_n be a $\mathcal{F}_n$-measurable random variable. Assume that*

$$|\,\mathbb{E}[M_n|\mathcal{F}_j] - \mathbb{E}[M_n|\mathcal{F}_{j-1}]\,| = |M_j - M_{j-1}| \leq d_j \ \ a.s.$$

for all $j = 1, \dots, n$. *Then for any* $t > 0$,

$$\mathbb{P}(|M_n - \mathbb{E}M_n| > t) \leq 2 \cdot \exp\left(-\frac{t^2}{4\sum_{j=1}^n d_j^2}\right).$$

Example 7.9. Suppose that $X_1, \dots X_n$ are centered independent random variables and for every $j \in \{1, \dots, n\}$ σ-algebra $\mathcal{F}_j$ is generated by $X_1. \dots, X_j$. Let $M_n = \sum_{j=1}^n X_j$. Then

$$M_j = \sum_{k=1}^j X_k \quad \text{and} \quad M_j - M_{j-1} = X_j.$$

Assume that every $|X_j| \leq d_j$ a.s. Then

$$\mathbb{P}\{|\sum_{i=1}^n X_i - \mathbb{E}\sum_{i=1}^n X_i| > t\} \leq 2 \cdot \exp\left(-\frac{t^2}{4\sum_{i=1}^k d_i^2}\right).$$

So, we obtain the Hoeffding inequality under an additional assumption that X_j are a.s. bounded.

Proof. (of Azuma) Let $\lambda > 0$, then

$$\begin{aligned} f(\lambda) = \mathbb{E}\exp(\lambda M_n) &= \mathbb{E}(\mathbb{E}[\exp(\lambda(M_n - M_{n-1}) + \lambda M_{n-1})|\mathcal{F}_{n-1})]) = \\ &= \mathbb{E}(\exp(\lambda M_{n-1}) \cdot \mathbb{E}[\exp(\lambda(M_n - M_{n-1}))|\mathcal{F}_{n-1}]). \end{aligned}$$

Note that for every $x \in \mathbb{R}$

$$e^x \leq x + e^{x^2}. \tag{2}$$

Indeed, this inequality is obvious for x with $|x| \geq 1$, and for small x ($|x| < 1$) we have the following estimate from the Taylor series comparison:

$$\frac{x^{2n}}{(2n)!} + \frac{x^{2n+1}}{(2n+1)!} \leq x^{2n}\left(\frac{1}{(2n)!} + \frac{1}{(2n+1)!}\right) \leq \frac{x^{2n}}{n!}.$$

Using inequality (2) for $x = \lambda(M_n - M_{n-1})$ we get that

$$\begin{aligned} \mathbb{E}[\exp(\lambda(M_n - M_{n-1}))|\mathcal{F}_{n-1}] &\leq \mathbb{E}[\lambda(M_n - M_{n-1})|\mathcal{F}_{n-1}] + \\ &+ \mathbb{E}[\exp(\lambda^2(M_n - M_{n-1})^2|\mathcal{F}_{n-1}] \leq 0 + \exp(\lambda^2 d_n^2). \end{aligned}$$

Therefore,

$$f(\lambda) = \mathbb{E}\exp(\lambda M_n) \leq \exp(\lambda^2 d_n^2) \cdot \mathbb{E}\exp(\lambda M_{n-1}).$$

Applying the same procedure to $\mathbb{E}\exp(\lambda M_{n-1})$, $\mathbb{E}\exp(\lambda M_{n-2})$ and so on, after n steps we get

$$f(\lambda) = \mathbb{E}\exp(\lambda M_n) \leq \prod_{j=1}^n \exp(\lambda^2 d_j^2) \cdot \mathbb{E}\exp(\lambda M_0).$$

Recall that $M_0 = \mathbb{E}M_n$. By Markov inequality,

$$\mathbb{P}\{M_n - \mathbb{E}M_n > t\} = \mathbb{P}\{e^{\lambda M_n} > e^{\lambda(t+M_0)}\} \leq$$

$$\leq \mathbb{E}e^{\lambda M_n}e^{-\lambda t}e^{-\lambda M_0} \leq \exp(\lambda^2 \sum_{j=1}^{n} d_j^2 - \lambda t).$$

Optimizing over λ, we take $\lambda = \frac{t}{2\sum {d_j}^2}$ and get that

$$\mathbb{P}(M_n - \mathbb{E}M_n > t) \leq \exp\left(-\frac{t^2}{4\sum_{j=1}^n d_j^2}\right).$$

Applying the same argument for $-M_n$ we get

$$\mathbb{P}(-M_n + \mathbb{E}M_n > t) \leq \exp\left(-\frac{t^2}{4\sum_{j=1}^n d_j^2}\right),$$

so absolute value of the difference cost us the multiple 2 and Azuma's inequality is proved.

□

Corollary 7.10 (Averaged bounded differences inequality)**.** *Let $X_1, \ldots,$ X_n be independent random variables with the values in some sets Ω_i. Let $f : \Omega_1 \times \ldots \times \Omega_n \to \mathbb{R}$ be a function, such that for every $(a_1, \ldots, a_n) \in$ Ω and every $j \in \{1, \ldots, n\}$,*

(3)

$$|\mathbb{E}[f(a_1, \ldots, a_{j-1}, a_j, X_{j+1}, \ldots, X_n) - f(a_1, \ldots, a_{j-1}, X_j, \ldots, X_n)]| \leq a_j \ \ a.s.$$

Then

$$\mathbb{P}\{|f(X_1, \ldots, X_n) - \mathbb{E}f(X_1, \ldots, X_n)| > t\} \leq 2 \cdot \exp\left(-\frac{t^2}{4\sum_{j=1}^n a_j^2}\right)$$

Remark 7.11. If for any $j \in \{1, \ldots, n\}$ and for any $X_1, \ldots, X_{j-1}$, $X_{j+1}, \ldots, X_n$, u and v we have that

$$|f(X_1, \ldots, X_{j-1}, u, X_{j+1}, \ldots, X_n) -$$

$$-f(X_1, \ldots, X_{j-1}, v, X_{j+1}, \ldots, X_n)| \leq a_{j-1},$$

then condition (3) holds. This is called *bounded differences condition.*

Proof. (of Corollary 7.10). Let $\mathcal{F}_j$ be a σ-algebra, generated by $X_1 \ldots,$ X_j, and $X = (X_1, \ldots, X_n)$. Let $M_n = f(X_1, \ldots, X_n)$. All that we need to check is that M_j's satisfies the condition of Azuma's theorem, namely, $|M_j - M_{j-1}| \leq a_j$ almost surely. Note that

$$\mathbb{E}[f(X)|X_1 = a_1, \ldots, X_{j-1} = a_{j-1}] = f(a_1, \ldots, a_{j-1}, X_j, \ldots, X_n).$$

Hence,

$$|M_j - M_{j-1}| = |\mathbb{E} f(a_1, \ldots a_j, X_{j+1}, \ldots, X_n) -$$
$$- \mathbb{E}(f(a_1, \ldots, a_{j-1}, X_j, X_{j+1}, \ldots, X_n))| \le a_j \text{ a.s.}$$

□

Example 7.12 (An application of the bounded differences inequality). Let $X_1, \ldots, X_n$ be independent random vectors in a normed space Y. If $\|X_j\|_Y \le a_j$ almost surely for any $j \in \{1, \ldots, n\}$, then

$$\mathbb{P}\left(\Big| \| \sum_{j=1}^{n} X_j \|_Y - \mathbb{E} \| \sum_{j=1}^{n} X_j \|_Y \Big| > t \right) \le 2 \cdot \exp\left(- \frac{ct^2}{4 \sum_i a_i^2} \right)$$

Indeed, define $f(X_1, \ldots, X_n) := \| \sum_{j=1}^n X_j \|_Y$. Then it satisfies bounded differences condition, as

$$|f(X_1, \ldots, X_{i-1}, X_j, X_{i+1}, \ldots, X_n) - f(X_1, \ldots, X_{i-1}, \tilde{X}_j, X_{i+1}, \ldots, X_n)| \le$$
$$\le \|X_j - \tilde{X}_j\|_Y \le 2a_j$$

by triangle inequality. So we can apply Corollary 7.10 to get the desired concentration estimate.

8. 10/01 Applications of Azuma inequality

Last time we proved Azuma's inequality:

Theorem 8.1. *(Azuma's inequality) Let $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_n$ be a filtration. Let M_n be a $\mathcal{F}_n$-measurable random variable. Assume that*

$$|\, \mathbb{E}[M_n | \mathcal{F}_j] - \mathbb{E}[M_n | \mathcal{F}_{j-1}] \,| = |M_j - M_{j-1}| \le d_j \;\; a.s.$$

for all $j = 1, \ldots, n$. Then for any $t > 0$,

$$\mathbb{P}(|M_n - \mathbb{E} M_n| > t) \le 2 \cdot \exp\left(- \frac{t^2}{4 \sum_{j=1}^n d_j^2} \right).$$

We also discussed Bounded Differences (BD) condition:

for every $X_1, \ldots, X_n$, every $j \in \{1, \ldots, n\}$ and every X'_j
$$|f(X_1, \ldots, X_{j-1}, X_j, X_{j+1}, \ldots, X_n) -$$
$$f(X_1, \ldots, X_{j-1}, X'_j, X_{j+1}, \ldots, X_n)| \le d_j \;\; a.s.$$

that implies the condition of Azuma and so leads to the same conclusion. One may notice that BD condition looks like discrete partial derivative boundedness condition, and this analogy is essential as we will see later in the course.

We will discuss several applications of Azuma inequality.

Concentration for the discrete cube. We consider the set $\{-1,1\}^n$ with the Hamming metric $d(x,y) = \frac{1}{2}\|x-y\|_1$.

Theorem 8.2. *Let $f : \{-1,1\}^n \to \mathbb{R}$ be a 1-Lipschitz function. Then*

$$\mathbb{P}\{|f(X) - \mathbb{E}f(X)| > t\} \le 2\exp(-\frac{t^2}{4n}).$$

Proof. If $X = (X_1, \dots, X_{j-1}, X_j, X_{j+1}, \dots, X_n)$ and $X^{(j)} = (X_1, \dots, X_{j-1}, X_j', X_{j+1}, \dots, X_n)$, then by Lipschitz condition

$$|f(X) - f(X^{(j)})| \le \operatorname{dist}(X, X^{(j)}) \le 1.$$

So, taking all $d_j := 1$ we have BD condition satisfied. □

Corollary 8.3. *Let $A \in \{-1,1\}^n$ and $\mu(A) = \mathbb{P}\{x \in A\} \ge \frac{1}{2}$. Let $t > 0$ and define $A_t = \{x \in \{-1,1\}^n : d(x,A) \le t\}$ (t-neibourhood of the set A). Then*

$$\mu(A_t^c) \le 2\exp(-ct^2/n).$$

Proof. Set $f(x) = \operatorname{dist}(x, A)$. Then f is a 1-Lipschitz function. Hence,

$$\frac{1}{2} \le \mu(A) \le \mathbb{P}\{f(x) = 0\} \le \mathbb{P}\{|f(x) - \mathbb{E}f(x)| \ge \mathbb{E}f(x)\} \le$$

$$\le 2\exp\left(-\frac{(\mathbb{E}f(X))^2}{4n}\right).$$

Hence, $\mathbb{E}f(X) \le c\sqrt{n}$. So,

$$\mathbb{P}\{f(x) \ge c\sqrt{n} + t\} \le 2\exp\left(-\frac{t^2}{4n}\right).$$

Without loss of generality, we can assume that $t \ge c\sqrt{n}$ (otherwise the right hand side is ay least 1, so does not give any meaningful estimate).

Then $t + c\sqrt{n} \le Ct$, so

$$\mu(A_t^c) = \mathbb{P}\{f(x) > t\} \le 2\exp(-ct^2/n).$$

□

Bin occupancy. Suppose that n balls are put randomly into m bins. Let X_j be the number of the bin containing the ball j, $j = 1, \dots, n$. Let Z denote the number of empty bins.

$$\mathbb{E}Z = \sum_{j=1}^{m} \mathbb{E}Z_j = m(1 - \frac{1}{m})^n \approx m e^{-n/m}.$$

The interesting regime is when $m \sim n$ (or $n/m = r \sim 1$).

Clearly, BD-condition is satisfied for the function Z: one ball can change the number of empty bins at most by one (i.e. all $d_j = 1$). Hence,

$$\mathbb{P}\{|Z(X) - \mathbb{E}Z(X)| > t\} \leq 2 \cdot \exp\left(-\frac{t^2}{4n}\right). \tag{4}$$

Here $\mathbb{E}Z(X) \sim me^{-r}$.

Now let us look to the averaged bounded differences (ABD) condition. Is the difference

$$|\mathbb{E}[Z(X)|X_1, \dots, X_j] - \mathbb{E}[Z|X_1, \dots, X_{j-1}]|$$

nicely bounded?

Recall that $Z = \sum_{k=1}^{m} Z_k$, where $Z_k = \mathbb{1}_{\{\text{bin k is empty}\}}$.

Fix an index j. Then for any k

$$\mathbb{E}[Z_k(X)|X_1, \dots X_j] = \begin{cases} \left(1 - \frac{1}{m}\right)^{n-j}, & \text{if } k \notin \{X_1, \dots, X_j\} \\ 0, & \text{if } k \in \{X_1, \dots, X_j\} \end{cases} =$$

$$= \left(1 - \frac{1}{m}\right)^{n-j} (1 - \mathbb{1}_{\{X_1, \dots, X_j\}}(k)).$$

Similarly,

$$\begin{aligned} \mathbb{E}[Z_k(X)|X_1, \dots, X_{j-1}] &= \left(1 - \frac{1}{m}\right)^{n-j+1} (1 - \mathbb{1}_{\{X_1, \dots, X_{j-1}\}}(k)) \\ &= \left(1 - \frac{1}{m}\right)^{n-j} \mathbb{E}_{X_j}(1 - \mathbb{1}_{\{X_1, \dots, X_j\}}(k)). \end{aligned}$$

Hence,

$$\left|\mathbb{E}[Z(X)|X_1, \dots, X_j] - \mathbb{E}[Z|X_1, \dots, X_{j-1}]\right|$$

$$= \left(1 - \frac{1}{m}\right)^{n-j} \cdot \Big| \sum_{k=1}^{m} (1 - \mathbb{1}_{\{X_1, \dots, X_j\}}(k)) - E_{\tilde{X}_j}(1 - \mathbb{1}_{\{X_1, \dots, X_{j-1}, \tilde{X}_j\}}(k)) \Big|$$

$$\leq \left(1-\frac{1}{m}\right)^{n-j} \cdot E_{\tilde{X}_j} \Big| \sum_{k=1}^{m} (1-\mathbb{1}_{\{X_1,\dots,X_j\}}(k)) - (1-\mathbb{1}_{\{X_1,\dots,X_{j-1},\tilde{X}_j\}}(k)) \Big|$$

$$\leq 2\left(1-\frac{1}{m}\right)^{n-j} =: a_j.$$

So we can apply Azuma inequality with these a_j. Let us estimate right hand side of the inequality in this case:

$$\sum_{j=1}^{n} a_j^2 = 4\sum_{j=1}^{n}\left(1-\frac{1}{m}\right)^{2(n-j)} = 4\frac{1-\left(1-\frac{1}{m}\right)^{2n}}{\left(1-\frac{1}{m}\right)^2} \approx 4\frac{1-e^{2r}}{1/2m}.$$

So, using the ABD-condition we have another bound from Azuma:

$$\mathbb{P}\{|Z(X)-\mathbb{E}Z(X)|>t\} \leq 2\cdot\exp\left(-\frac{t^2}{32m(1-e^{-2r})}\right). \tag{5}$$

Let us compare two estimates we got. When $r \gg 1$ the bound (5) is clearly better. In the opposite regime $r \ll 1$ the estimate (4) wins.

However, when r is very small (i.e. there are lots of bins and comparatively small amount of balls), we can observe that random variables Z_k are almost independent. If they were independent, we could have used the Bennet inequality and get a combination of exponential and subgaussian tails. And in the average regime ($t_* < t < sm$) exponential tail of Bennet gives better estimate than subgaussian tail of Azuma.

Note that the crucial fact that would allow us to get such improvement is that random variables in the sum are "almost independent" and we cannot take advantage of this condition by using the Laplace transform technique. So to justify this informally described tail refinement we will need different approach (information theory approach) We will discuss later in the course.

Summing up, we need another (not Laplace transform based) approach when we deal with non-Lipschitz function or Lipschitz one with the Lipschitz constant that is too large.

Before discussing alternative approaches, let us consider an application of Azuma's inequality to a probability space which does not have the product structure.

Concentration on the permutation group. Let Π_n denotes the group of permutations on the set $\{1,\dots,n\}$ with the metric

$$d(\pi,\pi') = \sum_{i=1}^{n} \mathbb{1}_{\pi(j)\neq\pi'(j)}.$$

Theorem 8.4 (Maurey). *Let $f : \Pi_n \to \mathbb{R}$ be a 1-Lipschitz function. Then*

$$\mathbb{P}\{|f(\pi) - \mathbb{E}f(\pi)| \geq t\} \leq 2\exp(-t^2/16n). \tag{6}$$

We will discuss its proof next time.

Corollary 8.5 (geometric concentration). *Let $A \subset \Pi_n$ be such that $\mathbb{P}\{\pi \in A\} \geq 1/2$. Then for any $t > 0$*

$$\mathbb{P}(A_t^c) \leq \exp(-ct^2/n).$$

Corollary 8.6. *Let $l(\pi)$ be the longest increasing sequence in π. Then*

$$|l(\pi) - l(\pi')| \leq d(\pi, \pi'),$$

so the function l satisfies (6).

This last question has its independent importance, as it is connected with Young tableaus and representations of finite groups.

As usual for concentration results, Maurey's theorem provides no information about $\mathbb{E}l(\pi)$. It was proved by Vershik and Kerov, that $\mathbb{E}f(\pi) \approx 2\sqrt{n}$. This means that Maurey's theorem applied to the function l provides a subgaussian *upper* tail, i.e. a bound for $\mathbb{P}\{f(\pi) - \mathbb{E}f(\pi) \geq t\}$, but fails to provide a *lower* tail, i.e. a non-trivial bound for $\mathbb{P}\{\mathbb{E}f(\pi) - f(\pi) \geq t\}$.

In this case we face the limitations of Bernstein's method again, i.e. we need to know more about the function except from being 1-Lipschitz to be able to get better tail estimate (and Bernstein method disregards other properties of the function). We will see later how information theory based technique improves Maurey's theorem considerably (see Talagrand inequality).

However the optimal tail for $l(\pi) =$ "longest increasing sequence" was established by J.Baik, P.Deift and K.Johansson in 1999, and their proof uses different methods.

9. 10/06 Maurey's theorem, Poincare inequality

Last time we stated Maurey's theorem. We will give its proof here.

Recall that Π_n denotes the group of permutations on the set $\{1, \ldots, n\}$ with the metric

$$d(\pi, \pi') = \sum_{i=1}^{n} \mathbb{1}_{\pi(j) \neq \pi'(j)}.$$

Theorem 9.1 (Maurey)**.** *Let $f : \Pi_n \to \mathbb{R}$ be a 1-Lipschitz function. Then*

$$\mathbb{P}\{|f(\pi) - \mathbb{E}f(\pi)| \geq t\} \leq 2\exp(-t^2/16n). \tag{7}$$

Proof. The main step of the proof is to generate a martingale structure suitable to apply Azuma. Let F_j be a σ-algebra, generated by the sets

$$A_{k_1,\ldots,k_j} = \{\pi \in \Pi_n : \pi(1) = k_1, \ldots, \pi(j) = k_j\}.$$

We need to check the Azuma's inequality condition, namely, to show that the quantities

$$|\mathbb{E}[f(\pi)|\mathcal{F}_j] - \mathbb{E}[f(\pi)|\mathcal{F}_{j-1}]|$$

are bounded for all j. The idea is to couple π with its copy π', preserving the uniform distribution of both permutations π and π' over Π_n and minimizing the distance above.

Let π be uniformly distributed in Π_n. Let $S = \{\pi(1), \ldots, \pi(j-1)\}$. Let y be a random index uniformly distributed in $\{1, \ldots, n\} \setminus S$ and let σ be a transposition, such that $\sigma(\pi(j)) = y$. Set $\pi' = \sigma \circ \pi$. Note that π and π' differs by at most one permutation, i.e. the distance $d(\pi, \pi')$ between them is at most 2.

Then

$$\big|\mathbb{E}[f(\pi)|F_j] - \mathbb{E}[f(\pi)|F_{j-1}]\big| = \big|\mathbb{E}[f(\pi)|F_j] - \mathbb{E}[f(\pi')|F_{j-1}]\big| \leq$$

$$\leq \mathbb{E}[|f(\pi) - \mathbb{E}_\sigma f(\pi')|\,|F_j] \leq 2,$$

as f is 1-Lipschitz function. Hence, we can apply Azuma with $d_j = 2$ and get that

$$\mathbb{P}\{|f(\pi) - \mathbb{E}f(\pi)| \geq t\} \leq 2\exp\left(-\frac{t^2}{\sum_{j=1}^n d_j^2}\right) \leq 2\exp(-t^2/16n).$$

□

Corollary 9.2 (geometric concentration)**.** *Let $A \subset \Pi_n$ be such that $\mathbb{P}\{\pi \in A\} \geq 1/2$. Then for any $t > 0$*

$$\mathbb{P}(A_t^c) \leq \exp(-ct^2/n).$$

The proof of this corollary is an exercise, very similar to the concentration on the discrete cube.

Similar concentration questions for the random matrices face the difficulty that even independent entries does not give enough independence to split Laplace transform into the product of exponents. Linear algebra techniques can help with these difficulties and get similar results for random matrices (see Golden-Thompson and Lieb's inequalities).

Now we will discuss how to get better concentration results using other properties of function, besides the Lipschitz condition. We will start with an important motivating example.

Poincare a.k.a. spectral gap inequality

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. For any $f \in L_1(\mu)$ with $\int f d\mu = 0$ we say that f satisfies Poincare inequality, if

$$\lambda_1 \int f^2 d\mu \leq \int \|\nabla f\|_2^2 d\mu$$

for some $\lambda_1 > 0$.

Remark 9.3. Note that depending on the space we can attach different meanings for the gradient in this formula (standard gradient in $\mathbb{R}^n$, or gradient with respect to the Riemannian metric on manifodls, or vector of differences on the adjacent vertices for graphs).

The parameter λ_1 is the first eigenvalue of the Laplacian, if we enumerate the eigenvalues starting from 0. Indeed, denoting by $\langle \cdot, \cdot \rangle$ the inner product in $L_2(\mu)$, we have

$$\int \|\nabla f\|_2^2 d\mu = \langle \nabla f, \nabla f \rangle = \langle -\Delta f, f \rangle = \sum_{j=1}^{\infty} \lambda_j \langle f, \varphi_j \rangle^2.$$

Here Δ is the Laplacian in the corresponding space. If there is no common definition of Laplacian in the considered space (like in the case of graphs), we can consider this to be definition of Laplacian. Then $-\Delta$ is a non-negative linear operator the corresponding space, say the Sobolev space, so we can write it in the eigenbasis $\{\varphi_j\}$ with the eigenvalues λ_j.

Suppose that eigenvalues are ordered

$$\lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \dots$$

Note that $\lambda_0 = 0$ and $\varphi_0 = 1$ since μ is a probability measure. The smallest non-zero eigenvalue (λ_1) is sometimes called a *spectral gap* of an operator, that explains the second name of the theorem. Using that $\lambda_0 = 0$,

$$\int \|\nabla f\|_2^2 d\mu = \sum_{\lambda=1}^{\infty} \lambda_i \langle f, \varphi_j \rangle^2 \geq \lambda_1 \sum_{\lambda=1}^{\infty} \langle f, \varphi_j \rangle^2 =$$

(as φ_0 is a constant, hence $\langle f, \varphi_0\rangle^2 = 0$)

$$= \lambda_1 \cdot \sum_{\lambda=0}^{\infty} \langle f, \varphi_j\rangle^2 \geq \lambda_1 \|f\|_2^2.$$

Remark 9.4. If the first moment of f is non-zero, applying the theorem to $f - \mathbb{E}f$, we obtain

$$\lambda_1 \operatorname{Var}(f(X)) \leq \mathbb{E}\|\nabla f\|^2$$

holds for every f such that right hand side is finite.

Poincare inequality can be extended to multiple dimensions with the same constant λ_1. This property of Poincare inequality is of general interest for us, as we will see in sequel.

Theorem 9.5 (Tensorization property)**.** *Let $X_1, \ldots, X_n$ be independent random variables satisfying Poincare inequality with some constant λ_1. Then $X = (X_1, \ldots, X_n)$ satisfies Poincare inequality with the same λ_1.*

Proof. Let $f : \Omega_1 \times \ldots \times \Omega_n \to \mathbb{R}$ be a measurable function. Denote

$$\mathbb{E}_{\geq j} f(X) := \mathbb{E}[f(X)|\mathcal{F}_j],$$

where $\mathcal{F}_j$ are σ-algebras, generated by $X_1, \ldots, X_j$. Since we deal with the product space, taking the conditional expectation with respect to $\mathcal{F}_j$ amounts to fixing the values of $X_1, \ldots, X_j$ and integrating over $X_{j+1}, \ldots, X_n$.

Hence,

$$f(X) - \mathbb{E}f(X) = \sum_{j=1}^{n} (\mathbb{E}_{\geq j} f(X) - \mathbb{E}_{\geq j-1} f(X)).$$

Denoting $\Delta_j := \mathbb{E}_{\geq j} f(X) - \mathbb{E}_{\geq j-1} f(X)$, squaring and and taking expectation of both parts, we get that

$$\mathbb{E}(f(X) - \mathbb{E}f(X))^2 = \mathbb{E}(\sum_{j=1}^{n} \Delta_j)^2 = \sum_{j=1}^{n} \mathbb{E}\Delta_j^2 + 2\sum_{j<k} \mathbb{E}\Delta_j \Delta_k.$$

Note that the second sum is zero by martingale property of the sequence. Then,

$$\operatorname{Var} f(X) = \mathbb{E}(f(X) - \mathbb{E}f(X))^2 = \sum_{j=1}^{n} \mathbb{E}(\mathbb{E}_{\geq j} f(X) - \mathbb{E}_{\geq j}\mathbb{E}_j f(X))^2.$$

Cauchy-Swartz inequality allows us to interchange squares and summation to get

$$\operatorname{Var} f(X) \leq \sum_{j=1}^{n} \mathbb{E}(f(X) - \mathbb{E}_j f(X))^2. \tag{8}$$

Now, let us consider the function f as a one variable function (by fixing all coordinates except X_j):

$$f_j^{(X_1 \ldots X_{j-i} X_{j+1} \ldots X_n)}(X) = f(X_1 \ldots X_{j-1}, X, X_{j+1} \ldots X_n).$$

Rewriting (8) with the new notation

$$\operatorname{Var} f(X) \leq \sum_{j=1}^{n} \mathbb{E}(\operatorname{Var}_j f_j^{(X_1 \ldots X_{j-i} X_{j+1} \ldots X_n)}(X)) \leq$$

(we can apply one dimensional Poincare inequality for $f_j^{(\ldots)}(X)$)

$$\leq \frac{1}{\lambda_1} \mathbb{E} \sum_{j=1}^{n} \mathbb{E}_j \left\| \frac{\partial f}{\partial X_j} \right\|_2^2 = \frac{1}{\lambda_1} \mathbb{E} \|\nabla f\|_2^2.$$

$\square$

Poincare inequality implies exponential concentration, as we will prove in the following theorem.

Theorem 9.6 (Gromov-Milman). *Let X be a random variable, satisfying Poincare inequality with some $\lambda_1 > 0$. Then for any 1-Lipschitz function f*

$$\mathbb{P}\{|f(X) - \mathbb{E} f(X)| > t\} \leq 2 \exp(-c\sqrt{\lambda_1} t).$$

Remark 9.7. Gromov and Milman used this result to establish concentration on maniflods. Also, Alon and Milman used Poincare inequality for graphs. In particular, they obtained a spectral characterization for expander graphs. A discussion of expanders and their properties can be found in Professor Barvinok's lectures (see recommended literature).

10. 10/08 Gromov-Milman theorem. Entropy

Theorem 10.1 (Gromov-Milman)**.** *Let X be a random variable, satisfying Poincare inequality with some $\lambda_1 > 0$. Then for any function f such that $\|\nabla f\|_2 \le 1$ a.s., one has*

$$\mathbb{P}\{|f(X) - \mathbb{E}f(X)| > t\} \le 2\exp(-c\sqrt{\lambda_1}t).$$

Proof. Assume that $\|f\|_\infty < \infty$ (if this is not the case, consider the sequence of truncations $f_n = f \cdot \mathbb{1}_{[-n,n]}(f)$, if we know that the theorem holds for every truncation, then it holds for the original function by continuity).

For every $\nu \in \mathbb{R}$ set

$$\varphi(\nu) := \mathbb{E}\exp(\nu f(X)).$$

Then by the Poincare inequality,

$$\lambda_1 \operatorname{Var}\varphi(\nu) = \lambda_1 \left(\mathbb{E}\exp(2\nu f(X)) - (\mathbb{E}\exp(\nu f(X)))^2\right) \le$$
$$\le \mathbb{E}\|\nabla \exp(\nu f(X))\|_2^2 = \nu^2 \mathbb{E}[\exp(2\nu f(X))\|\nabla f(X)\|_2^2].$$

The random variable $\|\nabla f(X)\|_2^2$ is uniformly bounded by 1 by the Lipschitz condition, so we got a recurrence

$$\lambda_1(\varphi(2\nu) - \varphi^2(\nu)) \le \nu^2 \varphi(2\nu).$$

Iterating it down to zero, we obtain for $\mu := 2\nu$

$$\varphi(\mu) \le \prod_{j=1}^{n} \left(1 - \frac{\mu^2}{\lambda_1} 2^{-2j}\right)^{-2^j} \cdot (\varphi(2^{-n}\mu))^{2^n},$$

provided that all terms in the product are positive, i.e., whenever $|\mu| \le c\sqrt{\lambda_1}$. Note that

$$\prod_{j=1}^{\infty} \left(1 - \frac{\mu^2}{\lambda_1} 2^{-2j}\right)^{-2^j} < +\infty$$

when $|\mu| \le c\sqrt{\lambda_1}$.

Also,

$$\lim_{t\to+0} (\varphi(\mu t))^{1/t} = \lim_{t\to 0+} (\mathbb{E}\exp(\mu t f(X)))^{1/t} =$$
$$= \lim_{t\to 0+} (1 + \mu t \mathbb{E}(f(X)) + o(t^2))^{1/t} = \exp(\mu \mathbb{E}f(X)).$$

Therefore,

$$\mathbb{E}\exp(\mu f(X)) \le C\exp(\mu \mathbb{E}f(X))$$

for all μ such that $|\mu| \le c\sqrt{\lambda_1}$.

And so, taking $\mu := c\sqrt{\lambda_1}$, by usual exponential Markov inequality,

$$\mathbb{P}\{f(X) > t + \mathbb{E}f(X)) = \mathbb{P}(\exp(\mu f(X)) > \exp(\mu(t + \mathbb{E}f(X)))\} \le$$

$$\le \frac{\mathbb{E}\exp(\mu f(X))}{\exp(\mu(t+\mathbb{E}f(X)))} \le Ce^{-\mu t} = Ce^{-c\sqrt{\lambda_1}t}.$$

As before, the case when $\mathbb{E}f(X) > f(X)$ can be treated in the same way (with $X := -X$) and gives an additional multiple 2 in probability estimate.

□

Remark 10.2. Poincare inequality technique cannot give concentration better than exponential.

Remark 10.3. (Concentration on permutation group) Let Σ_n denote the set of all transpositions and let Π_n be the permutation group equipped with the Hamming metric $d(\pi,\pi') = \sum_{i=1}^n \mathbb{1}_{\pi(j)\neq\pi'(j)}$. Define the gradient of function $f : \Pi_n \to \mathbb{R}$ as

$$\nabla f(\pi) = (f(\pi) - f(\sigma \circ \pi))_{\sigma\in\Sigma_n}.$$

Let $f : \Pi_n \to \mathbb{R}$ be a function with $\|\nabla f\|_2 = 1$, then by Gromov-Milman theorem,

$$\mathbb{P}\{|f(X) - \mathbb{E}f(X)| > t\} \le 2\exp(-c\sqrt{\lambda_1}t).$$

To be able to use this result, we have to estimate λ_1. The precise value of λ_1 was obtained by Diaconis. and Shahshahani using methods of harmonic analysis on Π_n. A weaker result can be obtained with the help of Markov chains. We can walk from a given permutation to a neighboring one by applying a randomly chosen transposition. This creates a reversible Markov chain called the *transposition walk*. The spectral gap of Laplacian is controlled by the mixing time of this chain. It turns out that known methods of the mixing time estimation give the right order (but not the value) of λ_1 : $\lambda_1 = O(n)$.

On the other hand, if the function f is 1-Lipschitz in Hamming metric, we can also apply Maurey's theorem tho get subgaussian concentration:

$$p := \mathbb{P}\{|f(\pi) - \mathbb{E}f(\pi)| \ge t\} \le 2\exp(-t^2/16n). \tag{9}$$

Let us compare the results of two theorems.

First, for large deviations subgaussian concentration is generally better than subexponential.

Second, if f is 1-Lipschitz, then

$$\|\nabla f\|_2^2 = \sum_{\sigma\in\Sigma_n} (f(\pi) - f(\sigma\circ\pi))^2 = n^2.$$

Hence, Gromov-Milman theorem and the estimate of λ_1 yield

$$p \le 2\exp(-ct/\sqrt{n}). \tag{10}$$

If $t \geq c'\sqrt{n}$, then (9) is better than (10). On the other hand, if $t \leq c'\sqrt{n}$ and c' is small enough, then both inequalities are vacuous. Thus, if we only know that the function f is Lipschitz, Maurey's theorem always yields a stronger bound.

So, can Gromov-Milman be better in some case? The answer is yes, and this case is actually very important in applications, as it models sampling without repetitions from some large set.

Let $f(\pi) = F(\{\pi(1), \ldots, \pi(k)\})$ for some $k < n$ and

$$F : \{1, \ldots, n\}^k \to \mathbb{R}$$

(so it is lower dimensional function in higher dimensional space Π_n). Then

$$\|\nabla f\|_2^2 \leq k(n-k),$$

hence, by Gromov-Milman theorem,

$$p \leq 2\exp\left(-c\sqrt{\frac{n}{n-k}\frac{1}{k}}t\right).$$

In the regime $k \ll n$

$$p \leq 2\exp(-ct/\sqrt{k}),$$

and for a reasonably small deviation t this is better then $\exp(-t^2/16n)$ given by Maurey's theorem.

Our next goal is another inequality that has tensorization property (log-Sobolev inequality). As a preparation to it, in the rest of this class we will discuss *entropy* of a random variable and its main properties.

Definition 10.4 (Entropy)**.** *Let X be a positive random variable. We define the entropy of X as*

$$\mathrm{Ent}(X) = \mathbb{E}X\log X - \mathbb{E}X \cdot \log \mathbb{E}X$$

Note that $t\log t \geq -e^{-1}$ for all positive t, i.e. negative part of $X\log X$ is bounded, hence, the expression is well-defined (and sometimes might be $+\infty$).

Some properties of the entropy:

- entropy is 1-homogenious:
$$\mathrm{Ent}(cX) = c\mathbb{E}X(\log c + \log X) - c\cdot\mathbb{E}X\log(c+\mathbb{E}X) =$$
$$= c\mathbb{E}X\log X - c\mathbb{E}X\log\mathbb{E}X = c\,\mathrm{Ent}(X).$$
- by Jensen's inequality, for any random variable X its entropy $\mathrm{Ent}(X) \geq 0$ and $\mathrm{Ent}(X) = 0$ if and only if $X = \mathrm{Const}$ almost surely.

- The function $\varphi(t) = t \log t$ is bounded below and convex, hence, it can be used to an Orlitz norm (it is not positive, so we would have to modify it). We will not be generally interested in the Banach structure of the space, but, as in the case of Banach spaces, we need the concept of duality. As we discussed before, the dual norm for φ-Orlitz norm is defined by the Legendre transform of φ:

$$L(\varphi)(x) = \sup_{y>0}(xy - \varphi(y)) = \sup_{y>0}(xy - y \log y).$$

The only critical point is $y = e^{x-1}$ (derivative condition is $x - 1 - \log y = 0$). It is easy to check that this gives maximum and then

$$L(\varphi)(x) = xe^{x-1} - (x-1)e^{x-1} = e^{x-1}.$$

Indeed, we have the following duality theorem for entropy:

Theorem 10.5 (Duality). *Let Y be a positive random variable. Then*

$$\mathrm{Ent}(Y) = \sup_{u:\mathbb{E}e^u \le 1} \mathbb{E}uY.$$

Moreover, if

- *U is an L_1-variable and*
- *for all $Y > 0$ with $\mathrm{Ent}(Y) \le 1$ we have $\mathbb{E}UY \le 1$*

then $\mathbb{E}e^U \le 1$.

Proof. Note that it is enough to check that entropy is a supremum over u such that $\mathbb{E}e^u = 1$. This allows us to consider new probability measure $\tilde{\mathbb{P}}$ with the density e^u with respect to $\mathbb{P}$: for every Borel set A, set

$$\tilde{P}(A) = \int_A e^u d\mathbb{P}.$$

Then

$$\mathbb{E}_{\mathbb{P}}Y = \mathbb{E}_{\tilde{\mathbb{P}}}e^{-u}Y.$$

We can estimate the difference

$$\mathrm{Ent}(Y) - \mathbb{E}uY = \mathbb{E}_{\mathbb{P}}Y \log Y - \mathbb{E}_{\mathbb{P}}Y \log \mathbb{E}_{\mathbb{P}}Y - \mathbb{E}_{\mathbb{P}}uY =$$

$$= \mathbb{E}_{\mathbb{P}}(Y(\log Y - u)) - \mathbb{E}_{\mathbb{P}}Y \log \mathbb{E}_{\mathbb{P}}Y =$$

$$= \mathbb{E}_{\tilde{\mathbb{P}}}(e^{-u}Y \log e^{-u}Y) - \mathbb{E}_{\tilde{\mathbb{P}}}e^{-u}Y \log \mathbb{E}_{\tilde{\mathbb{P}}}e^{-u}Y = \mathrm{Ent}_{\tilde{P}}(e^{-u}Y) \ge 0$$

by the general property of entropy. Moreover,

$$\mathrm{Ent}(Y) - \mathbb{E}uY = \mathrm{Ent}_{\tilde{P}}(e^{-u}Y) = 0$$

if and only if $e^{-u}Y$ is a constant. So, the first part of the theorem is proved.

Now, assume that U satisfies the conditions of the second part of the theorem. Assume in addition that $\mathbb{E}Ue^U$ is finite.

Set $\tilde{Y} := \lambda e^U$. Then

$$\mathrm{Ent}(\tilde{Y}) = \mathbb{E}\lambda e^U(\log\lambda + U) - \mathbb{E}\lambda e^U \log(\lambda \mathbb{E}e^U) = \lambda(\mathbb{E}Ue^U - \mathbb{E}e^U \log \mathbb{E}e^U).$$

Take

$$\lambda := (\mathbb{E}Ue^U - \mathbb{E}e^U \log \mathbb{E}e^U)^{-1}$$

to ensure that $\mathrm{Ent}(\tilde{Y}) = 1$. Then by the second assumption we know that

$$1 \geq \mathbb{E}U\tilde{Y} = \frac{\mathbb{E}Ue^U}{\mathbb{E}Ue^U - \mathbb{E}e^U \log \mathbb{E}e^U}.$$

To satisfy this, we must have $\log \mathbb{E}e^U \leq 0$ or, equivalently, $\mathbb{E}e^U \leq 1$.

To remove the assumption that $\mathbb{E}Ue^U$ is finite, it is enough to apply the previous argument to the random variable $U_n = U \wedge n$ and take a limit as $n \to +\infty$. $\square$

11. 10/13 Subadditivity of entropy. Log-Sobolev inequality

Last time we discussed the entropy function

$$\mathrm{Ent}(X) = \mathbb{E}X \log X - \mathbb{E}X \cdot \log \mathbb{E}X$$

and its duality property

$$\mathrm{Ent}(X) = \sup\{\mathbb{E}Yu : \mathbb{E}e^u = 1\}.$$

Corollary 11.1.

$$\mathrm{Ent}(Y) = \mathbb{E}Y(\log Y - \log \mathbb{E}Y) =$$

$$= \sup\{\mathbb{E}Y(\log Z - \log \mathbb{E}Z) : Z > 0, \mathbb{E}Z < +\infty\}.$$

Proof. Given Z, set $X = \log Z - \log \mathbb{E}Z$, then $\mathbb{E}e^X = 1$.

$\square$

Theorem 11.2 (Subadditivity of entropy)**.** *Let $X_1, \ldots, X_n$ be independent random variables. For $f : \mathbb{R}^n \to \mathbb{R}$ and $j \in \{1, \ldots, n\}$ set $\mathbb{E}_j f(X_1, \ldots, X_n) = \mathbb{E}[f(X_1, \ldots, X_n)|X_1, \ldots, X_{j-1}, X_{j+1}, \ldots, X_n]$. Ent_j denotes entropy with respect to $\mathbb{E}_j$. Then*

$$\mathrm{Ent}\, f(X_1, \ldots, X_n) \leq \mathbb{E} \sum_{j=1}^{n} \mathrm{Ent}_j\, f(X_1, \ldots, X_n).$$

Proof.

$$\mathrm{Ent}\, f(X_1, \ldots, X_n) = \mathbb{E}[f(X_1, \ldots, X_n) \cdot \\ \cdot (\log f(X_1, \ldots, X_n) - \log \mathbb{E} f(X_1, \ldots, X_n))].$$

Denote $\mathbb{E}_{>j}$ the expectation conditional on σ-algebra, generated by $X_1, \ldots, X_j$ [1] We can decompose the difference as a telescopic sum:

$$\log f(X_1, \ldots, X_n) - \log \mathbb{E} f(X_1, \ldots, X_n) =$$

$$= \sum_{j=1}^{n} (\log \mathbb{E}_{>j} f(X_1, \ldots, X_n) - \log \mathbb{E}_{>j-1} f(X_1, \ldots, X_n)) =$$

$$= \sum_{j=1}^{n} (\log \mathbb{E}_{>j} f(X_1, \ldots, X_n) - \log \mathbb{E}_j \mathbb{E}_{>j} f(X_!, \ldots, X_n)).$$

Setting $Z_j = \mathbb{E}_{>j} f(X_1, \ldots, X_n)$, we get by Corollary 11.1 that

$$\mathrm{Ent}\, f(X_1, \ldots, X_n) = \mathbb{E} \sum_{j=1}^{n} f(X_1, \ldots, X_n)(\log Z_j - \log \mathbb{E}_j Z_j) \leq$$

$$\leq \mathbb{E} \sum_{j=1}^{N} \mathrm{Ent}_j (f(X_1, \ldots, X_n)).$$

□

Our goal is to prove *the log-Sobolev inequality* (LSI):

$$\mathrm{Ent}\, f^2(X) \leq \alpha \mathbb{E} \|\nabla f\|_2^2.$$

First, let us obtain an easy corollary from subadditivity theorem, showing that log-Sobolev inequality is an another example of tensorizable inequality (like Poincare inequality).

Corollary 11.3 (Tensorization)**.** *Let $X_1, \ldots, X_n$ be independent random variables satisfying the log-Sobolev inequality with some parameter α. Then $X = (X_1, \ldots, X_n)$ satisfies the same inequality with the same constant α.*

[1]We used the notation $\mathbb{E}_{\geq j}$ in class. However, several students pointed out that the notation $\mathbb{E}_{>j}$ is more convenient.

Proof. By subadditivity and then by 1-dimensional log-Sobolev, we obtain

$$\operatorname{Ent} f(X_1,\dots,X_n) \le \mathbb{E}\sum_{j=1}^{n} \operatorname{Ent}_j f(X_1,\dots,X_n) \le$$
$$\le \mathbb{E}\sum_{j=1}^{n} \alpha\mathbb{E}_j\left|\frac{\partial f}{\partial X_j}\right|^2 = \alpha\mathbb{E}\|\nabla f\|_2^2.$$

□

Now we will prove log-Sobolev inequality in the simplest case (for Rademacher random variables)

Lemma 11.4. *X is a Rademacher random variable. Then $f(X)$ satisfies LSI with parameter $1/2$. Moreover, if $f : \{-1,1\} \to \{0,1\}$, then $f(X)$ satisfies LSI with parameter $\alpha = \log 2/2$.*

Proof. Let $f(-1) = a$, $f(1) = b$ for some $a, b \in \mathbb{R}$. For the first part we have to show that

$$\frac{1}{2}a^2\log a^2 + \frac{1}{2}b^2\log b^2 - \frac{a^2+b^2}{2}\log\frac{a^2+b^2}{2} \le \frac{1}{2}(a-b)^2.$$

This can be checked directly. The derivative with respect to a with fixed b has only one zero at b, moreover, it gives max by the second derivative test.

For the second part, note that the only $\{0,1\}$-valued function that gives non-zero left hand side is for $a = 0$, $b = 1$ or reverse. And these two functions satisfy

$$-\frac{a^2+b^2}{2}\log\frac{a^2+b^2}{2} \le \frac{\log 2}{2}(a-b)^2 \Leftrightarrow$$
$$\Leftrightarrow -\frac{1}{2}\log\frac{1}{2} = \frac{\log 2}{2}\cdot 1.$$

□

Applying tensorization property we can get multidimensional version of the same theorem:

Theorem 11.5. *Let $\{-1,1\}^n \to \mathbb{R}$. Then f satisfies LSI with parameter $1/2$. Moreover, Moreover, if $f : \{-1,1\} \to \{0,1\}$, then $f(X)$ satisfies LSI with parameter $\alpha = \log 2/2$.*

Corollary 11.6 (Edge isoperimetric inequality for discrete cube)**.** *Let $A \subset \{-1,1\}^n$. Denote*

$$I(A) := \sum_{j=1}^{n} \mathbb{P}(\mathbb{1}_A(X) \ne \mathbb{1}_A(\overline{X}^j)),$$

where X is a random variable, uniformly distributed in $\{-1,1\}^n$ and $\overline{X}^j = (X_!, \ldots, X_{j-1}, -X_j, X_{j+1}, \ldots, X_n)$. Then

$$2\mathbb{P}(A)\log_2 \frac{1}{\mathbb{P}(A)} \leq I(A).$$

Proof. Consider $f(X) = \mathbb{1}_A(X)$. Then

$$\operatorname{Ent} f^2(X) = -\mathbb{E}(\mathbb{1}_A(X)) \cdot \log \mathbb{E}(\mathbb{1}_A(X)) = \mathbb{P}(A) \log \frac{1}{\mathbb{P}(A)}.$$

Gradient on the right hand side of LSI is non-zero, if $X \in A$ and $\overline{X}^j \notin A$ or reverse, $X \notin A$ and $\overline{X}^j \in A$. Hence,

$$\mathbb{P}(A) \log \frac{1}{\mathbb{P}(A)} \leq \frac{1}{2} \log 2 \sum_{j=1}^{n} \mathbb{E}(\mathbb{1}_A(X) - \mathbb{1}_A(\overline{X}^j))^2 =$$

$$= \frac{1}{2} \log 2 \sum_{j=1}^{n} \mathbb{P}(\mathbb{1}_A(X) \neq \mathbb{1}_A(\overline{X}^j)).$$

By definition, we have

$$\mathbb{P}(A) \frac{1}{\log \mathbb{P}(A)} \leq \frac{1}{2} \log 2 \cdot I(A).$$

□

The quantity $I(A)$ is called *total influence* of the set A. Note that

$$I(A) = 2^{-n} \cdot 2 \cdot |\partial_E(A)|,$$

where $\partial_E(A)$ denotes the *edge boundary* of A, i.e. the set of all edges connecting A with A^c.

Corollary 11.7. *Let $1 < k < n$. Among all subsets of $\{-1,1\}^n$ of cardinality 2^k, the set with the smallest edge boundary is $\{-1,1\}^k$ (low-dimensional cube).*

Proof. If $|A| = 2^k$, then by LSI for boolean functions

$$\mathbb{P}(A) \log_2 \frac{1}{\mathbb{P}(A)} = (n-k) \cdot 2^{k-n} \leq \frac{1}{2} I(A).$$

On the other hand, if

$$A_0 = \{-1,1\}^k \subset \{-1,1\}^n,$$

then every vertex of A_0 is connected to $n-k$ vertices outside A_0, so

$$I(A_0) = 2^{-n} \cdot 2 \cdot (n-k)|A_0| = (n-k) \cdot 2^{k-n+1}.$$

Thus $I(A) \geq I(A_0)$, hence, $|\partial_E(A)| \geq |\partial_E(A_0)|$.

□

This example shows how the knowledge of precise constants in concentration inequalities might matter (here we got a delicate geometric result using the precise value of $\alpha = (1/2)\log 2$ for boolean functions).

Theorem 11.8 (Federbush, Gross)**.** *Let $X = (X_1, \dots, X_n)$, where $X_1, \dots, X_n$ are independent standard normal random variables ($X_j \sim N(0,1)$). Then X satisfies LSI with parameter 2.*

Proof. It is clearly enough to consider $n = 1$.

Any function f suitable to LSI is integrable and has its gradient in L_2, so it lies in W^2-Sobolev class (by definition). Such function can be approximated by infinitely differentiable function (from C^∞ class) by convolution, and all C^∞ functions can be in turn approximated by the C^∞ functions with the bounded support. So, it is enough to prove the theorem for all C^∞ functions with the bounded support.

Let f be such function and let $Y_1, \dots, Y_m$ be independent Rademacher random variables. Set

$$S_m = \frac{1}{\sqrt{m}} \sum_{j=1}^{m} Y_j.$$

Then, by the central limit theorem (CLT)

$$S_m \to X \sim N(0,1) \quad \text{weakly.}$$

We can extend the function $\varphi : (0, +\infty) \to \mathbb{R},\ \varphi(t) = t \log t$ continuously to 0 by setting $\varphi(0) = 0$. Then weak convergence allows us to conclude that

$$\operatorname{Ent} f(S_m) \to \operatorname{Ent} f(X).$$

By the LSI for Rademacher random variables,

$$\operatorname{Ent} f^2(S_n) \leq \frac{1}{2} \mathbb{E} \sum_{j=1}^{m} \left(f(S_m) - f(S_m - 2\frac{Y_j}{\sqrt{m}}) \right)^2.$$

Let us use Taylor approximation:

$$f(S_m) - f(S_m - 2\frac{Y_j}{\sqrt{m}}) = f'(S_m) \cdot 2\frac{Y_j}{\sqrt{m}} + O\left(\frac{1}{m}\right)$$

since $\|f''\|_\infty < +\infty$. Squaring the previous expression and summing over j, we obtain

$$\operatorname{Ent} f^2(S_n) \leq \mathbb{E}[2f'(S_m) + o(1)] \underset{CLT}{\to} 2 \cdot \mathbb{E}[f'(X)^2].$$

$\square$

12. 10/15 Concentration for discrete cube, gaussian and S^{n-1}

Recall that if X is a ± 1-vector in $\mathbb{R}^n$, we denote by $\overline{X}^{(i)}$ the vector X with the i-th coordinate flipped.

Theorem 12.1. *Let $f : \{-1,1\}^n \to \mathbb{R}$ (a function on a discrete cube). If there is some constant $\nu > 0$ such that for all $X \in \{-1,1\}^n$*

$$\sum_{i=1}^{n}[(f(X) - f(\overline{X}^{(i)}))_+]^2 \leq \nu.$$

Then

$$\mathbb{P}\{|f(X) - \mathbb{E}f(X)| > t\} \leq 2\exp(-t^2/\nu).$$

Remark 12.2. Note that the condition of the theorem reminds bounded differences condition for martingales, with the difference that we need to consider only the positive part of each term (this does not matter for Lipschitz condition, but it might matter if the other properties are in use).

Also note that, unlike previous time, here the coefficient 1 in front of t^2/ν is optimal.

Proof. The idea is to apply LSI to the Laplace transform in a way we applied Poincare inequality in Gromov-Milman theorem. Let

$$\varphi(x) = \mathbb{E}\exp(\lambda f(x)).$$

Applying LSI to $\exp\big((\lambda/2)f(x)\big)$ we get

$$\mathrm{Ent}(\varphi) \leq \frac{1}{2}\mathbb{E}\|\nabla \exp\big((\lambda/2)f(x)\big)\|_2^2,$$

which can be rewritten as

$$\mathbb{E}\exp(\lambda f(X))\lambda f(X) - \mathbb{E}\exp(\lambda f(X))\log\mathbb{E}(\exp(\lambda f(X)) \leq$$

$$\leq \frac{1}{2}\mathbb{E}\sum_{j=1}^{n}(\exp(\lambda f(X)/2) - \exp(\lambda f(\overline{X}^j)/2))^2.$$

Here, left hand side is $\lambda\varphi'(\lambda) - \varphi(\lambda)\log\varphi(\lambda)$. To simplify the right hand side, note that as both X and $\overline{X}^{(j)}$ are uniformly distributed on $\{-1,1\}^n$, we have

$$\mathbb{E}(\exp(\lambda f(X)) - \exp(\lambda f(\overline{X}^j))) = 0.$$

Then,

$$\mathbb{E}(\exp(\lambda f(X)/2) - \exp(\lambda f(\overline{X}^j)/2))^2 =$$
$$= 2\mathbb{E}[(\exp(\lambda f(X)/2) - \exp(\lambda f(\overline{X}^j)/2))_+]^2.$$

Assume that $\lambda > 0$.

For any $a < b$ by Lagrange theorem

$$(e^b - e^a)^2 = e^{2\theta}(b-a)^2 \leq e^{2b}(b-a)^2.$$

Hence,

$$\lambda\varphi'(\lambda) - \varphi(\lambda)\log\varphi(\lambda) \leq$$
$$\leq \frac{1}{2}\mathbb{E}\sum_{j=1}^{n}(\exp(\lambda f(X)/2) - \exp(\lambda f(\overline{X}^j)/2))^2 \leq$$
$$\leq \frac{1}{2}\cdot 2\mathbb{E}\sum_{j=1}^{n} e^{\lambda f(X)}(\lambda/2)^2[(f(X) - f(X^j))_+]^2 \leq \frac{\lambda^2}{4}\nu\varphi(\lambda).$$

Then,

$$\lambda\varphi'(\lambda) - \varphi(\lambda)\log\varphi(\lambda) \leq \frac{\lambda^2}{4}\nu\varphi(\lambda)$$

and

$$\lambda(\log\varphi(\lambda))' - \log\varphi(\lambda) \leq \frac{\lambda^2}{4}\nu,$$
$$\left(\frac{\log\varphi(\lambda)}{\lambda}\right)' \leq \frac{\nu}{4}.$$

Solving this differential inequality, we get

$$\frac{\log\varphi(\lambda)}{\lambda} \leq \frac{\nu}{4}\lambda + C_0,$$

where C_0 is the value of the function at zero, hence

$$\frac{\log\varphi(\lambda)}{\lambda} \leq \frac{\nu}{4}\lambda + \lim_{\lambda\to 0}\left(\frac{\log\varphi(\lambda)}{\lambda}\right).$$

Let us compute the constant. By definition of φ and Taylor expansion,

$$\lim_{\lambda\to 0}\frac{\log\varphi(\lambda)}{\lambda} = \lim_{\lambda\to 0}\frac{1}{\lambda}\log\mathbb{E}[1 + \lambda f(X) + O(\lambda^2)] =$$
$$= \lim_{\lambda\to 0}\frac{1}{\lambda}\log(1 + \lambda\mathbb{E}f(X) + O(\lambda^2)] = \mathbb{E}f(X).$$

We have

$$\varphi(\lambda) = \mathbb{E}\exp(\lambda f(X)) \le \exp(\lambda \mathbb{E} f + \frac{\lambda^2}{4}\nu).$$

Hence, by exponential Markov inequality,

$$\mathbb{P}\{f(X) - \mathbb{E}f(X) > t\} = \mathbb{P}\{\exp(\lambda f(X)) \ge \exp(\lambda \mathbb{E} f(X) + \lambda t)\} \le$$
$$\le \exp(-\lambda t + \frac{\lambda^2}{4}\nu).$$

Taking $\lambda = \frac{2t}{\nu}$ we will get the desired estimate

$$\mathbb{P}\{f(X) - \mathbb{E}f(X) > t\} \le \exp(-t^2/\nu).$$

Absolute value in the left hand side costs us the multiple 2, as usual.

□

Theorem 12.3 (Gaussian concentration)**.** *Let $g \in \mathbb{R}^n$ be a standard gaussian vector and let $f : \mathbb{R}^n \to \mathbb{R}$ be any 1-Lipschitz function. Then*

$$\mathbb{P}\{|f(g) - \mathbb{E}f(g)| > t\} \le 2\exp(-\frac{t^2}{2}).$$

Remark 12.4. Note that this concentration statement is dimension independent.

Remark 12.5. In the previous lecture, we derived the isoperimetric inequality on the discrete cube from the LSI. Gaussian concentration is also related to the *gaussian isoperimetric inequality*. It states that among all sets of given Gaussian measure in $\mathbb{R}^n$, half-spaces have the minimal Gaussian boundary measure.

The gaussian isoperimetric inequality was originally proved by Sudakov and Tsirelson, and independently by Borell. They used Levy isoperimetric inequality for the unit sphere (the fact that among all sets on S^{n-1} with the same uniform measure, spherical cap has the smallest boundary measure) along with Maxwell's theorem (saying that the projections to $\mathbb{R}^n$ of the uniform measure on $\sqrt{N}S^{N-1}$ tend to gaussian measure on $\mathbb{R}^n$ when $N \to \infty$). As projections of spherical caps tend to half-spaces when $N \to \infty$, Levy isoperimetric inequality will imply gaussian isoperimetric inequality. After the gaussian isoperimetric inequality is established, Theorem 13.1 can be easily derived from it.

The LSI provides much simpler proof of the same statement (avoiding the hard result by Levy). It is, however, insufficient to derive the gaussian isoperimetric inequality. To obtain it by the functional method, one needs a tensorizable inequality which is stronger than the LSI. Such inequality has been recently proved by Bobkov.

Proof of the gaussian concentration inequality. Let $\varphi(\lambda) = \mathbb{E}\exp(\lambda f(g))$. Applying the gaussian LSI to $\exp(\lambda f(g)/2)$, we get

$$\lambda\varphi'(\lambda) - \varphi(\lambda)\log\varphi(\lambda) \le \mathbb{E}\|\nabla\exp(\lambda f(g)/2)\|_2^2.$$

Here,

$$\mathbb{E}\|\nabla\exp(\lambda f(g)/2)\|_2^2 = 2\mathbb{E}[(\lambda/2)^2\|\nabla f(g)\|_2^2\exp(\lambda f(g))] \le$$

(if $\|f\|_{\mathrm{Lip}} \le L$)

$$\le \frac{\lambda^2}{2}L^2\varphi(\lambda).$$

Then, exactly like in the proof of Theorem 12.1, we conclude that

$$\mathbb{P}\{|f(g) - \mathbb{E}f(g)| > t\} \le 2\exp(-\frac{t^2}{2L^2}).$$

□

Theorem 12.6 (Concentration on S^{n-1})**.** *Let $f : S^{n-1} \to \mathbb{R}$ be a 1-Lipschitz function. Then or any $\varepsilon > 0$*

$$\mathbb{P}\{|f(X) - \mathbb{E}f(X)| > \varepsilon\} \le 2\exp(-c\varepsilon^2 n).$$

Proof. Without loss of generality, we can assume that $\mathbb{E}f = 0$ (otherwise consider the function $\tilde{f} := f - \mathbb{E}f$). Define the 1-homogeneous extension of our function to the whole space, i.e. the function $F : \mathbb{R}^n \to \mathbb{R}$ such that

$$F(x) = \begin{cases} \|x\|_2 f\left(\frac{x}{\|x\|_2}\right), & \text{if } x \ne 0, \\ 0, & \text{if } x = 0. \end{cases}$$

Let $x, y \in \mathbb{R}^n$. Then, if $\|y\|_2 > \|x\|_2$,

$$|F(x)-F(y)| \le \|x\|_2 \cdot \left| f\left(\frac{x}{\|x\|_2}\right) - f\left(\frac{y}{\|y\|_2}\right)\right| + f\left(\frac{y}{\|y\|_2}\right) \cdot \big|\|x\|_2 - \|y\|_2\big| \le$$

$$\le \|x\|_2 \cdot \left\| \frac{x}{\|x\|_2} - \frac{y}{\|y\|_2}\right\|_2 + \|f\|_\infty \|x - y\|_2 \le 3\|x - y\|_2.$$

In the last step we used that as f is Lipschitz and $\mathbb{E}f = 0$, there exists $x_0 \in S^{n-1}$, such that $x_0 = 0$. Hence, for any $z \in S^{n-1}$ we can estimate $|f(z)|$ above by the longest possible distance between z and x_0, i.e. $\|f\|_\infty \le 2$.

So, $F(x)$ is 3-Lipschitz and so it concentrates

$$\mathbb{P}\{|F(x)| > t\} \le 2\exp(-ct^2).$$

We need the following claim to conclude the theorem: *there exist constants $a, b > 0$ such that*

$$\mathbb{P}\{\|g\|_2 \leq a\sqrt{n}\} \leq \exp(-bn).$$

This inequality can be checked directly (exercise).

Fix some $\varepsilon > 0$. Since the diameter of the sphere is 2, we may assume that $\varepsilon \leq 2$. Then,

$$\mathbb{P}(|f(x)| > \varepsilon a\} = \mathbb{P}\left\{\|g\|_2 \cdot \left|f\left(\frac{g}{\|g\|_2}\right)\right| > \|g\|_2 \cdot \varepsilon a\right\} \leq$$

$$\leq \mathbb{P}(|F(g)| \geq \varepsilon\sqrt{n}\} + \mathbb{P}\{\|g\| \leq a\sqrt{n}\} \leq$$

$$\leq 2\exp(-c\varepsilon^2 n) + \exp(-bn) \leq 3\exp(-c\varepsilon^2 n),$$

whenever $c\varepsilon^2 \leq b$ or $\varepsilon \leq c_0$.

For $c_0 < \varepsilon \leq 2$, we use the previous inequality to obtain

$$\mathbb{P}(|f(x)| > \varepsilon a\} \leq \mathbb{P}(|f(x)| > c_0 a\} \leq 3\exp(-cc_0^2 n) \leq 3\exp(-\tilde{c}\varepsilon^2 n)$$

for an appropriately chosen constant $\tilde{c} > 0$.

□

13. 10/22 Concentration of $\mathbb{E}\|A\|$ for subgaussian matrix A

Last time we discussed gaussian concentration as a corollary of log-Sobolev inequality (LSI). We proved the following theorem.

Theorem 13.1 (Gaussian concentration)**.** *Let $g \in \mathbb{R}^n$ be a standard gaussian vector and let $f : \mathbb{R}^n \to \mathbb{R}$ be any 1-Lipschitz function. Then*

$$\mathbb{P}\{|f(g) - \mathbb{E}f(g)| > t\} \leq 2\exp(-\frac{t^2}{2}).$$

Corollary 13.2. *If $\|.\|_Y$ is some norm in $\mathbb{R}^n$ and let g be a standard gaussian vector, then*

$$\mathbb{P}\{\big|\|g\|_Y - \mathbb{E}\|g\|_Y\big| > t\} \leq 2\exp(-\frac{t^2}{2L^2}),$$

where

$$L = \max_{y \in S^{n-1}} \|g\|_Y.$$

Proof.

$$\big|\|x\|_Y - \|y\|_Y\big| \leq \|x - y\|_Y \leq L\|x - y\|_2,$$

so the function

$$f(x) = \frac{\|x\|_Y}{L}$$

is 1-Lipschitz and we can apply Therem 13.1 to it. □

Let us consider several particular cases of this corollary.

- For ℓ_2 norm we have
$$\mathbb{P}\{\big|\|g\|_2 - \sqrt{n}\big| > t\} \leq 2\exp(-\frac{t^2}{4}).$$
- For the operator norm on $n \times m$ matrices (it is 1-Lipschitz, because
$$\|A\| \leq \|A\|_F := \|A\|_2^{n\times m})$$
and for G being a Gaussian matrix,
$$\mathbb{P}\{\big|\|G\| - \mathbb{E}\|G\|\big| > t\} \leq 2\exp(-\frac{t^2}{2}).$$
Integrating the tail, we can see that this implies boundedness of the variance:
$$\mathrm{Var}(\|G\|) \leq \int_0^\infty 4t\exp(-t^2/2)dt = C.$$
- Note that if the matrix A is not gaussian, but has independent bounded entries, we can apply Azuma inequality to get some concentration of its operator norm:
$$\mathbb{P}\{\big|\|A\| - \mathbb{E}\|A\|\big| > t\} \leq 2\exp(-c\frac{t^2}{nm}).$$
We can see that this result is much weaker than in gaussian case.

To see how strong the gaussian concentration is, let compare $\mathrm{Var}(\|G\|)$ and $\mathbb{E}\|G\|$.

Theorem 13.3. *Let A be $n\times m$ matrix with subgaussian entries. Then*
$$c(\sqrt{n}+\sqrt{m}) \leq \mathbb{E}\|A\| \leq C(\sqrt{n}+\sqrt{m}).$$

The first step in the proof is the discretization of the sphere. We need a couple of lemmas to proceed.

Lemma 13.4. *Let $\mathcal{M} \subset S^{m-1}$ and $\mathcal{N} \subset S^{n-1}$ be a $\frac{1}{2}$-nets , then for any $n \times m$ matrix A,*
$$\|A\| \leq 4 \max_{x\in\mathcal{M}, y\in\mathcal{N}} \langle Ax, y\rangle.$$

Proof. Let $\tilde{x} \in S^{m-1}$. Take $x \in \mathcal{M}$, such that $\|\tilde{x} - x\| < \frac{1}{2}$. Then
$$\|A\tilde{x}\|_2 \leq \|Ax\|_2 + \|A\| \cdot \|x - \tilde{x}\|_2 \leq \max_{x\in\mathcal{M}} \|Ax\|_2 + \frac{1}{2}\|A\|.$$
Hence,
$$\|A\| = \max_{\tilde{x}\in S^{m-1}} \|A\tilde{x}\|_2 \leq \max_{x\in\mathcal{M}} \|Ax\| + \frac{1}{2}\|A\|$$

and so
$$\|A\| \leq 2 \max_{x\in\mathcal{M}} \|Ax\|.$$
Then, for any $\tilde{y} \in S^{n-1}$ there is $y \in \mathcal{N}$, such that $\|\tilde{y} - y\| < \frac{1}{2}$, and so
$$\max_{x\in\mathcal{M}} \langle Ax, \tilde{y}\rangle \leq \max_{x\in\mathcal{M}} \langle Ax, y\rangle + \|y - \tilde{y}\| \cdot \max_{x\in\mathcal{M}} \|Ax\| \leq$$
$$\leq \max_{x\in\mathcal{M}, y\in\mathcal{N}} \langle Ax, y\rangle + \frac{1}{2} \cdot \max_{x\in\mathcal{M}} \|Ax\|.$$
Taking supremum over S^{n-1}, we get
$$\max_{x\in\mathcal{M}} \|Ax\| \leq \max_{x\in\mathcal{M}, y\in\mathcal{N}} \langle Ax, y\rangle + \frac{1}{2} \cdot \max_{x\in\mathcal{M}} \|Ax\|.$$
Hence,
$$\|A\| \leq 2 \max_{x\in\mathcal{M}} \|Ax\| \leq 4 \max_{x\in\mathcal{M}, y\in\mathcal{N}} \langle Ax, y\rangle.$$
□

Lemma 13.5 (Volumetric estimate)**.** *Let $\varepsilon > 0$. There exists an ε-net on S^{n-1} of the cardinality $|\mathcal{N}| \leq (1 + \frac{2}{\varepsilon})^n$.*

Proof. Let $\mathcal{N}$ be a maximal $\varepsilon/2$-separated subset on B_2^n. By definition, for any $x, x' \in \mathcal{N}$
$$(x + \frac{\varepsilon}{2} B_2^n) \cap (x' + \frac{\varepsilon}{2} B_2^n) = \emptyset.$$
All $\varepsilon/2$-balls lie inside $(1 + \varepsilon/2)$-ball (as the centers lie inside the unit ball).
$$\cup_{x\in\mathcal{N}} (x + \frac{\varepsilon}{2} B_2^n) \subset (1 + \frac{\varepsilon}{2}) B_2^n.$$
Comparing volumes of these two sets, we get that
$$|\mathcal{N}| \cdot \frac{\varepsilon^n}{2^n} \operatorname{Vol}(B_2^n) \leq (1 + \frac{\varepsilon}{2})^n \operatorname{Vol}(B_2^n),$$
so
$$|\mathcal{N}| \leq \frac{2^n}{\varepsilon^n} (1 + \frac{\varepsilon}{2})^n = (1 + \frac{2}{\varepsilon})^n.$$
□

Proof. (of Theorem 13.3) Take $x \in S^{m-1}$ and $y \in S^{n-1}$. Then
$$\langle Ax, y\rangle = \sum_{j=1}^{m} \sum_{k=1}^{n} a_{jk} x_j y_k$$
is a linear combination of subgaussians. Let us apply Hoeffding inequality to it:
$$\mathbb{P}\{|\langle Ax, y\rangle| \geq t\} \leq 2 \exp\left(- \frac{ct^2}{\sum_{j,k} x_i^2 y_k^2} \right) \leq 2 \exp(-ct^2).$$

Hence,

$$\mathbb{P}\{\|A\| \geq t(\sqrt{n}+\sqrt{m})\} \leq \mathbb{P}\{|\max_{x\in\mathcal{M},y\in\mathcal{N}}\langle Ax,y\rangle| \geq \frac{t}{4}(\sqrt{n}+\sqrt{m})\} \leq$$

$$\leq |\mathcal{M}|\cdot|\mathcal{N}|\cdot 2\exp(-ct^2(n+m)) \leq 3^m\cdot 3^n\cdot 2\exp(-ct^2(n+m)).$$

Integrating the tail bound, we get that

$$\mathbb{E}\|A\| \leq C(\sqrt{n}+\sqrt{m}).$$

To get the lower bound, note that

$$\|A\| \geq \|Ae_i\|_2 = (\sum_{k=1}^n a_{ik}^2)^{1/2}.$$

Hence,

$$\mathbb{P}\{\|A\| \geq c\sqrt{n}\} \geq \mathbb{P}\{\sum_{k=1}^n a_{ik}^2 > c^2 n\} > 1/2$$

for some small $c = c_1$ (for example, by the Law of Large numbers). Then, by Markov inequality,

$$\frac{\mathbb{E}\|A\|}{c_1\sqrt{n}} \geq \mathbb{P}\{\|A\| \geq c_1\sqrt{n}\} > 1/2.$$

Repeating this for A^T, we get that $\mathbb{E}\|A\| \geq c_2\sqrt{m}$. And so

$$\mathbb{E}\|A\| \geq \frac{\min\{c_1,c_2\}}{2}(\sqrt{n}+\sqrt{m}).$$

□

Since the LSI implies strong concentration properties, it is important to know which measures satisfy it. For probability measures on $\mathbb{R}$, a complete characterization has been obtained by Bobkov and Gotze (see supplementary materials). For measures in $\mathbb{R}^n$, one has the following sufficient condition.

Theorem 13.6 (Bakry-Emery)**.** *Let* $\nu : \mathbb{R}^n \to [0,+\infty)$ *be a function, such that* $\mathrm{Hess}(\nu) \geq c\,\mathrm{Id}$*,* $c > 0$ *(strongly positive definite). Then the measure* $\exp(-\nu(x))dx$ *satisfies LSI.*

(No proof)

Bakry-Emery theorem is especially interesting as it gives access to non-product measures. Informally speaking, it asserts thateverything that is stronger than gaussian, satisfies LSI.

But what if log-Sobolev inequality is not satisfied in some case?

For the bounded random variables, there is Talagrand inequality:

$$\mathbb{P}\{\big|\|G\| - \mathbb{E}\|G\|\big| > t\} \leq 2\exp(-\frac{t^2}{2})$$

A functional approach to this inequality based on a modification of the LSI was proposed by Ledoux. We will now discuss more general version of Ledoux's inequality, which is also a generalization of LSI: *modified log-Sobolev inequality* (proved by Massart, Boucheron, Bousquet and Lugosi). We will start with an elementary lemma.

Lemma 13.7. *Let $I \subset \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a convex function. Let X be a ranom variable, taking values in I. Then for all $a \in I$,*

$$\mathbb{E}f(X) - f(\mathbb{E}X) \leq \mathbb{E}(f(X) - f(a) - f'(a)(X - a)).$$

Proof. Note that

$$\begin{aligned} f(\mathbb{E}X) + \mathbb{E}(-f(a) - f'(a)(X - a)) = \\ = f(\mathbb{E}X) - f(a) - f'(a)\mathbb{E}(X - a) \geq 0 \end{aligned}$$

as f is convex. And this is precisely what we need to prove. $\square$

Theorem 13.8 (modified LSI)**.** *Let $X_1, \ldots, X_n$ be independent random variables. Let $f : \Omega_1 \times \ldots \times \Omega_n \to \mathbb{R}$ be some function and let*

$$f_j : \Omega_1 \times \ldots \Omega_{j-1} \times \Omega_{j+1} \times \Omega_n \to \mathbb{R}$$

for every for $j = 1, \ldots, n$. Set

$$Y := f(X_1, \ldots, X_n)$$

and

$$Y := f_j(X_1, \ldots, X_{j-1}, X_{j+1}, \ldots, X_n)$$

. Let $\varphi(t) := e^t - t - 1$. Then for any $\lambda \in \mathbb{R}$,

$$\mathrm{Ent}(e^{\lambda Y}) \leq \mathbb{E}\sum_{j=1}^{n} \mathbb{E}_j(e^{\lambda Y} \cdot \varphi(-\lambda(Y - Y_j))).$$

Note that this inequality is similar to log-Sobolev, it also estimates the entropy of the function, but with exponential type function φ instead of the gradient.

Unlike the LSI, the argument of this inequality is an exponential type function $e^{\lambda Y}$. For our purposes this would be enough, since we plan to combine it with Herbst argument to establish concentration.

Also, in contrast to the LSI, this inequality holds for all probability measures. It suggests the main difficulty in deriving the concentration from the modified LSI is hidden now in further estimation of the right hand side for a certain function.

14. 10/27 Modified log-Sobolev and Herbst argument

We are about to prove the modified log-Sobolev inequality.

Theorem 14.1 (modified LSI). *Let $X_1, \dots, X_n$ be independent random variables. Let $f : \Omega_1 \times \dots \times \Omega_n \to \mathbb{R}$ be some function and let*

$$f_j : \Omega_1 \times \dots \Omega_{j-1} \times \Omega_{j+1} \times \Omega_n \to \mathbb{R}$$

for every for $j = 1, \dots, n$. Set

$$Y := f(X_1, \dots, X_n)$$

and

$$Z_j := f_j(X_1, \dots, X_{j-1}, X_{j+1}, \dots, X_n)$$

. Let $\varphi(t) := e^t - t - 1$. Then for any $\lambda \in \mathbb{R}$,

$$\mathrm{Ent}(e^{\lambda Y}) \le \mathbb{E} \sum_{j=1}^{n} \mathbb{E}_j(e^{\lambda Y} \cdot \varphi(-\lambda(Y - Z_j))).$$

Before we start the proof, recall that the following lemma from the previous class:

Lemma 14.2. *Let $I \subset \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a convex function. Let X be a random variable, taking values in I. Then for all $a \in I$,*

$$\mathbb{E}f(X) - f(\mathbb{E}X) \le \mathbb{E}(f(X) - f(a) - f'(a)(X - a)).$$

Proof. (of Theorem 14.1) Let $v(t) = t \log t$. Apply Lemma 14.2 to v (which is convex) to get that

$$\mathbb{E}v(U) - v(\mathbb{E}U) \le \mathbb{E}[v(U) - v(a) - v'(a)(U - a)].$$

It means that

$$\mathrm{Ent}(U) \le \mathbb{E}[U \log U - a \log a - (\log a + 1)(U - a)] =$$

$$= \mathbb{E}[U(\log U - \log a) - (U - a)].$$

Combining this with subadditivity of entropy, we obtain

$$\mathrm{Ent}(e^{\lambda Y}) \le \mathbb{E} \sum_{j=1}^{n} \mathrm{Ent}_j(e^{\lambda Y}) \le$$

(we can take any constant in place of a, and $a = e^{\lambda Z_j}$ is a constant inside j-th term, as it is conditioned on X_j)

$$\le \mathbb{E} \sum_{j=1}^{n} \mathbb{E}_j[e^{\lambda Y}(\lambda Y - \lambda Z_j) - e^{\lambda Y}(1 - e^{\lambda(Z_j - Y)})] =$$

$$= \mathbb{E}\sum_{j=1}^{n}\mathbb{E}_j[e^{\lambda Y}\cdot\varphi(-\lambda(Y-Z_j))].$$

□

So, given some random variable X, if we can invent functions f_j providing the nice estimate for the right hand side, then this lemma can serve as a first step of the Herbst argument (leading to the gaussian concentration). We will see it in the example right now.

Theorem 14.3 (Generalized bounded differences inequality)**.** *Let X_1, $\ldots, X_n$ be independent random variables, and let $X_1', \ldots, X_n'$ be their independent copies. Set $X = (X_1, \ldots, X_n)$ and*

$$X^{(j)} = (X_1, \ldots, X_{j-1}, X_j', X_{j+1}, \ldots, X_n).$$

Let $f : \Omega_1 \times \ldots \times \Omega_n \to \mathbb{R}$.

(1) *Assume that*

$$\sum_{j=1}^{n}(f(X) - f(X^{(j)}))_+^2 \leq \nu \quad a.s.$$

Then

$$\mathbb{P}\{f(X) - \mathbb{E}f(X) > t\} \leq \exp(-\frac{t^2}{2\nu}).$$

(2) *Assume that*

$$\sum_{j=1}^{n}(f(X) - f(X^{(j)}))_-^2 \leq \nu \quad a.s.$$

Then

$$\mathbb{P}\{\mathbb{E}f(X) - f(X) > t\} \leq \exp(-\frac{t^2}{2\nu}).$$

Remark 14.4. The conclusion is split into two sub cases, as for many reasonable functions only one of two conditions is satisfied.

Proof. Set $\tilde{X} = (X_1, \ldots, X_{j-1}, X_{j+1}, \ldots, X_n)$ and

$$f_j(\tilde{X}) := \inf_{x_j \in \Omega_j} f(X_1, \ldots, X_{j-1}, x_j, X_{j+1}, \ldots, X_n).$$

Then

$$f(X) - f_j(\tilde{X}) \geq 0$$

and

$$\sum_{j=1}^{n}(f(X) - f_j(\tilde{X}))^2 \leq \nu.$$

Note that for $\lambda > 0$ and $Z_j = f_j(\tilde{X})$

$$\varphi(-\lambda(Y - Z_j)) \leq \frac{\lambda^2}{2}(Y - Z_j)^2$$

(because $\varphi(-t) \leq \frac{t^2}{2}$ for every $t > 0$).

By the modified LSI,

$$\mathrm{Ent}(e^{\lambda Y}) \leq \frac{\lambda^2 \nu}{2}.$$

Denoting $\psi(\lambda) = \mathbb{E}e^{\lambda Y}$, we get that

$$\lambda\psi'(\lambda) - \psi(\lambda)\log\psi(\lambda) \leq \frac{\lambda^2}{2}\nu\psi(\lambda).$$

This is the same differential inequality that we saw before in the Herbst argument (see lecture 12, page 2). Solving it in the same way, we obtain subgaussian concentration:

$$\mathbb{P}\{f(X) - \mathbb{E}f(X) > t\} \leq \exp(-t^2/2\nu).$$

□

Obviously, the standard bounded differences inequality follows from Theorem 14.3.

Let us now consider several applications.

(1) *Empirical distributions of eigenvalues of a Hermitian (symmetric) random matrix.*

Let A be an $n \times n$ random matrix with independent columns above the main diagonal. Define a random measure ν_A on $\mathbb{R}$ by

$$\nu_A = \frac{1}{n}\sum_{j=1}^{n} \delta_{\lambda_j(A)},$$

where $\lambda_j(A)$ are the eigenvalues of A (it is called an empirical distribution of eigenvalues).

Let $\alpha : \mathbb{R} \to \mathbb{R}$ be a test function, $\alpha \in C^1(\mathbb{R})$. Set

$$f(A) := \int \alpha d\nu_A.$$

Let $X = (a_{1j}, \ldots, a_{jj})^T$. Consider $f(A) - f(A^{(j)})$, where A is defined by $(X_1, \ldots, X_n)$, $A^{(j)}$ is defined by $(X_1, \ldots, X_j', \ldots, X_n)$. Then

$$|f(A) - f(A^{(j)})| \leq |\int_{\mathbb{R}} \alpha d(\nu_A - \nu_{A^{(j)}})| =$$

(function is continuously differentiable, so we can integrate by parts)

$$= |\int_{\mathbb{R}} (F_{\nu_A}(t) - F_{\nu_{A^{(j)}}}(t))\alpha'(t)dt| \leq \|F_{\nu_A} - F_{\nu_{A^{(j)}}}\|_\infty \cdot \|\alpha'\|_1.$$

By the Cauchy interlacing theorem,

$$\|F_{\nu_A} - F_{\nu_{A^{(j)}}}\|_\infty \leq \frac{2}{n},$$

hence,

$$\mathbb{P}\{f(A) - \mathbb{E}f(A) > t\} \leq \exp\left(-\frac{t^2 n}{8\|\alpha\|_1^2}\right).$$

In this example, we used only the standard bounded differences inequality, so it could have been derived from Azuma's inequality as well.

(2) *Concentration for the operator norm of the matrix.*

Recall that for the matrix with bounded (but not gaussian) entries, Azuma's inequality gives absolutely unsatisfactory concentration result ($\exp(-ct^2/nm) \sim 1$ for large dimensions n and m).

Theorem 14.5 (Talagrand)**.** *Let A be an $n \times n$ symmetric matrix with independent above diagonal entries a_{ij}, such that $|a_{ij}| \leq 1$ almost surely. Then*

$$\mathbb{P}\{\|A\| - \mathbb{E}\|A\| > t\} \leq \exp(-ct^2).$$

And similarly for the lower tail:

$$\mathbb{P}\{\mathbb{E}\|A\| - \|A\| > t\} \leq \exp(-ct^2).$$

Remark 14.6. Note that the result is dimension independent, like in gaussian case.

Proof. We will prove only the upper estimate here. The lower estimate will be proved later. Recall that $\|A\| = \sup_{x \in S^{n-1}} |\langle Ax, x\rangle|$. Hence,

$$(\|A\| - \|A^{(j,k)}\|)_+ = (\sup_{x \in S^{n-1}} |\langle x, Ax\rangle| - \sup_{y \in S^{n-1}} |\langle y, A^{(j)}y\rangle|)_+ \leq$$

(setting $y = x$ can only increase the positive part)

$$\leq \sup_{x \in S^{n-1}} |\langle x, (A - A^{(j)})x\rangle| \leq \begin{cases} 2|x_j||x_k| & \text{if } j \neq k, \\ x_j^2 & \text{if } j = k. \end{cases}$$

Thus,

$$\sum_{j=1}^{n}\sum_{k=1}^{j}(\|A\|-\|A\|^{(j,k)})_+^2 \leq 4\sum_{j=1}^{n}\sum_{k=1}^{j-1}x_j^2x_k^2+\sum_{j=1}^{n}x_j^4 \leq$$

$$\leq 2(\sum_{j=1}^{n}x_j^2)^2 \leq 2.$$

□

(3) *Minimal weight spanning trees*

Let $G(n)$ be the complete graph on n vertices with independent weights X_{ij} for each edge (i,j). Assume that the weights X_{ij} are uniformly distributed on $[0,1]$.

Let $X=(X_{ij})_{1\leq i<j\leq n^2}$ and let

$$F(X)=\min_T w(T),$$

where T are the spanning trees and the weight of the tree is

$$w(T)=\sum_{(i,j)\in T}X_{ij}.$$

Lemma 14.7. *Let (G,X) be a weighted complete graph. Assume that there is a spanning tree with the weights of edges less than p_0. Then the spanning tree of the minimum weight contains only edges with the weight less than p_0.*

Proof. Suppose minimal spanning tree T_1 contains some edge with the weight more than p_0. If we delete this edge spanning tree will decompose into two connected components. In the other spanning tree with all weights less than p_0 there is an edge connecting these two components. We can add this edge getting a new spanning tree with the weight smaller than T_1. Contradiction proves the lemma. □

Let $p_0 \geq \frac{c\log n}{n}$. Consider the new weights obtained by truncation:

$$Y_{ij}=X_{ij}\mathbb{1}_{[0,p_0]}(X_{ij})+p_0\mathbb{1}_{(p_0,1]}(X_{ij}).$$

Note that new graph with weights Y_{ij} will coincide with the original one with exponentially high probability (if not, then the minimal spanning tree contained an edge with the weight greater than p_0, which is equivalent to the event that the Erdos-Renyi $G(n,p_0)$ is disconnected, and this last probability is less than $\exp(cnp_0)$ for $p_0 \leq \frac{c\log n}{n}$).

Truncated weights satisfy bounded differences condition, so the subgaussian concentration takes place. However, even if we set $p_0 = \frac{c \log n}{n}$, the estimate obtained from the bounded differences inequality will be $\exp\left(-\frac{ct^2}{\log^2 n}\right)$. This yields some upper tail bound, but no lower tail, as $\mathbb{E}F(X) = O(1)$. Moreover, adding the probability of the event that $F(X) \neq F(Y)$ will make the upper tail bound loose as well. We will discuss tighter bounds for the minimal weight spanning tree next time.

15. 10/29 LSI corollaries. Self-bounding functions

Throughout the lecture if $X = (X_1, \dots, X_n)$ is a random vector, then $X^{(i)} = (X_1, \dots, X_{i-1}, X_i', X_{i+1}, \dots, X_n)$, where X_i' is an independent copy of X_i.

Minimal spanning trees

Last time we discussed minimal weight spanning trees and saw that bounded differences condition gives some bound for the upper tail, but not for the lower tail. We will see now how generalized bounded differences inequality solves this problem.

Let $\overline{X}_{ij}$ be truncated weights of edges (bounded by p_0). The weight of the tree we denote by

$$F(\overline{X}) = \min_T \sum_{(i,j)\in T} \overline{X}_{(i,j)}$$

(minimum is taken over all spanning trees).

Note that if $T(\overline{X})$ is a minimal spanning tree for $\overline{X}$, and we change $\overline{X}_{i,j}$, then the minimal weight can increase only if $(i,j) \in T(\overline{X})$. Hence,

$$\sum_{i<j} (F(\overline{X}) - F(\overline{X}^{((i,j))}))_-^2 = \sum_{(i,j)\in T(\overline{X})} (\overline{X}_{(i,j)} - \overline{X}'_{(i,j)})^2 \le p_0^2 \cdot n$$

(because all the weights are bounded by p_0 and any spanning tree contains $n-1$ edges). So,

$$\mathbb{P}\{\mathbb{E}F(\overline{X}) - F(\overline{X}) > t\} \le \exp(-\frac{t}{np_0^2}),$$

and using that $\overline{X} = X$ with the probability $1 - \exp(-cp_0 n)$ for $p_0 \geq C\frac{\log n}{n}$ (as we discussed last time)

$$\mathbb{P}\{\mathbb{E}F(X) - F(X) > t\} \leq \exp(-\frac{t}{np_0^2}) + \exp(-cp_0 n).$$

Taking the minimizing $p_0 = t^{2/3}/n^{2/3}$, we get the estimate

$$\mathbb{P}\{\mathbb{E}F(X) - F(X) > t\} \leq 3\exp(-t^{2/3}n^{1/3}).$$

Note that the inequality $p_0 \geq C\frac{\log n}{n}$ holds automatically for all relevant t.

In the last two cases (the norm of a random matrix and the minimal weight spanning tree), we derived only one tail estimate from the generalized bounded differences inequality. this was due to the fact that only the positive or the negative parts of $F(X) - F(X^{(j)})$ was easy to bound. We will now obtain the inequality for the other tail in the case when only one bound is available.

Theorem 15.1. *Let $X_1, \dots, X_n$ be independent random variables. Let $F : \Omega_1 \times \ldots \times \Omega_n \to \mathbb{R}$ be a function. Assume that for every $j \in \{1, \dots, n\}$,*

$$(F(X) - F(X^{(j)}))_+ \leq 1 \quad \textit{and} \tag{11}$$

$$\sum_{j=1}^{n} (F(X) - F(X^{(j)}))_+^2 \leq \nu. \tag{12}$$

Then

$$\mathbb{P}\{\mathbb{E}F(X) - F(X) \geq t\} \leq \exp(-\nu h(\frac{t}{\nu})),$$

where

$$h(x) = (1 + x)\log(1 + x) - x.$$

Remark 15.2. If $x \in [0, c]$ then $h(x) \geq c'x^2$, so

$$\exp(-\nu h(\frac{t}{\nu})) \leq \exp(-c\frac{t^2}{\nu}) \quad \text{- gaussian tail.}$$

If $x > c > e$ then $h(x) \geq c'x \log x$, so

$$\exp(-\nu h(\frac{t}{\nu})) \leq \exp\left(-c't\log\left(\frac{t}{\nu}\right)\right) \quad \text{- Poissonian tail}$$

So, we get a Bennett tail here. For the norm of a random matrix, we see only the subgaussian part, as we use it for the lower tail, and the norm is non-negative. For the spanning trees, we used this theorem for the upper tail, and see the full Bennett tail.

Proof. Set $Y := F(X)$ and

$$Y_j := \inf_{x_j \in \Omega_j} F(X_1, \dots, X_{j-1}, x_j, X_{j+1}, \dots, X_n).$$

By the modified LSI,

$$\mathrm{Ent}(e^{\lambda Y}) \le \mathbb{E} \sum_{j=1}^{n} \mathbb{E}_j (e^{\lambda Y} \varphi(-\lambda(Y - Y_j))),$$

where

$$\varphi(x) = e^x - x - 1.$$

Let $\lambda < 0$. If $y > 0$, then $\varphi(y)/y^2$ is an increasing function. Then for any $\alpha \in (0, 1]$,

$$\frac{\varphi(\alpha y)}{\alpha^2 y^2} \le \frac{\varphi(y)}{y^2}.$$

Therefore, using condition (1),

$$\mathrm{Ent}(e^{\lambda Y}) \le \mathbb{E} \sum_{j=1}^{n} \mathbb{E}_j (e^{\lambda Y} (Y - Y_j)_+^2 \varphi(-\lambda))$$

and by condition (2)

$$\mathrm{Ent}(e^{\lambda Y}) \le \nu \varphi(-\lambda) \mathbb{E} e^{\lambda Y}.$$

Let

$$\psi(\lambda) = \mathbb{E} e^{\lambda Y}.$$

Then the last inequality can be rewritten as

$$\lambda \psi'(\lambda) - \psi(\lambda) \log \psi(\lambda) \le \nu \psi(-\lambda) \psi(\lambda).$$

Let

$$g(\lambda) = \frac{\log \psi(\lambda)}{\lambda}.$$

Then

$$g'(\lambda) \le \nu \frac{\varphi(-\lambda)}{\lambda^2}.$$

Hence,

$$g(\lambda) - g(\lambda_0) \le \nu \int_{\lambda_0}^{\lambda} \frac{\varphi(-s)}{s^2} ds.$$

If $\lambda_0 \to 0$, then $g(\lambda_0) \to \mathbb{E} Y$. So,

$$\frac{\log(\psi(\lambda))}{\lambda} \le \mathbb{E} Y + \nu \int_0^{\lambda} \frac{\varphi(-s)}{s^2} ds \le \mathbb{E} Y + \nu \frac{\varphi(-\lambda)}{\lambda^2} \lambda,$$

where in the last inequality, we used the fact that the function $\varphi(-s)/s$ is increasing on the interval $(0, \lambda)$. We have shown that

$$\log \psi(\lambda) \le \lambda \mathbb{E} Y + \nu \varphi(-\lambda),$$

or

$$\mathbb{E}e^{\lambda Y - \lambda \mathbb{E}Y} = \exp(\log \psi(\lambda) - \lambda \mathbb{E}Y) \leq \exp(\nu\varphi(-\lambda)).$$

Similar to what we did earlier in Cramer's theorem, by Markov inequality,

$$\mathbb{P}(\mathbb{E}Y - Y > t) \leq \exp(-\Lambda(t)),$$

where Λ is the Legendre transform of $\nu\varphi(-\lambda)$. A simple explicit calculation of Λ finishes the proof.

□

Self-bounding functions.

Definition 15.3 (configuration property)**.** *A property of a finite sequence is called a configuration property, if any subsequence of a sequence satisfies it as well.*

For a configuration property Π, we can define a function

$$F_\Pi : \Omega_1 \times \ldots \Omega_n \to \mathbb{N} \cup \{0\}.$$

such that $F_\Pi(X_1, \ldots, X_n)$ is the largest length of a subsequence of $(X_1, \ldots, X_n)$ satisfying Π. Let us now list several examples such functions, defined by a configuration property.

Example 15.4. Let $(X_1, \ldots, X_n) \in X$. Let $F_1(X_1, \ldots, X_n)$ be the number of different values of the sequence $X_1, \ldots, X_n$. (We saw this function before in bin occupancy problem).

Example 15.5. Let $X_1, \ldots, X_n$ be i.i.d. random variables uniformly distributed in $\{1, \ldots, n\}$. Let $F_2(X_1, \ldots, X_n)$ be the largest length of an increasing sequence in $X_1, \ldots, X_n$. We discussed a similar problem for a random permutation. The permutation case lacks independence, and the Maurey's theorem estimate is sub-optimal for this function. In contrast to it, in the independent case, we will obtain the optimal tail estimate.

Example 15.6. If $G = G(n, p)$ is a graph and $F(G)$ is a maximal degree of its vertex.

Definition 15.7 (self-bounding function)**.** *Let $F : \Omega_1 \times \ldots \Omega_n \to \mathbb{R}$ be a function. The function is called self-bounding, if there exist functions*

$$F^{(j)} : \Omega_1 \times \ldots \times \Omega_{j-1} \times \Omega_{j+1} \times \ldots \times \Omega_n \to \mathbb{R},$$

such that

$$0 \le F(X) - F(X^{(j)}) \le 1, \quad \text{and}$$
$$\sum_{j=1}^{n}(F(X) - F(X^{(j)})) \le F(X).$$

Remark 15.8. Any configuration function is self-bounding.

Theorem 15.9 (Boucheron, Bousquet, Lugosi, Massart)**.** *Let $F \ge 0$ be a self-bounding function. Then for any $t > 0$*

(1)
$$\mathbb{P}\{F(X) - \mathbb{E}F(X) > t\} \le \exp\left(-\mathbb{E}F(X)h(\frac{t}{\mathbb{E}F(X)})\right),$$
where $h(x) = (1+x)\log(1+x) - x$.

(2)
$$\mathbb{P}\{\mathbb{E}F(X) - F(X) > t\} \le \exp(-\frac{t^2}{2\mathbb{E}F(X)}).$$

We will discuss its proof next time.

Example 15.10. (Rademacher complexity, or R-functional) It is an example of a self-bounding function, which is not a configuration function. Let $\mathcal{F} : \Omega \to \mathbb{R}$ be a set of functions such that $\forall f \in \mathcal{F}$

$$\sup_{x\in\Omega} |f(x)| \le 1.$$

Let $X_1, \dots, X_n$ be i.i.d. random variables with the values in Ω. Let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademachers. Set

$$r(X) = \mathbb{E}_t[\sup_{f\in\mathcal{F}} \sum_{j=1}^{n} \varepsilon_j f(X_j)|X].$$

Or, equivalently, let $Y_1, \dots, Y_n$ be i.i.d. random vectors in a Banach space E. Assume that $\|Y_j\|_E \le 1$ almost surely. Set

$$r(Y) = \mathbb{E}\|\sum_{j=1}^{n} \varepsilon_j Y_j\|_E.$$

Next time, we will prove that r is self-bounding.

16. 11/3 Concentration for self-bounding functions

The concept of self-bounding functions was invented by Bousquet, Boucheron, Lugosi and Massart (BBLM) in order to put Talagrand's theorem about configuration functions into a general framework. Let us recall the definition of a self-bounding function.

Definition 16.1 (self-bounding function)**.** *Let $F : \Omega_1 \times \ldots \times \Omega_n \to \mathbb{R}$ be a function. The function is called self-bounding, if there exist functions*

$$F^{(j)} : \Omega_1 \times \ldots \times \Omega_{j-1} \times \Omega_{j+1} \times \ldots \times \Omega_n \to \mathbb{R},$$

such that

(1)
$$0 \le F(X) - F(X^{(j)}) \le 1 \quad a.s.$$

(2)
$$\sum_{j=1}^{n} (F(X) - F(X^{(j)})) \le F(X) \quad a.s.$$

Remark 16.2. Usually we don't need to invent anything subtle for $F^{(j)}$ (for "good" functions restriction on $n-1$ coordinates work). For example, this is true for all configuration functions.

Theorem 16.3 (BBLM)**.** *Let $f : \Omega_1 \times \ldots \Omega_n \to \mathbb{R}$ be a self-bounding function. Then*

(1) *for any $t > 0$,*
$$\mathbb{P}\{f(X) - \mathbb{E}f(X) > t\} \le \exp\left(-\mathbb{E}f(X)h(\frac{t}{\mathbb{E}f(X)})\right),$$

(2) *and for any $0 < t \le \mathbb{E}f(X)$,*
$$\mathbb{P}\{\mathbb{E}f(X) - f(X) > t\} \le \exp\left(-\mathbb{E}f(X)h(\frac{t}{\mathbb{E}f(X)})\right),$$

where $h(x) = (1+x)\log(1+x) - x$.

Proof. Set $Y = f(X)$ and

$$Y_j = f_j(X_1, \ldots, X_{j-1}, X_{j+1}, \ldots, X_n).$$

Recall that modified LSI gives

$$\mathrm{Ent}(e^{\lambda Y}) \le \mathbb{E} \sum_{j=1}^{n} \mathbb{E}_j e^{\lambda Y} \varphi(-\lambda(Y - Y_j))$$

As before, $\varphi(x) = e^x - x - 1$. Note that $\varphi(0) = 0$ and φ is convex, so by simple calculus the function $\frac{\varphi(x)}{x}$ is increasing.

Hence, using condition (1) of self-bounding functions,

$$\varphi(-\lambda(Y - Y_j)) \le (Y - Y_j)\varphi(-\lambda)$$

and, using condition (2),

$$\mathrm{Ent}(e^{\lambda Y}) \le \mathbb{E}(Ye^{\lambda Y}) \cdot \varphi(-\lambda)$$

Recall that the entropy function is 1-homogeneous, so let us multiply both sides by $e^{-\lambda \mathbb{E}Y}$ to get

$$\mathrm{Ent}(e^{\lambda(Y-\mathbb{E}Y)}) \le \frac{\mathbb{E}(Ye^{\lambda Y})}{e^{\lambda \mathbb{E}Y}} \cdot \varphi(-\lambda) =$$

$$= \left(\frac{\mathbb{E}((Y-\mathbb{E}Y)e^{\lambda Y})}{e^{\lambda \mathbb{E}Y}} + \frac{\mathbb{E}(\mathbb{E}Ye^{\lambda Y})}{e^{\lambda \mathbb{E}Y}} \right) \cdot \varphi(-\lambda).$$

Set

$$g(\lambda) = \mathbb{E}e^{\lambda(Y-\mathbb{E}Y)}.$$

Then

$$\lambda(\log(g(\lambda))' - g(\lambda) \le ((\log(g(\lambda))' + \mathbb{E}Y)\varphi(-\lambda).$$

So,

$$(-e^{-\lambda} + 1)(\log g(\lambda))' - \log g(\lambda) \le \mathbb{E}Y\varphi(-\lambda).$$

We follow the Herbst argument here. We want to write the left hand side as a derivative of some function (then we can solve this differential inequality):

$$\frac{(e^{-\lambda} - 1)(\log g(\lambda))' - e^{\lambda} \log g(\lambda)}{e^{\lambda}} \le \mathbb{E}Y\varphi(-\lambda),$$

or

$$\left(\frac{\log g(\lambda)}{e^{\lambda} - 1} \right)' \cdot \frac{(e^{\lambda} - 1)^2}{e^{\lambda}} \le (e^{-\lambda} + \lambda - 1)\mathbb{E}Y,$$

$$\left(\frac{\log g(\lambda)}{e^{\lambda} - 1} \right)' \le \left(\frac{-\lambda}{e^{\lambda} - 1} \right)' \mathbb{E}Y.$$

The only problem of the inequality is that both sides are undefined at $\lambda = 0$, so we integrate from ε to λ and then consider a limit as $\varepsilon \to 0$. We have the following:

$$\frac{\log g(\lambda)}{e^{\lambda} - 1} - \frac{\log g(\varepsilon)}{e^{\varepsilon} - 1} \le \mathbb{E}Y \cdot \left(\frac{\varepsilon}{e^{\varepsilon} - 1} - \frac{\lambda}{e^{\lambda} - 1} \right).$$

Here,

$$\lim_{\varepsilon \to 0} \frac{\varepsilon}{e^{\varepsilon} - 1} = 1$$

and

$$\lim_{\varepsilon \to 0} \frac{\log \mathbb{E}e^{\varepsilon(Y-\mathbb{E}Y)}}{e^{\varepsilon} - 1} = 0.$$

Thus,

$$\frac{\log g(\lambda)}{e^\lambda - 1} \le \mathbb{E}Y \cdot \left(1 - \frac{\lambda}{e^\lambda - 1}\right),$$

or

$$\log \mathbb{E}e^{\lambda(Y - \mathbb{E}Y)} \le (e^\lambda - 1 - \lambda) \cdot \mathbb{E}Y = \varphi(\lambda) \cdot \mathbb{E}Y.$$

The proof finishes by using Markov's inequality and optimizing by λ. Note that the argument is the same for $\lambda > 0$ and $\lambda < 0$, so we got both upper and lower tails this way.

□

Example 16.4. (Rademacher complexity) Let $\mathcal{F} : \Omega \to \mathbb{R}$ be a set of functions such that $\forall f \in \mathcal{F}$

$$\sup_{x \in \Omega} |f(x)| \le 1.$$

Let $X_1, \ldots, X_n$ be i.i.d. random variables with the values in Ω. Let $\varepsilon_1, \ldots, \varepsilon_n$ be independent Rademachers. Set

$$r(X) = \mathbb{E}_\varepsilon[\sup_{f \in \mathcal{F}} \sum_{j=1}^n \varepsilon_j f(X_j) | X].$$

If the right hand side is not well-defined, usually it is enough to consider suprema over all finite subsets and take the supremum over them.

Or, equivalently, let $Y_1, \ldots, Y_n$ be i.i.d. random vectors in a Banach space E. Then assume that $\|Y_j\|_E \le 1$ almost surely. Set

$$r(Y) = \mathbb{E}\| \sum_{j=1}^n \varepsilon_j Y_j \|_E.$$

Obviously, the Rademacher complexity is not a configuration function, but it is self-bounding (if $\mathcal{F}$ is good enough), as we will show in the next lemma.

Lemma 16.5. *Assume that*

$$\forall f \in \mathcal{F} \ \sup_{x \in \Omega} |f(x)| \le 1. \tag{13}$$

In the Banach space formulation, (13) should be replaced with $\|Y_i\|_2 \le 1$.

Proof. Let

$$r_k(X_1, \ldots, X_{j-1}, X_{j+1}, \ldots, X_n) = \mathbb{E}[\sup_{f \in \mathcal{F}} \sum_{j \ne k} \varepsilon_j f(X_j) | X].$$

Then

$$r(X) \geq \mathbb{E}[\sup_{f\in\mathcal{F}} \sum_{j=1}^{n} \mathbb{E}_k \varepsilon_j f(X_j)|X] = r_k(X_1, \ldots, X_{k-1}, X_{k+1}, \ldots, X_n).$$

On the other hand, by the condition on f,

$$r(X) - r_k(X_1, \ldots, X_{k-1}, X_{k+1}, \ldots, X_n) \leq 1$$

So, the first self-bounding property is checked. For the second one, let us consider $\varepsilon \in \{-1, 1\}^n$ and $f_{X,\varepsilon}$ such that the supremum is achieved for this f:

$$r(X) = \mathbb{E}_\varepsilon[\sum_{j=1}^{n} \varepsilon_j f_{X,\varepsilon}(X_j)|X].$$

(we assume that $\mathcal{F}$ is finite so that this supremum is clearly attained on some function). Then

$$r(X) - r_k(X) \leq \mathbb{E}[\sum_{j=1}^{n} \varepsilon_j f_{X,\varepsilon}(X_j)|X] - \mathbb{E}[\sum_{j\neq k} \varepsilon_j f_{X,\varepsilon}(X_j)|X] =$$

(in the second term we can drop supremum as we estimate it from below)

$$= \varepsilon_k f_{X,\varepsilon}(X_k).$$

So,

$$\sum_{k=1}^{n} (r(X) - r_k(X)) \leq \mathbb{E}[\sum_{k=1}^{n} \varepsilon_k f_{X,\varepsilon}(X_k)|X] = r(X)$$

and the second condition is checked.

□

Our next goal is to prove another classical theorem by Talagrand: Talagrand convex concentration theorem.

Let $X = (X_1, \ldots, X_n)$ be independent random variables taking values in $\Omega = \Omega_1 \times \cdots \times \Omega_n$. Let $A \subset \Omega$. Recall that the Hamming distance is defined by

$$d_H(x, y) = \sum_{j=1}^{n} \mathbb{1}_{x\neq y}(j).$$

The function $f(x) = d_H(x, A) = \min_{y\in A} d_H(x, y)$ satisfies the BD (bounded differences) condition. Hence,

$$\mathbb{P}(d_H(X, A) - \mathbb{E}d_H(X, A) > t) \leq e^{-t^2/2n}.$$

Assume that $\mathbb{P}(A) \geq 1/2$. Then

$$\mathbb{P}(d_H(X, A) > t) \leq 2e^{-t^2/4n}.$$

Let us rewrite this inequality in the dimension independent form. To this end, take a vector $a = (a_1, \ldots, a_n) \in \mathbb{R}^n_+$ and define the weighted Hamming distance by

$$d^{(a_1,\ldots,a_n)}(x, y) = \sum_{j=1}^{n} a_j \mathbb{1}_{x \neq y}(j).$$

Clearly, if all $a_i = 1$, we get usual Hamming distance. Hence,

$$\mathbb{P}\{d^{(\frac{1}{\sqrt{n}},\ldots,\frac{1}{\sqrt{n}})}(X, A)\} \leq 2e^{-t^2/4}.$$

Similarly,

$$\mathbb{P}\{d^{(a)}(X, A) \geq t\} \leq 2e^{-t^2/4}$$

for any a, such that $\|a\|_2 = 1$, so

$$\sup_{a \in \mathbb{R}^n_+, \|a\|_2 = 1} \mathbb{P}\{d^{(a)}(X, A) \geq t\} \leq 2e^{-t^2/4}.$$

Talagrand convex concentration theorem (that we will se next time) justifies the fact that this supremum can be pulled inside the expectation.

17. 11/5 and 11/10 Talagrand's convex concentration theorem

As we discussed in the end of the previous class, if $x, y \in \Omega^n$ (two points in any space) and $a \in S^{n-1}$, we can define generalized Hamming distance between the points by

$$d^{(a_1,\ldots,a_n)}(x, y) = \sum_{j=1}^{n} a_j \mathbb{1}_{x \neq y}(j).$$

This sum concentrates:

$$\mathbb{P}\{d^{(a)}(x, A) \geq t\} \leq 2e^{-ct^2},$$

provided that $\mathbb{P}(A) \geq \frac{1}{2}$. We can take supremum over all a, such that $\|a\|_2 = 1$, so

$$\sup_{a \in \mathbb{R}^n_+, \|a\|_2 = 1} \mathbb{P}\{d^{(a)}(X, A) \geq t\} \leq 2e^{-t^2/4}.$$

Definition 17.1. *Let $A \subset \Omega^n$. Define*

$$d_T(x, A) = \sup_{a \in \mathbb{R}^n_+, \|a\|_2 = 1} \inf_{y \in A} \sum_{j=1}^n a_j \mathbb{1}_{x \neq y}(j).$$

Theorem 17.2 (Talagrand)**.** *Let $X = (X_1, \ldots, X_n) \in \Omega^n$ be a sequence of independent random variables. Let $A \subset \Omega^n$ be a measurable set. Then for any $t > 0$*

$$\mathbb{P}(A) \wedge \mathbb{P}(A_t^c) \leq \exp(-ct^2),$$

where

$$A_t = \{y \in \Omega^n : d_T(y, A) \leq t\}.$$

Remark 17.3. We will discuss two proofs of this theorem. Second one is the original proof made by Talagrand (and works in full generality). It is actually not too hard (being one of the simplest proofs for concentration results), but it is 'made "by trick" and thus not helpful in other similar cases (cannot be turned into machine).

First proof uses our usual bounded differences technique, and works well under the additional assumption that $\mathbb{P}(A) > \frac{1}{2}$ (we need this to beat exponential lower tail that inevitably appears with the bounded-differences inequality). In principle, the proof can be completed in full generality by this way as well, by self-bounding properties of the distance function, this part is heavy and and we omit it.

Proof. (First proof, where we assume that $\mathbb{P}(A) \geq \frac{1}{2}$)

Denote $S_+ = \{a \in \mathbb{R}^n_+ : \|a\|_2 = 1\}$. For $x \in \Omega^n$, define

$$\varphi_x : \Omega^n \to \{0, 1\}^n$$

by

$$\varphi_x(y) := \mathbb{1}_{x \neq y}.$$

Then

$$d_T(x, A) = \sup_{a \in S_+} \inf_{y \in \varphi_x(A)} \langle a, y \rangle =$$

$$= \mathrm{dist}_2(0, \mathrm{conv}(\varphi_x(A))) = \inf_{z \in \mathrm{conv}(\varphi_x(A))} \sup_{a \in S_+} \langle z, a \rangle =$$

(discrete cube is compact, so inf and sup are attained)

$$= \min_{z \in \mathrm{conv}(\varphi_x(A))} \max_{a \in S_+} \langle z, a \rangle = \min \max_{a \in S_+} \int \langle a, \mathbb{1}_{x=y} \rangle d\mu(y),$$

where the minimum is taken over all probability measures μ such that $\mathrm{supp}(\mu) \subset A$.

We will derive the theorem from the generalized BD condition. Let X and $X^{(j)}$ be as in BD inequality. Then we have to bound

$$(d_T(X,A) - d_T(X^{(j)},A))_+.$$

Let μ_x and a_x be the optimal measure and vector for $d_T(X,A)$. Then

$$(d_T(X,A) - d_T(X^{(j)},A))_+ \leq$$

$$\leq \int \langle a_x, \mathbb{1}_{x\neq y}\rangle d\mu_x(y) - \inf_{\mu} \int \langle a_x, \mathbb{1}_{X^{(j)}\neq y}\rangle d\mu(y) \leq$$

(let $\mu_{X^{(i)}}$ be an optimal measure for the second term)

$$\leq \inf \langle a_x, \mathbb{1}_{x\neq y}\rangle d\mu_{X^{(i)}}(y) - \int \langle a_x, \mathbb{1}_{X^{(j)}\neq y}\rangle d\mu_{X^{(i)}}(y) \leq a_{X,j}.$$

Then,

$$\sum_{j=1}^{n} (d_T(X,A) - d_T(X^{(j)},A))_+^2 \leq \sum_{j=1}^{n} a_{X,j}^2 = 1.$$

Denote $F_A = \mathbb{E} d_T(X,A)$. By the generalized BD inequality,

$$\mathbb{P}\{d_T(X,A) > F_A + t\} \leq \exp(-t^2/2). \tag{14}$$

and

$$\mathbb{P}\{d_T(X,A) \leq F_A - t\} \leq \exp(-F_A h(\frac{t}{F_A})) \leq \exp(-c(t^2 \wedge t)),$$

as $h(u) = (1+u)\log(1+u) - u$.

By the assumption we made, it is easy to check that subgaussian tail wins:

$$\frac{1}{2} \leq P(A) \leq \mathbb{P}\{d_T(X,A) = 0\} \leq \exp(-c[F_A^2 \wedge F_A]),$$

hense $F_A \leq c'$ and

$$\mathbb{P}\{A\} \leq \exp(-cF_A^2). \tag{15}$$

Now if $t > 2F_A$, then by (14)

$$\mathbb{P}\{d_T(X,A) \geq \frac{t}{2}\} \leq \exp(-\frac{t^2}{2}).$$

If $t \leq 2F_A$, then by (15)

$$\mathbb{P}\{A\} \leq \exp(-\frac{c}{4}t^2).$$

$\square$

Remark 17.4. Following the lines of the proof we can find the constant c and it will be some number less then 10. However it is not the optimal one. The optimal constant can be obtained by the original Talagrand's proof and it is

$$\mathbb{P}(A) \wedge \mathbb{P}(A_t^c) \leq \exp(-\frac{t^2}{4}).$$

Proof. (Second proof, due to Talagrand). First, let us check that it is enough to prove the estimate

$$\mathbb{P}(A) \cdot \mathbb{E} \exp\big(\frac{d_T^2(X,A)}{4}\big) \leq 1. \tag{16}$$

Indeed, by Markov,

$$\mathbb{P}(A_t^c) = \mathbb{P}\{\exp\big(\frac{d_T^2(X,A)}{4}\big) \geq \exp(\frac{t^2}{4})\} \cdot \exp(-t^2/4) \leq$$

$$\leq \mathbb{E} \exp\big(\frac{d_T^2(X,A)}{4}\big) \leq \frac{1}{\mathbb{P}(A)}.$$

So, we are concentrated on proving (16). We proceed by induction.

Base: $n = 1$, then

$$d_T(X,A) = \begin{cases} 1, & \text{if } x \notin A, \\ 0, & \text{if } x \in A. \end{cases}$$

For any set A,

$$\mathbb{P}(A) \cdot (\mathbb{P}(A^c) e^{1/4} + \mathbb{P}(A)) \leq 1$$

(as the function $g(x) := x^2(1 - e^{1/4}) - x$ is increasing on [0,1] and $g(1) < 0.72 < 1$).

Inductive step: assume that (16) holds for $n-1$. We say that $y \in \Omega$ *witnesses* $u \in \{0,1\}^n$ for $x \in \Omega$, if

$$\varphi_x(y) := \mathbb{1}_{x \neq y} = u.$$

Let $X = (Y, z)$, where $y \in \Omega_1 \times \ldots \times \Omega_{n-1}$, $z \in \Omega$. Fix $z \in \Omega_n$. Define

$$A_z := \{u : (u,z) \in A\}.$$

Case 1. Assume that $A_z \neq \emptyset$.

If $\alpha \in A_z$ witnesses $u \in \{0,1\}^{n-1}$ for y, then $\alpha_0 = (\alpha, z)$ witnesses $(u, 0)$ for X (the last coordinate coincides). Also, if $\beta \in P_{n-1}A$ witnesses $v \in \{0,1\}^n$, then (β, z') witnesses $(v, 1)$ (for some $z \neq z' \in A$).

Distance between x and A is bounded by the biggest witness, so for any $\lambda \in [0,1]$

$$d_T^2(X,A) \leq \operatorname{dist}_2^2((\lambda \tilde{u} + (1-\lambda)\tilde{v}, 0 \cdot \lambda + 1 \cdot (1-\lambda)), 0),$$

where $\tilde{u} \in \mathrm{conv}(\varphi_y(A_z))$ and $\tilde{v} \in \mathrm{conv}(\varphi_y(P_{n-1}A))$.

Then,

$$d_T^2(X, A) \leq (\lambda d_T(Y, A_z) + (1-\lambda) d_T(Y, P_{n-1}A))^2 + (1-\lambda)^2 \leq$$

(by convexity of square function)

$$\leq \lambda \cdot d_T^2(Y, A_z) + (1-\lambda) \cdot d_T^2(Y, P_{n-1}A)) + (1-\lambda)^2.$$

Therefore,

$$\mathbb{E}[\exp(\frac{d_T^2(X, A)}{4} | X_n = z] \leq$$

$$\leq \mathbb{E}[\exp \frac{\lambda d_T^2(Y, A_z)}{4} \cdot \exp \frac{(1-\lambda) d_T^2(Y, P_{n-1}A)}{4} \cdot \exp \frac{(1-\lambda)^2}{4} | X_n = z] \leq$$

(Holder)

$$\leq \mathbb{E}[\exp \frac{d_T^2(Y, A_z)}{4} | X_n = z]^{\lambda} \cdot \mathbb{E}[\exp \frac{d_T^2(Y, P_{n-1}A)}{4} | X_n = z]^{(1-\lambda)} \cdot e^{\frac{(1-\lambda)^2}{4}} \leq$$

(induction hypothesis)

$$\leq \left(\frac{1}{\mathbb{P}(A_z)}\right)^{\lambda} \cdot \left(\frac{1}{\mathbb{P}(P_{n-1}A)}\right)^{1-\lambda} \cdot e^{\frac{(1-\lambda)^2}{4}} \leq$$

$$\leq \left(\frac{1}{\mathbb{P}(P_{n-1}A)}\right) \cdot \left(\frac{\mathbb{P}(A_z)}{\mathbb{P}(P_{n-1}A)}\right)^{-\lambda} \cdot e^{\frac{(1-\lambda)^2}{4}}.$$

We need the following fact to proceed (exercise): let $s \in (0,1)$, then

$$\inf_{\lambda \in [0,1]} s^{-\lambda} e^{(1-\lambda^2/4} \leq 2 - s.$$

(The idea behind this claim is that we would like to apply Fubini's theorem, so we need to bound our function by something nicely integrable).

Thus,

$$\mathbb{E}[\exp(\frac{d_T^2(X, A)}{4}) | X_n = z] \leq \frac{1}{\mathbb{P}(P_{n-1}(A))} \cdot \left(2 - \frac{\mathbb{P}(A_z)}{\mathbb{P}(P_{n-1}A)}\right).$$

Case 2. Assume that $A_z = \emptyset$. Clearly, intersection with A_z is empty, so we forced to use projection:

$$d_T^2(X, A) \leq d_T^2(X, P_{n-1}A) + 1.$$

Then

$$\mathbb{E}[\exp \frac{d_T^2(X, A)}{4} | X_n = z] \leq \mathbb{E}[\exp(\frac{d_T^2(X, P_{n-1}A)}{4}) \cdot e^{1/4} | X_n = z] \leq$$

$$\leq \frac{1}{\mathbb{P}(P_{n-1}A)} e^{1/4} \leq \frac{1}{\mathbb{P}(P_{n-1}A)} \cdot \left(2 - \frac{\mathbb{P}(A_z)}{\mathbb{P}(P_{n-1}A)}\right)$$

(in the last step we used that $e^{1/4} \leq 2$ and $\mathbb{P}(A_z) = 0$). So, we arrived to the same estimate as in Case 1.

Taking expectation with respect to X_n, we get

$$\mathbb{E}\exp(\frac{d_T^2(X,A)}{4}) \leq \frac{\mathbb{P}(A_z)}{\mathbb{P}(P_{n-1}A)} \cdot \left(2 - \frac{\mathbb{P}(A_z)}{\mathbb{P}(P_{n-1}A)}\right) \cdot \frac{1}{\mathbb{P}(A_z)} \leq \frac{1}{\mathbb{P}(A_z)},$$

as $x(2-x) \leq 1$ for all $x \in \mathbb{R}$. Theorem is proved.

□

Lemma 17.5. *Let $A = [0,1]^n$ be a convex set. Then for any $x \in [0,1]^n$,*

$$d_2(x,A) \leq d_T(x,A),$$

where $d_2(x,y) = \inf_y(\|x-y\|_2)$.

Proof. Let $M(A)$ be the set of all probability measures with $\text{supp}(\mu) \subset A$. Then

$$d_2(x,A) = \min_{\mu \in M(A)} \|x - \int y d\mu(y)\|_2 \leq$$

(the first equality is trivial, it is just saying that infimum is attained at some point of A, actually, minimizing measure is a delta-measure)

$$\leq \int \big(\sum_{j=1}^n (x_j - y_j)^2\big)^{1/2} d\mu(y) \leq$$

$$\leq \min_{\mu \in M(A)} \int \big(\sum_{j=1}^n \mathbb{1}_{x \neq y}(j)\big)^{1/2} d\mu(y) \leq d_T(x,A).$$

□

Definition 17.6. *Let $U \subset \mathbb{R}^n$ be a convex set. A function $f : U \to \mathbb{R}$ is called quasi-convex, if for any $u \in \mathbb{R}$ the set*

$$\{x \in U : f(x) \leq u\}$$

is convex.

Such functions satisfy sub-gaussian concentration, as we will see in the next theorem.

Theorem 17.7 (Talagrand for quasi-convex functions)**.** *Let $f : [0,1]^n \to \mathbb{R}$ be a 1-Lipschitz quasi-convex function. Let $X \subset [0,1]^n$ be a random vector with independent coordinates. Then*

$$\mathbb{P}\{|f(X) - Mf(X)| \geq t\} \leq 2\exp(-ct^2),$$

where $Mf(X)$ is the median of f.

Proof. Let $u \in \mathbb{R}$. Let

$$A_u = \{X \in [0,1]^n : f(X) \le u\}.$$

Let $t > 0$, $y \in [0,1]$ be such that $f(Y) > u + t$. Then for any $x \in A_u$

$$u + t < f(Y) \le f(X) + \|X - Y\|_2.$$

Hence,

$$u + t < \inf_{X \in A_n} (f(X) + \|X - Y\|_2) \le u + d_2(Y, A) \le u + d_T(Y, A).$$

We have shown that for all Y $f(Y) > u + t$ implies $d_T(Y, A) \ge t$.
Set $u = Mf(X)$. Then

$$\mathbb{P}(A_{M(f)}) \wedge \mathbb{P}(f(X) > Mf(X) + t) \le \exp(-ct^2),$$

$$\frac{1}{2} \wedge \mathbb{P}(f(X) - Mf(X) > t) \le \exp(-ct^2),$$

and upper tail is proved.
Set $u = Mf(X) - t$. Then

$$\mathbb{P}(A_u) = \mathbb{P}\{f(X) \le M(f) - t\},$$

so

$$\mathbb{P}\{f(X) \le M(f) - t\} \wedge \frac{1}{2} \le \exp(-ct^2)$$

and the lower tail is proved.

□

The first application of the concentration of quasi-convex functions is the concentration of the norm of a random matrix with the bounded elements.

Corollary 17.8. *Let V be a $n \times n$ random matrix with independent entries, such that $|v_{ij}| \le 1$ a.s. Then*

$$\mathbb{P}\{\big|\|V\| - M(\|V\|)\big| > t) \le 2\exp(-ct^2).$$

Remark 17.9. Recall that before we had a subgaussian concentration for gaussian matrices (it was around expectation, but, as we saw in homework, it is equivalent to the concentration around median).

The application of the original Talagrand convex concentration is the following Alon-Krivelevich-Vu theorem about k-th eigenvalue of a bounded random matrix.

Theorem 17.10 (Alon, Krivelevich, Vu)**.** *Let V be an $n \times n$ random Hermitian matrix with independent entries $(v_{ij})_{i \le j}$ such that $|v_{ij}| \le 1$ almost surely. Let $\lambda_1 \le \cdots \le \lambda_n$ be the eigenvalues of V. Then for every $k = 1, \dots, n$*

$$\mathbb{P}\{\big|\lambda_k(V) - \mathrm{Med}(\lambda_k(V))\big| > t) \le 2\exp(-\frac{ct^2}{k^2}).$$

Proof. Let $s \in \mathbb{R}$, $t > 0$ and W is an independent copy of the matrix V. Assume that $\lambda_k(W) \le s$ and $\lambda_k(V) \ge s + t$.

Let $v_1, \dots, v_k$ be unit eigenvectors of V corresponding to $\lambda_1(V)$, $\dots, \lambda_k(V)$. Similarly, let $w_1, \dots, w_k$ be unit eigenvectors of W corresponding to $\lambda_1(W), \dots, \lambda_k(W)$.

Then there exists $u \in \mathrm{span}(v_1, \dots, v_k)$ such that $\|u\|_2 = 1$ and u is orthogonal to $k-1$ first eigenvectors of W (by dimension consideration). Let $u = \sum_{j=1}^k \nu_j v^j$.

Then, by geometric interpretation of eigen spaces,

$$\lambda_k(V) - \lambda_k(W) \le \langle u, Vu\rangle - \langle u, Wu\rangle =$$

$$= \sum_{i,j=1}^n (v_{ij} - w_{ij})(\sum_{l=1}^k \nu_l v_i^l)(\sum_{m=1}^k \nu_m v_j^m) \le$$

(by the condition $|v_{ij}| \le 1$ a.s.)

$$\le 2\sum_{i,j=1}^n \mathbb{1}_{v \ne w}(i,j)\big|\sum_{l=1}^k \nu_l v_i^l\big| \cdot \big|\sum_{m=1}^k \nu_m v_j^m\big| \le$$

(by Cauchy-Swartz and condition $\|u\|_2 = 1$)

$$\le 2\sum_{i,j=1}^n \mathbb{1}_{v \ne w}(i,j)\big(\sum_{l=1}^k (v_i^l)^2\big)^{1/2}\big(\sum_{m=1}^k (v_j^m)^2\big)^{1/2}.$$

The right hand side can define generalized Hamming distance, if we take as weights

$$a_{ij} := 2\big(\sum_{l=1}^k (v_i^l)^2\big)^{1/2}\big(\sum_{m=1}^k (v_j^m)^2\big)^{1/2}.$$

Indeed,

$$4\sum_{i,j=1}^n a_{ij}^2 = 4\sum_{i,j=1}^n \big(\sum_{l=1}^k (v_i^l)^2\big)^2\big(\sum_{m=1}^k (v_j^m)^2\big)^2 = 4\big(\sum_{i=1}^n \sum_{l=1}^k (v_i^l)^2\big)^2 = 4k^2$$

(we used that v^l are unit vectors in $\mathbb{R}^n$). So, we can apply Talagrand convex concentration theorem after normalization by $2k$.

Take $s = \mathrm{Med}(\lambda_k)$, then for $A = \{V : \lambda_k(V) \le s\}$, we have $\mathbb{P}(A) \ge \frac{1}{2}$ and

$$\mathbb{P}(A) \wedge \cdot \mathbb{P}\left(V : \frac{1}{2k}\lambda_k(V) \ge \frac{\mathrm{Med}(\lambda_k(V))}{2k} + \frac{t}{2k}\right) \le \exp(-\frac{ct^2}{k^2}),$$

so we have an upper bound.

For the lower bound, take $s + t = \mathrm{Med}(\lambda_k)$ and the same set A, then

$$A_t^c = \{V : \lambda_k(V) \ge \mathrm{Med}(\lambda_k(V))\},$$

and so, by Talagrand's theorem,

$$\mathbb{P}\left(V : \frac{\lambda_k(V)}{2k} \le \frac{\mathrm{Med}(\lambda_k(V)) - t}{2k}\right) \wedge \frac{1}{2} \le \exp(-\frac{ct^2}{k^2}).$$

Theorem is proved. $\square$

18. 11/12 Applications of Talagrand's convex concentration

The first example we started to discuss in previous class (Alon-Krivelevich-Vu theorem). Recall that our choice of weights for the distance function was the most non-trivial part of the proof (and this is usual for Talagrand's inequality applications).

Remark 18.1. If A is an $n \times n$ Hermitian matrix with independent entries $(a_{ij})_{i<j}$ such that $|a_{ij}| \le 1$. Let $\lambda_n \le \cdots \le \lambda_1$ be eigenvalues of A. Then

$$\mathbb{P}\{|\lambda_k(A) - \mathrm{Med}(\lambda_k(A))| > t\} \le 2\exp(-\frac{ct^2}{k^2}) \le$$

$$\le 2\exp(-\frac{ct^2}{(k \wedge (n-k))^2}).$$

Mark Meckes proved that the right hand side can be improved to

$$\ldots \le 2\exp(-\frac{ct^2}{k \wedge (n-k)}).$$

Note that these results can be also extended to the singular values of rectangular matrices by a standard trick: let B be an $n \times m$ matrix. Then the eigenvalues of the block matrix

$$\begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}$$

are equal to the singular values of B and we can apply concentration results for Hermitian matrices.

Example 18.2. (Bin packing)

Let $X_1 \dots X_n \in [0,1]$ be independent random variables. We want to pack $X_1, \dots, X_n$ to the minimal number of bins, such that the sum of the elements in one bin is at most 1.

Let $F(X)$ be the minimal number of bins needed. Further, let

$$A_s = \{x \in [0,1]^n, F(x) \leq s\}$$

and $s, t > 0$. Let $X \in A_{s+t}$, $Y \in A_s$. Then

$$t \leq F(X) - F(Y) \leq 2 \sum_{j=1}^{n} X_j \mathbb{1}_{X \neq Y}(j) + 1 =$$

(the weight of every bin except 1 is at least 1/2 (otherwise we can merge two smallest bins), so the number of these bins is at most twice their weight)

$$= 2\|X\|_2 \sum_{j=1}^{n} \frac{X_j}{\|X\|_2} \mathbb{1}_{X \neq Y}(j) + 1 \leq 2\|X\|_2 \cdot d_T(X, A) + 1.$$

Hence,

$$\mathbb{P}\{A_{s+t}^c\} \leq \mathbb{P}\{F(X) \geq t + F(Y) \ \forall Y \in A_s\} \leq$$
$$\leq \mathbb{P}\{d_T(X, A_s) \geq \frac{t-1}{2\|X\|_2}\}.$$

We would like to apply Talagrand's inequality now, and the only problem is that we have a random variable in the denominator. Let us use that $\|X\|_2$ is concentrated around its expectation, and so it is suitably bounded. Let

$$\Sigma^2 = \sum_{j=1}^{n} \mathbb{E} X_i^2.$$

Then

$$\mathbb{P}(A_s) \cdot \mathbb{P}(A_{s+t}^c) \leq \mathbb{P}(A_s) \cdot \Big[\mathbb{P}\big(d_T(X, A) \leq \frac{t-1}{2(\Sigma + u)}\big) +$$
$$+ \mathbb{P}\big(\|X\|^2 > (\Sigma + u)^2\big)\Big] \leq$$

(first term by Talagrand's inequality, and second term by the trivial fact $\mathbb{P}(A_s) \leq 1$)

$$\leq \exp\left(-\frac{(t-1)^2}{16(\Sigma + u)^2}\right) + \mathbb{P}(\|X\|_2^2 \geq \Sigma^2 + u^2).$$

Note that
$$\|X\|_2^2 - \Sigma^2 = \sum_{j=1}^n (X_j^2 - \mathbb{E}X_j^2) =: \sum_{j=1}^n Z_j,$$
and $\mathbb{E}Z_j = 0$, $|Z_j| \le 1$. So we are in the position to apply Bennet. Let
$$\Delta = \sum_{j=1}^n \mathrm{Var}(X_j^2).$$
By Bennet inequality,
$$\mathbb{P}\{\|X\|_2^2 \ge \Sigma^2 + u^2\} \le \begin{cases} \exp(-c\frac{u^4}{\Delta}), & \text{if } u^2 \le c'\Delta, \\ \exp(-cu^2 \log \frac{cu^2}{\Delta}), & \text{if } u^2 > c'\Delta. \end{cases}$$
Note that $\mathrm{Var}\, X_i^2 \le \mathbb{E}X_i^4 \le \mathbb{E}X_i^2$, so $\Delta \le \Sigma^2$. Hence, it does not make sense to choose $u^2 < c'\Delta$ (it does not improve the first term and makes the second term bigger). So, for all good choices of u, we will have Poissonian tail. Thus
$$\mathbb{P}(A_s)\mathbb{P}(A_{s+t}^c) \le \exp(-\frac{(t-1)^2}{16(\Sigma+u)^2}) + 2\exp(-cu^2 \log \frac{cu^2}{\Delta})$$
and $u^2 > c'\Delta$. Optimizing over u, we can obtain a good estimate.

Log-concave measures.

Definition 18.3. *A Borel measure μ in $\mathbb{R}^n$ is called log-concave, if for any compact sets $A, B \subset \mathbb{R}^n$ and any $\lambda \in [0,1]$,*
$$\mu(\lambda A + (1-\lambda)B) \ge (\mu(A))^\lambda (\mu(B))^{1-\lambda}.$$
Here
$$\lambda A + (1-\lambda)B = \{\lambda x + (1-\lambda)y : x \in A, y \in B\}.$$

Theorem 18.4 (Borell)**.** *Let μ be a finite Borel measure in $\mathbb{R}^n$. Assume that*
$$\dim(\mathrm{supp}(\mu)) = n.$$
The following are equivalent:

(1) *μ is log-concave;*
(2) *$\mu \ll m$ and if $f = \frac{d\mu}{dm}$, then $\log f$ is concave ($d\mu = e^{-u(x)}dx$, where $u : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is convex).*

We will prove and use only the implication (2) $\Rightarrow$ (1). To this end, we need the following result.

Theorem 18.5 (Prekopa-Leindler inequality)**.** *Let $f, g, h : \mathbb{R}^n \to [0, +\infty)$ be measurable functions, and $\lambda \in [0, 1]$. Assume that for any $x, y \in \mathbb{R}^n$,*

$$h(\lambda x + (1 - \lambda)y) \geq (f(x))^\lambda (g(y))^{1-\lambda}.$$

Then

$$\int_{R^n} h dx \geq (\int_{R^n} f dx)^\lambda (\int_{R^n} g dx)^{1-\lambda}$$

This is a functional version of Brunn-Minkowsky inequality. Its main advantage is that it can be tensorized (one more example of tensorizable inequality after log-Sobolev and Poincare inequalities).

Proof. We proceed by induction. Consider $n = 1$.

Case 1. Assume that $f, g \in C(\mathbb{R})$ and $f, g : \mathbb{R} \to (0, +\infty)$. Define a function $\alpha : (0, 1) \to \mathbb{R}$ by

$$\int_{-\infty}^{\alpha(t)} f(x)dx = t \int_{\mathbb{R}} f(x)dx.$$

Then α is differentiable on $(0, 1)$ and

$$\lim_{t \to 0} \alpha(t) = -\infty,$$

$$\lim_{t \to 1} \alpha(t) = +\infty$$

and, differentiating both sides,

$$\alpha'(t) f(\alpha(t)) = \int_{\mathbb{R}} f dx.$$

Similarly, define $\beta : (0, 1) \to \mathbb{R}$ by

$$\int_{-\infty}^{\beta(t)} g(y)dy = t \int_{\mathbb{R}} g dy.$$

Then β posesses similar properties and

$$\beta'(t) = \frac{\int_{\mathbb{R}} g dy}{g(\beta(t))}.$$

Define $z(t) := \lambda\alpha(t) + (1 - \lambda)\beta(t)$. This is a strictly monotone function mapping $\mathbb{R}$ onto $(0, 1)$. Let us use it to change the variables. We have

$$\int_{\mathbb{R}} h(z)dz = \int_0^1 h(\lambda\alpha(t) + (1 - \lambda)\beta(t))(\lambda\alpha'(t) + (1 - \lambda)\beta'(t))dt \geq$$

(by the condition on functions)

$$\geq \int_0^1 (f(\alpha(t))^\lambda (g(\beta(t))^{1-\lambda} \cdot (\lambda \frac{\int f dx}{f(\alpha(t))} + (1 - \lambda) \frac{\int g dy}{g(\beta(t))})dt \geq$$

(by arithmetic-geometric mean)

$$\geq \int_0^1 (f(\alpha(t))^{\lambda}(g(\beta(t))^{1-\lambda} \cdot (\frac{\int f dx}{f(\alpha(t))})^{\lambda} \cdot (\frac{\int g dy}{g(\beta(t))})^{1-\lambda} dt =$$

$$= (\int f dx)^{\lambda} (\int g dy)^{1-\lambda}.$$

This proves the Prekopa-Leindler inequality in the one-dimensional case for continuous strictly positive functions. To complete the proof for $n = 1$, we have to remove the assumptions of continuity and strict positivity. We can try to use the density of continuous functions. This idea can be implemented, but we have to perform the extension accurately. For any $f, g : \mathbb{R} \to [0, +\infty)$, we can easily construct sequences of continuous strictly positive functions $\{f_n\}_{n=1}^{\infty}$ and $\{g_n\}_{n=1}^{\infty}$ such that $f_n \to f$ and $g_n \to g$ both almost everywhere and in L_1. However, for each pair (f_n, g_n), we also have to construct a function h_n satisfying the assumption of the Prekopa-Leindler inequality. It is not clear a priori, whether such functions h_n would converge to h.

We will complete the proof of the Prekopa-Leindler inequality in the next lecture consecutively extending it to broader and broader classes of functions. Inductive step is easier and will be also proved next time. □

19. 11/17 Prekopa-Leindler inequality and its corollaries

Last time we started to prove Prekopa-Leindler inequality,

Theorem 19.1 (Prekopa-Leindler inequality). *Let $f, g, h : \mathbb{R}^n \to [0, +\infty)$ be measurable functions, and $\lambda \in [0, 1]$. Assume that for any $x, y \in \mathbb{R}^n$,*

$$h(\lambda x + (1-\lambda)y) \geq (f(x))^{\lambda}(g(y))^{1-\lambda}.$$

Then

$$\int_{R^n} h dx \geq (\int_{R^n} f dx)^{\lambda} (\int_{R^n} g dx)^{1-\lambda}$$

Proof. We proceed by induction.

Consider $n = 1$. If $f, g \in C(\mathbb{R})$ and $f, g : \mathbb{R} \to (0, +\infty)$, we are in the case discussed last time.

To complete the proof for $n = 1$, we have to remove the assumptions of continuity and strict positivity. For any $f, g : \mathbb{R} \to [0, +\infty)$, we can easily construct sequences of continuous strictly positive functions $\{f_n\}_{n=1}^{\infty}$ and $\{g_n\}_{n=1}^{\infty}$ such that $f_n \to f$ and $g_n \to g$ both almost everywhere and in L_1. However, for each pair (f_n, g_n), we also have to construct a function h_n satisfying the assumption of the Prekopa-Leindler inequality. It is not clear a priori, whether such functions h_n would converge to h.

First, let us remove the assumption of strict positivity, but add a condition on domain. Assume that $f, g \in C(\mathbb{R})$ and

$$\operatorname{supp}(f) \cup \operatorname{supp}(g) \subset [-N, N].$$

Then we can replace h by $\tilde{h}$, such that

$$\tilde{h} = \sup_{z=\lambda x+(1-\lambda)y} f^{\lambda}(x) g^{1-\lambda}(y).$$

Let $\varepsilon > 0$. Set

$$f_\varepsilon(x) = f(x) + \varepsilon \exp(-\frac{x^2}{2}),$$

$$g_\varepsilon(x) = g(x) + \varepsilon \exp(-\frac{x^2}{2}).$$

Define

$$\tilde{h}_\varepsilon(z) = \sup_{z=\lambda x+(1-\lambda)y} f_\varepsilon^{\lambda}(x) g_\varepsilon^{1-\lambda}(y).$$

Assume that $\|f\|_\infty, \|g\|_\infty \leq M$. Note that

$$\tilde{h}(z) \leq \tilde{h}_\varepsilon(z) \leq \sup_{z=\lambda x+(1-\lambda)y} (f^{\lambda}(x) + \varepsilon^{\lambda})(g^{1-\lambda}(y) + \varepsilon^{1-\lambda}) \leq \tag{17}$$

$$\leq \tilde{h}(z) + \varepsilon^{\lambda} M^{1-\lambda} + \varepsilon^{1-\lambda} M^{\lambda} + \varepsilon,$$

so

$$\tilde{h}_\varepsilon(z) \to \tilde{h}(z).$$

If $z \in [-N, N]$, then (17) shows that $\tilde{h}_\varepsilon(z)$ is uniformly bounded. If $z \notin [-N, N]$, then $x \notin [-N, N]$ or $y \notin [-N, N]$.

Hence,

$$\tilde{h}_\varepsilon(z) \leq \sup_{z=\lambda x+(1-\lambda)y} [\exp(-\frac{x^2}{2})^{\lambda}(M+1)^{1-\lambda} \vee \exp(-\frac{y^2}{2})^{1-\lambda}(M+1)^{\lambda}] \leq$$

$$\leq \exp(-\frac{z^2}{2})^{\lambda}(M+1)^{1-\lambda} \vee \exp(-\frac{z^2}{2})^{1-\lambda}(M+1)^{\lambda},$$

if we take $z > N$. This is an integrable majorant for h.

For a bounded domain continuity assumption can be easily removed. Indeed, we can extend the Prekopa-Leindler inequality to $f, g \in L_\infty(\mathbb{R})$ and

$$\operatorname{supp}(f) \cup \operatorname{supp}(g) \subset [-N, N],$$

using the dominated convergence theorem.

Finally, we can approximate non-negative measurable functions from below by bounded functions with bounded domains and apply the monotone convergence theorem completing the proof of the one-dimensional Prekopa-Leindler inequality.

Inductive step. Assume that P-L inequality holds in $\mathbb{R}^n$. Denote

$$F_{x_{n+1}}(\overline{x}) = f(\overline{x}, x_{n+1}),$$

where $x_{n+1} \in \mathbb{R}$, $\overline{x} \in \mathbb{R}^n$. Similarly,

$$G_{y_{n+1}}(\overline{y}) = g(\overline{y}, y_{n+1}),$$

$$H_{z_{n+1}}(\overline{z}) = h(\overline{z}, z_{n+1}).$$

Then, for any $(\overline{x}, x_{n+1})$, $(\overline{y}, y_{n+1}) \in \mathbb{R}^{n+1}$.

$$H_{\lambda x_{n+1}+(1-\lambda)y_{n+1}}(\lambda\overline{x} + (1-\lambda)\overline{y}) \geq F^{\lambda}_{x_{n+1}}(\overline{x}) G^{1-\lambda}_{y_{n+1}}(\overline{y}).$$

Hence, by the inductive assumption,

$$\int_{\mathbb{R}^n} H_{\lambda x_{n+1}+(1-\lambda)y_{n+1}}(\overline{z}) d\overline{z} \geq (\int_{\mathbb{R}^n} F_{x_{n+1}}(\overline{x}) d\overline{x})^{\lambda} \cdot (\int_{\mathbb{R}^n} G_{y_{n+1}}(\overline{y}) d\overline{y})^{1-\lambda}.$$

Using P-L inequality in $\mathbb{R}$, we complete the proof.

$\square$

Definition 19.2. *A random vector X is called log-concave, if its desity has the form*

$$f_X(y) = \exp(-u(y)),$$

where $u : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is a convex function.

Corollary 19.3. *Let X be a log-concave vector in $\mathbb{R}^n$. Then for any comparct (Borel) sets A and B,*

$$\mathbb{P}\{X \in \lambda A + (1-\lambda)B\} \geq \mathbb{P}^{\lambda}\{X \in A\} \cdot \mathbb{P}^{1-\lambda}\{X \in B\}.$$

To prove this corollary, set

$$f(x) = \exp(-u(x))\mathbb{1}_A(x),$$

$$g(y) = \exp(-u(y))\mathbb{1}_B(y),$$

$$h(z) = \exp(-u(z))\mathbb{1}_{\lambda A+(1-\lambda)B}(z).$$

Then the triple f, g, h satisfies the assumption of the Prekopa-Leindler inequality.

Applying this corollary to the normalized lebesgue measure in a large cube, we derive the multiplicative Brunn-Minkowski inequality: for any compact sets $A, B \subset \mathbb{R}^n$,

$$m_0(\lambda A + (1-\lambda)B) \geq m_0^{\lambda}(A) m_0^{1-\lambda}(B).$$

(Here, m_n denotes Lebesgue measure in $\mathbb{R}^n$.)

Exercise: derive the classical Brunn-Minkowski inequality from it, that is

$$m_n^{1/n}(A+B) \geq m_n^{1/n}(A) + m_n^{1/n}(B).$$

Corollary 19.4. *Let X, Y be independent log-concave vectors in $\mathbb{R}^n$. Then $X+Y$ is log-concave.*

Corollary 19.5. *If $X \in \mathbb{R}^n$ is a log-concave random vector and $E \subset \mathbb{R}^n$ is a subspace, then $P_E X$ is a log-concave vector in E.*

We leave the proofs of these corollaries as exercises.

Let $K \subset \mathbb{R}^n$ be a convex body. Then the vector X uniformly distributed in K is log-concave.

Indeed, let $f_X(y) = (m_n(K))^{-1} \cdot \exp(-u(y))$, where

$$u(y) = \begin{cases} 0, & \text{if } y \in K, \\ +\infty, & \text{if } y \notin K. \end{cases}$$

In a sense, this is the only example of a log-concave random vector. More precisely, any absolutely continuous log-concave measure can be obtained as a limit of marginals of the uniform measures over some sequence of high-dimensional convex bodies.

Maurey infimum convolution method.

We will reprove the gaussian concentration using the P-L inequality now.

Let A be a Borel set in $\mathbb{R}^n$. Set

$$h(x) = \exp(-\|x\|_2^2/2),$$

$$f(x) = \exp(-\frac{\|x\|_2^2}{2}) \cdot \mathbb{1}_A(x)$$

and let us choose g satisfying the P-L condition with $\lambda = \frac{1}{2}$. We will get it in the form

$$g(x) = \exp(v(x)).$$

We have to check that for all $x, y \in \mathbb{R}^n$ that

$$\exp\left(-\frac{1}{2}\left\|\frac{x+y}{2}\right\|_2^2\right) \geq \exp\left(-\frac{\|x\|_2^2}{4}\right) \cdot \mathbb{1}_A(x) \cdot \exp\left(\frac{1}{2}v(y)\right),$$

Since this inequality holds trivially for $x \notin A$, we have to ensure that for all $y \in \mathbb{R}^n$ and all $x \in A$,

$$\left\|\frac{x+y}{2}\right\|_2^2 \leq \frac{\|x\|_2^2}{2} - v(y).$$

Fix $y \in \mathbb{R}^n$. We can choose

$$v(y) = \inf_{x \in A}\left[\frac{\|x\|_2^2}{2} - \left\|\frac{x+y}{2}\right\|_2^2\right] = \inf_{x \in A}\left[\left\|\frac{x-y}{2}\right\|^2 - \frac{\|y\|_2^2}{2}\right] =$$

(parallelogram law)

$$= \frac{\operatorname{dist}^2(y, A)}{4} - \frac{\|y\|_2^2}{2}.$$

Hence, by the P-L inequality,

$$\int_{\mathbb{R}^n} \exp(-\frac{\|z\|_2^2}{2})dz$$

$$\geq \left(\int_A \exp(-\frac{\|x\|_2^2}{2})dx\right)^{1/2} \cdot \left(\int_{\mathbb{R}^n} \exp(\frac{\operatorname{dist}^2(y, A)}{4}) \exp(-\frac{\|y\|_2^2}{2})dy\right)^{1/2},$$

or,

$$1 \geq \gamma_n(A) \cdot \int_{\mathbb{R}^n} \exp(\frac{\operatorname{dist}^2(y, A)}{4})d\gamma(y).$$

This is gaussian concentration on set A. Recall that we proved this for every Borel set A.

A more delicate application of such technique proves the concentration of the n-dimensional exponential measure

$$d\nu_n(x) = 2^{-n}\exp(-\|x\|_1)dx.$$

Namely, it allows to establish the following theorem of Talagrand.

Theorem 19.6. *Let $f \in C^1(\mathbb{R}^n)$ be a function satisfying*

$$\forall x \in \mathbb{R}^n \quad \|\nabla f(x)\|_2 \leq \alpha \quad \text{and} \quad \|\nabla f(x)\|_\infty \leq \beta.$$

Let $X \in \mathbb{R}^n$ be a random vector with the density $f_X(x) = 2^{-n}\exp(-\|x\|_1)$. Then,

$$\mathbb{P}(f(X) \geq \operatorname{Med}(f(X)) + t) \leq \exp\left(-C\left[\frac{t^2}{\alpha^2} \wedge \frac{t}{\beta}\right]\right).$$

Also, Maurey's infimum convolution method allows to prove the Talagrand convex concentration inequality.

20. 11/19 Borel's lemma and its corollaries

Recall that a random variable X is called *log-concave*, if its density

$$f_X(x) = C\exp(-u(x)),$$

where $u : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is convex.

Theorem 20.1 (Borel)**.** *Let $X \in \mathbb{R}^n$ be a log-concave random vector. Let $K \subset \mathbb{R}^n$ be a convex symmetric closed set. Then for any $t > 1$,*

$$\mathbb{P}((tK)^c) \leq \mathbb{P}(K) \cdot \left(\frac{1-\mathbb{P}(K)}{\mathbb{P}(K)}\right)^{\frac{t+1}{2}}.$$

Proof. Set $\|x\|_K = \inf\{\lambda > 0 : x \in \lambda K\}$.

We want to choose $\lambda \in (0,1)$, such that for all $\overline{x} \in (tK)^c$ and for all $y \in K$ we have $\lambda\overline{x} + (1-\lambda)y \in K^c$. In other words, for all $x, y \in \mathbb{R}^n$ $\|x\|_K > 1$, $\|y\|_K \leq 1$ we want to have $\|\lambda tx + (1-\lambda)y\|_K > 1$.

It is enough to require that

$$\lambda t\|x\|_K - (1-\lambda)\|y\|_K \geq 1.$$

We need to take λ satisfying

$$\lambda t - (1-\lambda) \geq 1,$$

or

$$\lambda \geq \frac{2}{t+1}.$$

Then, since the measure $\mathbb{P}(A)$ is log-concave,

$$\mathbb{P}^{\lambda}((tK)^c) \cdot \mathbb{P}^{1-\lambda}(K) \leq \mathbb{P}(K^c),$$

so

$$\left(\frac{\mathbb{P}(tK)^c)}{\mathbb{P}(K)}\right)^{\lambda} \leq \frac{1-\mathbb{P}(K)}{\mathbb{P}(K)}.$$

□

Corollary 20.2. *Let $f : \mathbb{R}^n \to [0, +\infty)$ be an even, quasi-convex 1-homogeneous function. Let X be a log-concave random variable Then for any $p > 1$*

$$(\mathbb{E}f^p(X))^{1/p} \leq Cp\mathbb{E}f(X).$$

Proof.

$$\mathbb{E}f^p(X) = \int_0^\infty pt^{p-1}\mathbb{P}(f(X) > t)dt.$$

Let

$$K := \{x \in \mathbb{R}^n : f(x) \leq 3\mathbb{E}f(X)\}.$$

By Markov inequality, $\mathbb{P}(K) \geq \frac{2}{3}$. For any $t = \lambda 3\mathbb{E}f(X)$ with $\lambda > 1$, $f(x) > t$, we have $x \notin \lambda K$ (by 1 - homogenuity). So, we can apply Borell's lemma:

$$\mathbb{P}(f(X) > \lambda \cdot 3\mathbb{E}f(X)) \leq \frac{2}{3} \cdot \left(\frac{1}{2}\right)^{\frac{\lambda+1}{2}}.$$

Hence,

$$\mathbb{E}f^p(X) = \int_0^{3\mathbb{E}f(X)} pt^{p-1}dt + \int_{3\mathbb{E}f(x)}^\infty pt^{p-1}\left(\frac{1}{2}\right)^{t/(6\mathbb{E}f(X))} dt \leq$$

$$\leq (3\mathbb{E}f(X))^p + (6\mathbb{E}f(X))^p\Gamma(cp) \leq (Cp\mathbb{E}f(X))^p.$$

□

Remark 20.3. (Khinchine-type inequality for log-concave random variables)

Note that the function

$$f(x) = |\langle x, u\rangle|,$$

for $u \in \mathbb{R}^n$ satisfies the conditions of the lemma. We obtain

$$\mathbb{E}|\langle x, u\rangle| = (\mathbb{E}|\langle x, u\rangle|^p)^{1/p} \leq Cp\mathbb{E}|\langle X, u\rangle|.$$

Theorem 20.4 (Kahane's inequality)**.** *Let $p > 1$. Let Y be a normed space. Then for any $x_1, \ldots, x_n \in Y$,*

$$\mathbb{E}\|\sum_{i=1}^n \varepsilon_i x_i\|_Y \leq (\mathbb{E}\|\sum_{j=1}^n \varepsilon_j x_j\|_Y^p)^{1/p} \leq B_p \cdot \mathbb{E}\|\sum_{i=1}^n \varepsilon_i x_i\|_Y.$$

Proof. Lower bound is true by Holder. For the upper bound, let us replace ε_i by i.i.d. random variables, uniformly distributed in $[-1, 1]$. If $A_1, \ldots, A_n \sim \mathrm{Unif}([-1, 1])$, then $A = (A_1, \ldots, A_n)$ is a log-concave random measure. Set

$$f(a_1, \ldots, a_n) = \|\sum_{i=1}^n a_i x_i\|_Y.$$

Then f is even, 1-homogeneous and convex. Hence, the original inequality holds from the Corollary 20.2.

Now we need to return to the discrete version (with the Rademacher ε_i). We have to show that

$$\mathbb{E}\| \sum A_i x_i \|_Y \le \mathbb{E}\| \sum \varepsilon_i x_i \|_Y \tag{18}$$

and

$$\mathbb{E}\| \sum A_i x_i \|_Y^p \ge 2^{-p}\mathbb{E}\| \sum \varepsilon_i x_i \|_Y^p. \tag{19}$$

Note that

$$\mathbb{E}\| \sum A_i x_i \|_Y^p = \mathbb{E}\| \sum A_i \varepsilon_i X_i \|_Y^p = \mathbb{E}\| \sum |A_i| \varepsilon_i x_i \|_Y^p.$$

By convexity, for any $u, v \in Y$ and $a \in [0,1]$,

$$\|au + v\|_Y^p + \| -au + v\|_Y^p \le \|u+v\|_Y^p + \| -u + v\|_Y^p.$$

Hence,

$$\mathbb{E}\||A_1|\varepsilon_1 x_1 + \sum_{i=2}^n |A_i|\varepsilon_i x_i\|_Y \le \mathbb{E}\|\varepsilon_1 x_1 + \sum_{i=2}^n |A_i|\varepsilon_i x_i\|_Y \le \mathbb{E}\| \sum \varepsilon_i x_i \|_Y.$$

and (18) is checked.

Then, by Jensen's inequality,

$$\mathbb{E}\| \sum_{j=1}^n |A_j|\varepsilon_j x_j\|_Y^p \ge \mathbb{E}\| \sum_{j=1}^n \mathbb{E}|A_j|\varepsilon_j X_j\|_Y^p = \left(\frac{1}{2}\right)^p \mathbb{E}\| \sum \varepsilon_j x_j \|_Y$$

and (19) is checked as well.

□

The next theorem historically was the first place where concentration of measure (on the sphere) played the crucial role. Dvoretzky's theorem shows that every normed space has a high-dimensional subspace, which is almost Euclidean.

Theorem 20.5 (Milman's Dvvoretzky theorem)**.** *Let $\|.\|_X$ be a norm on $\mathbb{R}^n$. Set*

$$M(X) = \int_{S^{n-1}} \|y\|_X d\mu(y)$$

and

$$b(X) = \max_{y \in S^{n-1}} \|y\|_X.$$

Let

$$k = c(\varepsilon) n \left(\frac{M(X)}{b(X)}\right)^2.$$

Let $E \subset \mathbb{R}^n$ be a random subspace of $\dim E = k$, uniformly distributed over the Grassmanian. Then, with high probability, for all $x \in E$,

$$(1-\varepsilon) M(X) \|x\|_2 \le \|x\|_X \le (1+\varepsilon) M(X) \|x\|_2.$$

Remark 20.6. Let G be an $n \times k$ Gaussian matrix. Then $G\mathbb{R}^k$ is a random subspace uniformly distributed over $\mathbb{R}^n$. Next time we will prove Milman's theorem with the gaussian measure $\mu = \gamma$, then $M(X) = \mathbb{E}\|g\|_2$ (g is a gaussian vector) and we can estimate $\|Gx\|_X$ instead of $\|x\|_X$.

21. 11/24 Shechtman proof of the Milman-Dvoretzky's theorem

Theorem 21.1 (Dvoretzky)**.** *Let $\|.\|_Y$ be a norm in $\mathbb{R}^n$. For any $\varepsilon > 0$, there exists $k > \varphi(\varepsilon, n)$ and a linear operator $T : l_2^k \to (\mathbb{R}^n, \|.\|_Y)$, such that for any $x \in l_2^k$*

$$(1-\varepsilon)\|x\|_2 \leq \|Tx\|_Y \leq (1+\varepsilon)\|x\|_2.$$

Remark 21.2. The estimates for $\varphi(\varepsilon, n)$ were established by

- Dvoretzky: $\varphi(\varepsilon, n) \geq C(\varepsilon)\sqrt{\log n}$
- V.Milman: $\varphi(\varepsilon, n) \geq \frac{C\varepsilon^2}{\log 1/\varepsilon} \log n$
- Gordon: $\varphi(\varepsilon, n) \geq C\varepsilon^2 \log n$

Remark 21.3. Note that D theorem as it is has no probability in formulation, but all known proofs goes via probability.

Theorem 21.4. *Let $\|.\|_Y$ be a norm on R^n. Let $g \in \mathbb{R}^n$ be the standard gaussian vector. Set*

$$b(Y) = \max_{x \in S^{n-1}} \|x\|_Y.$$

Then a random subspace $E \subset \mathbb{R}^n$ of dimension

$$k = C\varepsilon^2 \frac{\mathbb{E}\|g\|_Y}{b(Y)}$$

satisfies

$$\forall x \in E \;\; (1-\varepsilon)\|x\|_2 \leq \|x\|_Y \leq (1+\varepsilon)\|x\|_2.$$

Proof. (of Theorem 21.4 by Schehtman)

Let G be an $n \times k$ Gaussian matrix. Set $E = G\mathbb{R}^k$. Then E is uniformly distributed over the Grassmanian. We will show that for any unit vector x, with high probability

$$(1-\varepsilon)l(Y) \leq \|Gx\|_Y \leq (1+\varepsilon)l(Y) \tag{20}$$

for some constant $l(Y) > 0$ (l and b are parameters of the space).

An important twist, added in the proof by Schechtman, is a better concentration estimate for $\|Ga\|_Y,\ a \in S^{n-1}$, based on the independence properties of the marginals of the Gaussian random vector.

Lemma 21.5. *Let $a, b \in S^{k-1}$. Then*

$$\mathbb{P}\{\|Ga\|_Y - \|Gb\|_Y| \geq t\} \leq 2\exp\left(-\frac{ct^2}{b^2(Y)\|a-b\|_2^2}\right).$$

Proof. WLOG, assume that $a \neq b$ and $a \neq -b$. Let $e = \frac{a+b}{\|a+b\|_2}$. Then

$$P_e a = P_e b = \frac{\|a\|_2^2 + \langle a, b\rangle}{\|a+b\|^2} e = \frac{1 + \langle a, b\rangle}{\|a+b\|^2} e.$$

Set $u = a - P_e a$. Then

$$\mathbb{P}\{|\|Ga\|_Y - \|Gb\|_Y| > t\} = \tag{21}$$

$$\mathbb{P}\{|\|GP_e a + Gu\|_Y - \|GP_e a - Gu\|_Y| > t\}.$$

Since $P_e a$ is orthogonal to u and both $GP_e a$ and Gu are gaussian vectors, they are uncorrelated, and so, independent. We can condition on $GP_e a$. Consider a function

$$f(G) = \|y + Gu\|_Y$$

for some $y \in \mathbb{R}^n$. Let W, V be $n \times k$ matrices. Then

$$|f(W) - f(V)| \leq \|(W-V)u\|_Y = \sup_{y^* \in B_{y^*}} \sum_{i=1}^{n} \sum_{j=1}^{k} (w_{ij} - v_{ij}) u_j y_i^* \leq$$

(Cauchy-Swartz)

$$\leq (\sum_{i,j} (W_{ij} - V_{ij})^2)^{1/2} \sup_{y^* \in B_{y^*}} (\sum_{i,j} u_j^2 (y_i^*)^2)^{1/2} =$$

$$= \|W - U\|_F \cdot \|u\|_2 \sup_{y^* \in B_{y^*}} \|y^*\|_2.$$

Here,

$$\sup_{y^* \in B_{y^*}} \|y^*\|_2 = \sup_{y^* \in B_{y^*}} \sup_{x \in S^{n-1}} \langle x, y^* \rangle \leq \sup_{x \in S^{n-1}} \|x\|_Y = b(Y).$$

This shows that f is a Lipschitz function with the Lipschitz constant $\frac{\|a-b\|_2}{2} b(Y)$. Hence, by the Gaussian concentration,

$$\mathbb{P}\{|\|y + Gu\|_Y - \mathbb{E}\|y + Gu\|_Y| \geq t\} \leq 2\exp\left(-\frac{ct^2}{b^2(Y)\|a-b\|_2^2}\right).$$

Similarly,

$$\mathbb{P}\{|\|y - Gu\|_Y - \mathbb{E}\|y - Gu\|_Y| \geq t\} \leq 2\exp\left(-\frac{ct^2}{b^2(Y)\|a-b\|_2^2}\right).$$

Note that $\mathbb{E}\|y+Gu\|_Y = \mathbb{E}\|y-Gu\|_Y$. The proof finishes by using (21) with $y = GP_e a$ conditionally on y and removing the condition.

$\square$

We will derive the proof of Milman's theorem from this lemma. (This is an application of Dudley's inequality, that we did not discuss before. So we will prove it now for the particular case). We consider sequence of nets.

Let N_l be a 2^{-l}-net in S^{k-1}. By the volumetric estimate, we can assume that

$$|N_l| \le (1+\frac{2}{2^{-l}})^k \le (3\cdot 2^l)^k.$$

Let $\delta_l > 0$ be such that

$$\sum_{l=1}^{\infty} \delta_l = 1.$$

For any $x \in S^{n-1}$ let $\pi_l(x) \in N_l$ be a point such that

$$\|x-\pi_l(x)\|_2 < 2^{-l}.$$

We can write

$$x = \sum_{j=1}^{\infty}(\pi_{j+1}(x) - \pi_j(x)) + \pi_1(x).$$

Recall (20): we have to show that for any $x \in S^{n-1}$,

$$||\|Gx\|_Y - l(Y)| < 2\varepsilon l(Y). \tag{22}$$

Assume that there exists $x \in S^{k-1}$, such that (22) does not hold. We claim that in this case

$$\exists j \in \mathbb{N}\ \ ||\|G\pi_{j+1}(x)\|_Y - \|G\pi_j(x)\|_Y| \ge \varepsilon\delta_j l(Y),$$

or

$$||\|G\pi_1(x)\|_Y - l(Y)| \ge \varepsilon\delta_1 l(Y).$$

Indeed, if it doesn't hold, then

$$||\|Gx\| - l(Y)| \le$$

$$\le ||\|G\pi_1(x)|_Y - l(Y)| + \sum_{j=1}^{\infty} ||\|G\pi_{j+1}(x)\|_Y - \|G\pi_j(x)\|_Y| \le$$

$$\le \varepsilon\delta_1 l(Y) + \varepsilon l(Y)\sum_{j=1}^{\infty}\delta_j \le 2\varepsilon l(Y).$$

So, we need to check countably many conditions, i.e. can apply union bound:

$$p := \sum_{j=1}^{\infty} \mathbb{P}(|\|G\pi_{j+1}(x)\|_Y - \|G\pi_j(x)\|_Y| \geq \varepsilon\delta_j l(Y)) \leq$$

(by Lemma 21.5)

$$\leq \sum_{j=1}^{\infty} |N_j| \exp\left(-\frac{c\varepsilon^2\delta_j^2 l^2(Y)}{b^2(Y)(2\cdot 2^{-j})^2}\right) = \sum_{j=1}^{\infty} \exp\left(kCj - \frac{c\varepsilon^2\delta_j^2 l^2(Y)}{b^2(Y)} 2^{2j}\right).$$

Set $\delta_j = c' 2^{-j}\sqrt{j}$ choosing c' such that $\sum_1^\infty \delta_j = 1$. Then

$$p \leq \sum_{j=1}^{\infty} \exp\left(Ckj - \frac{c\varepsilon^2 l^2(Y)}{b^2(Y)} \cdot j\right) \leq \exp(-ck),$$

if we take

$$Ck - \frac{\varepsilon^2 l^2(Y)}{b^2(Y)} \leq -k,$$

or

$$k \leq \frac{c\varepsilon^2 l^2(Y)}{b^2(Y)}.$$

□

22. 12/1 More on Dvoretzky's theorem

(Dvoretzky dimension of l_p spaces, Dvoretzky-Rogers lemma without proof, John's theorem.)

Was typed by Feng Wei. Please write to Feng to get it.

23. 12/3 Vertices and facets of a symmetric polytope Concentration for non-linear functions

Corollary 23.1. *Let Y be a normed space,* $\dim Y = n$. *Then there exists a Euclidean structure on Y, such that*

$$k(Y)\cdot k(Y^*) \geq C\varepsilon^4 n$$

(A Euclidean structure on Y defines a Euclidean structure on the dual space Y^* : if $y \in Y$ and $y^* \in Y^*$, then $y^*(y) = \langle y, y^* \rangle$.)

Application to combinatorial geometry.

Definition 23.2. *Let $K \subset \mathbb{R}^n$ be a convex body. Assume that its origin is in the interior of K. Define the polar body K° as*

$$\mathrm{K}^\circ = \{x \in \mathbb{R}^n : \forall y \in K \ \ \langle x, y \rangle \leq 1\}.$$

If K is the unit ball of Y, then K° is the unit ball of Y^*.

Theorem 23.3. *Let $K \subset \mathbb{R}^n$ be a convex **symmetric** polytope. Let $V(K)$ denote the number of vertices of K. Then*

$$\log V(K) \cdot \log(V(\mathrm{K}^\circ)) \geq cn.$$

Equivalently, if $f(K)$ is the number of facets of K, then

$$\log V(K) \cdot \log(f(K)) \geq cn.$$

The second formulation is purely combinatorial. Indeed, it does not refer to the metric in $\mathbb{R}^n$. Yet, the only known proof of this deterministic combinatorial result goes via Milman's Dvoretzky theorem, whose proof is probabilistic.

The theorem means that a symmetric polytope cannot have too few facets and vertices simultaneously. The symmetry of the polytope plays a crucial role here. Consider the following example.

Example 23.4. If $K = \Delta_n \subset \mathbb{R}^n$ is the standard simplex, then $V(K) = n+1$ and $f(K) = n+1$, so

$$\log V(K) \cdot \log(f(K)) = O(\log^2 n).$$

To prove this theorem, we need the following lemma showing that a polytope close to the Euclidean ball must have exponentially many facets.

Lemma 23.5. *Let $Y = (\mathbb{R}^k, \|.\|_Y)$ be a normed space with*

$$\|x\|_Y = \max_{j=1,\dots,N} |\langle x, y \rangle|.$$

If for any $x \in Y$

$$\frac{1}{2}\|x\|_2 \leq \|x\|_Y \leq 2\|x\|_2, \tag{23}$$

then

$$\log N \geq ck = c \dim Y.$$

Proof. By (23), for every $l \in \{1, \ldots, N\}$,
$$\|y_l\|_2^2 = \max_{j=1,\ldots,N} |\langle y_j, y_l \rangle| \leq \|y\|_2,$$
so
$$\|y_l\|_2 \leq 2.$$
Also by (23),
$$\frac{1}{4}\sqrt{k} \leq \frac{1}{2}\mathbb{E}\|g\|_2 \leq \mathbb{E}\|g\|_Y = \mathbb{E} \max_{j=1,\ldots,N} |\langle g, y_j \rangle| \leq$$
$$\leq \mathbb{E}(\sum_{j=1}^{N} |\langle g, y_j \rangle|^p)^{1/p} \leq C\sqrt{p}N^{1/p} \leq C\sqrt{\log N}.$$
(if we take $p = \log N$ in the last step). $\square$

Proof. (of Theorem 23.3) Let Y be a normed space with the unit ball K. Then there exist a subspace $E \subset \mathbb{R}^n$, such that $\dim E \geq k(Y)$ and for all $x \in E$
$$\frac{1}{2}\|x\|_2 \leq \|x\|_Y \leq 2\|x\|_2.$$
Note that
$$f(K \cap E) \leq f(K) = V(\mathrm{K}^\circ).$$
By the Lemma 23.5,
$$\log V(\mathrm{K}^\circ) \geq ck(Y).$$
Similarly,
$$\log V(K) \geq ck(Y^*).$$
By the corollary,
$$\log V(K) \log V(\mathrm{K}^\circ) \geq c^2 k(Y)k(Y^*) \geq Cn.$$
$\square$

We would like to have some results for concentration of non-linear functions of log-concave random vectors. There is no independence (which would allow using LSI or Poincare inequality). But something still can be done. We will discuss Bourgain's result for polynomial functions.

Note that by Borell's lemma, if X is a log-concave vector, then
$$\mathbb{P}\{|\langle X, y \rangle| > t\mathbb{E}|\langle X, y \rangle|\} \leq e^{-ct}.$$
Hence,
$$\mathbb{P}\{|\langle X, y \rangle|^p > t\mathbb{E}|\langle X, y \rangle|^p\} \leq e^{-ct^{1/p}}.$$
If p is an even integer, we can view this as a nice concentration for one particular polynomial. Bourgain's theorem claims that the same estimate holds in the general case.

Theorem 23.6 (Bourgain)**.** *Let $p : \mathbb{R}^n \to \mathbb{R}$ be a polynomial of degree d. Let X be a log-concave random vector. Then for any $t > 1$,*

$$\mathbb{P}\{|p(X)| \geq t\mathbb{E}|p(X)|\} \leq \exp(-ct^{1/d}).$$

Idea of the proof.

We can assume that X is uniformly distributed over a convex body K, $\mathrm{Vol}(K) = 1$ (in general, log-concave measures can be represented as marginals of measures distributed over convex bodies, volume condition comes from the assumptions that our measure is a probability measure).

Define a set

$$K_\lambda = \{x \in K : |p(X)| > \lambda\}.$$

It is enough to show that the measure of K_λ decays fast enough as λ increases. More precisely, we will show that

$$m_n(K_{c_0^{-d}\lambda}) \geq c[m_n(K_\lambda)]^{2/3} \quad \text{for some } c_0 > 1. \tag{24}$$

Here, m_n denotes the n-dimensional Lebesgue measure.

To complete the proof, we iterate this inequality and observe that the product of constants converges to a new constant $(c^{1+2/3+(2/3)^2+\dots})$.

The proof of (24) is based on the following measure transportation result.

Theorem 23.7 (Knothe transformation)**.** *Let $A, B \subset \mathbb{R}^n$ be open sets. Assume that B is convex. Then there exists a C^1-function $\varphi : A \to B$, such that*

(1)
$$\varphi_1(x) = \varphi_1(x_1),$$
$$\varphi_2(x) = \varphi_2(x_1, x_2),$$
$$\dots$$
$$\varphi_n(x) = \varphi_n(x_1, \dots, x_n).$$

(2)
$$\forall x \in A \quad \partial_j \varphi_j(x) > 0.$$

(3)
$$\forall x \in A \quad J_\varphi(x) = \prod \partial_j \varphi_j(x) = \frac{m_n(B)}{m_n(A)}.$$

Here, J_φ denotes the Jacobian of φ.

Proof. For $x_1, \dots, x_m$, let

$$E_{x_1,\dots,x_m} = \{y \in \mathbb{R}^n : y_j = x_j \text{ for } j \leq m\}.$$

Define φ by

$$\frac{1}{m_n(A)}\int_{-\infty}^{x} m_{n-1}(A\cap E_y)dy = \frac{1}{m_n(B)}\int_{-\infty}^{\varphi_i(x)} m_n(B\cap E_y)dy.$$

Then

$$\partial_1\varphi_1(x) = \varphi'(x) = \frac{m_n(B)}{m_n(A)}\cdot\frac{m_{n-1}(A\cap E_x)}{m_{n-1}(B\cap E_{\varphi_1}(x))}.$$

Assume that $\varphi_1,\dots,\varphi_k$ are already constructed. Define φ_{k+1} by

$$\frac{1}{m_{n-k+1}(A\cap E_{x_1,\dots,x_{k-1}})}\int_{-\infty}^{x} m_{n-k}(A\cap E_{x_1,\dots,x_{k-1},y})dy =$$

$$= \frac{1}{m_{n-k+1}(B\cap E_{\varphi_1(x_1),\dots,\varphi_{k-1}(x_1,\dots,x_{k-1})})}\cdot$$

$$\cdot\int_{-\infty}^{\varphi(x_1,\dots,x_{k-1},x)} m_{n-k}(A\cap E_{\varphi_1(x_1),\dots,\varphi_{k-1}(x_1,\dots,x_{k-1}),y})dy.$$

Then $\partial_j\varphi_j$ can be calculated as above, which verifies (1) and (2). To check (3), notice that $\prod\partial_j\varphi_j(x)$ is a telescopic product.

$\square$

Lemma 23.8. *There exists $c_0 > 1$, such that for any polynomial $f : [0,1] \to \mathbb{R}$ of degree d,*

$$m_1\{f\in[0,1] : \quad |f(t)| \geq c_0^{-d}\|f\|_\infty\} > \frac{1}{2}.$$

We leave the proof as an exercise.

Proof. (of Theorem 23.6). Let $\lambda > 1$. Let φ be a Knothe transform $\varphi : K_\lambda \to K$. Set

$$\varphi_t(x) = tx + (1-t)\varphi(x)$$

and

$$p_x(t) = p(\varphi_t(x)).$$

Then φ_t is one-to-one. Indeed, if $\varphi_t(x) = \varphi_t(y)$, then $x_1 = y_1$ since $\partial_1(\varphi_t)_1 > 0$. After establishing this, we can consider $\partial_2(\varphi_t)_2$ and show that $x_2 = y_2$ etc.

Note that p_x is a polynomial of t of degree at most d. By Lemma 23.8, for any $x \in K_\lambda$,

$$m_1(t : |p_x(t)| > c_0^{-d}\lambda) \geq m_1(t : |p_x(t)| > c_0^{-d}\cdot\|p_x\|_\infty) > \frac{1}{2}.$$

Hence,

$$\frac{1}{m_n(K_\lambda)}\int_{K_\lambda} m_1(t : |p_x(t)| > c_0^{-d}\lambda)dx > \frac{1}{2},$$

which can be rewritten as

$$\int_0^1 \frac{m_n(x \in K_\lambda : |p_x(t)| > c_0^{-d} x)}{m_n(K_\lambda)} dt > \frac{1}{2}.$$

Therefore,

$$\exists t \leq \frac{2}{3} : \quad m_n(x \in K_\lambda : |p_x(t)| > c_0^{-d}\lambda) \geq \frac{1}{4} m_n(K_\lambda). \tag{25}$$

Fix such t and define a set

$$S := \{x \in K_\lambda : |p_x(t)| > c_0^{-d}\lambda\}.$$

Then we have

$$m_n(\varphi_t(S)) = \int_S |J_{\varphi_t}(x)| dx = \int_S \prod_{j=1}^n |\partial_j \varphi_{t,j}(x)| dx =$$

$$= \int_S \prod_{j=1}^n (1 \cdot t + (1-t)\partial_j \varphi_j(x)) dx \geq$$

(by the arithmetic-geometric mean inequality)

$$\geq \int_S \prod_{j=1}^n |\partial_j \varphi_j(x)|^{1-t} dx = \left(\frac{m_n(K)}{m_n(K_\lambda)} \right)^{1-t} \cdot m_n(S) \geq$$

(by (25))

$$\geq \frac{1}{4}(m_n(K_\lambda))^t \geq \frac{1}{4}(m_n(K_\lambda))^{2/3}.$$

By definition of S, $\varphi_t(S) \subset K_{c_0^{-d}\lambda}$. So, we proved that

$$m_n(K_{c_0^{-d}\lambda}) \geq \frac{1}{4}(m_n(K_\lambda))^{2/3}.$$

$\square$

I hope this notes were helpful to you!
For further reading see suggested literature on page 1 and don't forget
to send all corrections and suggestions to erebrova@umich.edu :)
Cheers, Liza